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非侵入式卵內雞蛋性別鑑定之前瞻性研究

A pioneer study on a non-invasive method for in ovo chicken egg sexing.

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Abstract

Globally, it is estimated that 7 billion of day-old male chicks from laying-hen hatcheries are killed annually. Due to the different production purposes of the meat and poultry strains, males are only preferred as breeding objects. This has become one of the biggest ethical problems for both the egg-producing industry and the humanity. Laws have been made in order to reduce the immorality of this doing but without a proper fully pledged technology to determine the chick sex before its life's been formed, the law has not been able to restrict industries from doing what they have been doing for decades.

In 2019, a new method has been developed for the detection of chicken egg gender. The Seleggt method is where a small amount of the egg content was extract through a laser-cut-hole on the egg shell. This egg content will then be use for Estrogen measurement. Since the amount of estrone sulfate in the female embryos is significantly higher than that in males at the time hormone has already presented in the egg, it can be used as a basis for identification. Although this method is said to have the same accuracy with the manual identification method on newly hatch chicks, due to the eggshell perforation sampling work, higher sanitary conditions are required for subsequent incubation facilities and environments to avoid possible pollution and incubation losses.

In this study, the aim is to develop a new non-invasive, non-chemical analysis method for the detection of chicken egg gender. By the use of the bioimpedance spectroscopy technique to distinguished the difference in chicken egg which has been previous confirmed by Burke and Sharp researches, it is believed that the egg gender can be identify. This method will not only solve the ethical problems of male chicks culling but also opens a new opportunities for animal studies to evaluate other species egg gender, viability, and quality.

The study also tries to build a measurement system specifically for the chicken egg gender measurement but without a correct integrated circuit and components, it has been proved to be non-successful. For future study, more accurate components should be tested in order to develop a fully function chicken egg gender identification device.

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Abbreviations

LRIC	Livestock Research Innovation Corporation
EBI	Electrical bioimpedance
BCA	body composition assessment
V	voltage
I	current
R	resistance
X	inductive reactance
BIA	Bioelectrical impedance analysis
FFM	Fat Free Mass
TBW	Total Body Water
ICW	Intra Cellular Water
ECW	Extra Cellular Water
BIS	Bioelectrical Spectroscopy
BIVA	Bioelectrical Impedance Vector Analysis
ECT	Electrical conductive tapes
EMI	electromagnetic interference

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CHAPTER 1. INTRODUCTION

Due to the different production purposes of the poultry industry, the demand for hatching hens or roosters also varies. In the production of meat and poultry strains, males are only preferred as breeding objects because of their higher feed exchange rate, and lower management and production costs [1]. One of the biggest ethical problems in modern egg-producing poultry farm is culling several billions of day-old male chicks every year. Poultry farms don't rear male domesticated birds because they can neither lay eggs nor put on sufficient flesh for meat production. Globally, it is estimated that 7 billion of day-old male chicks from laying-hen hatcheries are killed annually [1,2].

Traditionally, newly hatched chicks can be identified for genital morphology at the age of one day [3,4]. Trained and experienced chicken sexers can achieve a success rate of 95 ~ 98%, but it takes two to six years to train and accumulate experience, and there is a lack of sufficient human resources globally for this identification work [5]. Also, male and female chicks that are not suitable for production after gender identification need a euthanasia plan as well as chemical treatment equipment. These manpower allocations and processing work have become additional operating costs for the poultry industry, and on the other hand, a major ethical problem in killing lives that have been formed. At the moment, animal welfare issues has become increasingly important in developing countries. For example, EU legislation has facilitated market segmentation through trade standards for eggs (enriched cages, floor system, and free-range) and standardization of organic livestock production. Industries may introduce higher animal welfare standards for the goal of developing a national profile. Examples are Danish, and German egg production, which have committed to stop the beak trimming of layers [6]. EU has also discussed whether law enforcement should be used to restrict the poultry industries from continuing to adopt the gender elimination strategy. But without a proper fully pledged technology to determine the chick sex before its life's been formed, the law has not been able to restrict industries from doing what they have been doing for decades.

At the beginning of 2018, a study from the University of Leipzig [7] has stated that Fluorescence and Raman spectroscopy of blood offers the potential for precise and contact less in ovo sex determination of the chicken eggs during the fourth incubation day. Although this is a

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very fast and precise method, only 3.5 days of incubation process and the accuracy is said to be above 90% [7], such kind of spectroscopy requires a window on the eggshell (Figure 1.1), meaning this is an invasive method and will affects the embryo, which leads to decrease in hatching rates.

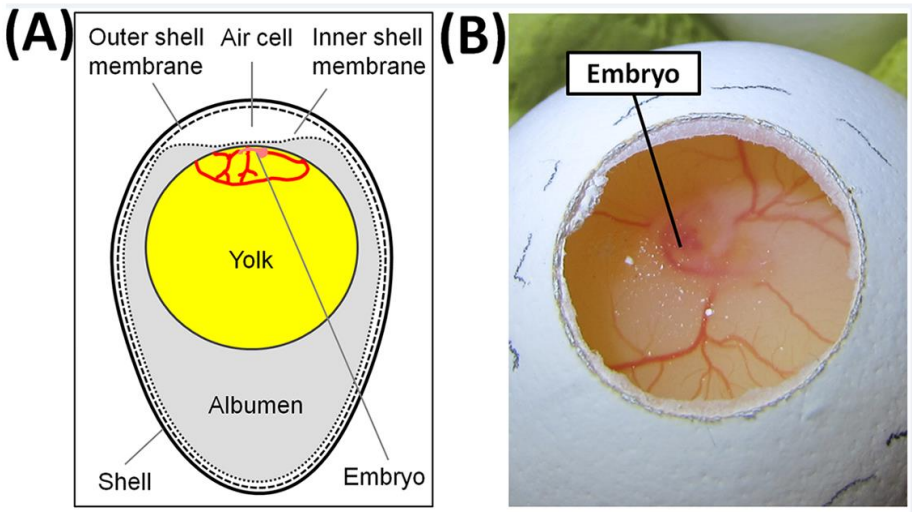

Figure 1.1. A window is required to perform Fluorescence and Raman spectroscopy to identify the gender [7]. (**A**) Schematic of the structures. (**B**) Embryo and extra embryonic vessels made accessible by opening the eggshell at the blunt pole.

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The Canadian is also developing a patented non-invasive egg sex identification tool called Hypereye. This technology, which uses hyper spectral imaging to identify infertility and gender on the day the egg is laid instead of testing fluid inside of the egg for the presence of hormones, was funded since 2007 by Egg Farmers of Ontario and conducted at the University of McGill, Montreal. Although it has been in developing for more than 10 years, this technology has yet to be available commercially for large-scale operations because on commercial scale, it needs to perform at 30,000 – 50,000 eggs per hour. Tim Nelson, Chief Executive Officer of the Livestock Research Innovation Corporation (LRIC), stated that the Hypereye method is nowhere near this speed [8]. Another issue is the method has only been developed for a typical flock of hens, which means, age of the flock, what the flock is being fed and especially the genetics of the flock may affect the already complex mathematical analysis [8]. In July 2018, Lawrence MacAulay, Canadian minister of agriculture and agri-food, visited the farm of Egg Farmers of Ontario

chairman Roger Pelissero, where he announced to invest $844,000 in support for Hypereye research [9].

Around the end of 2018, the first fertilized egg's gender detection technology has been introduced to the commercial production of laying hens, the Seleggt method. This technology is derived from another detection technology also developed by the University of Leipzig, Germany. On the blunt end, the eggshells were cut open using laser after 8 to 10 days, estimated time when the hormone can be found in the egg [9-12], of incubating, then a small amount of egg content solution was extracted and are used to measure the Estrogen content. Since the amount of estrone sulfate in the female embryos is significantly higher than that in males at the time hormone has already presented in the egg, it can be used as a basis for identification [9-12]. Although comparing to the manual identification methods this method is considered to have the same identification rate, due to the eggshell perforation sampling work, higher sanitary conditions are required for subsequent incubation facilities and environments to avoid possible pollution and incubation losses. Also, this commercial technology currently using radioimmunoassay for the determination of estrin, so relative technical cost and analysis time are not easy to reduce and shorten. In other words, the current commercialized technology still needs to complete the gender identification process through intrusive sampling and biochemical analysis. The direction of future technology development is to replacing intrusion sampling with non-invasive methods and fast real-time analysis, photoelectric analysis equipment is an example.

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Impedance spectroscopy is a proven method for many applications such as bacterial detection, biomarker detection, microorganism viability determination [13-18]. The principle of this technique is to inject a constant into the object to be measured and calculate its voltage to find out the electrical impedance respond of the object. As the eggshell can be imagined as the cell membrane, the protein can be imagined as the cytoplasm, and the egg yolk can be imagined as the nucleus, the theory of impedance spectroscopy can be logically apply for this case. Based on a study by Burke and Sharp in 1989, compared to hens, the eggs that hatched into roosters have larger yolks [19], so it is a reasonably belief that those eggs will have different electrical impedance value. By using a commercially available high-precision impedance analyzer (WK6420C) to perform egg impedance spectrum measurement, the electrical impedance

spectrum of the eggs can be obtained, and then the statistical method is used to identify the gender of the egg. This method can be performed early for prediction during the incubation days.

This work present the development process of a method for non-invasive in ovo chicken eggs gender identification applying the impedance spectroscopy technique. The gender will be confirmed after the incubation process by using standard PCR chicken egg gender identification process. The presented work involve an integrated system of measuring electrodes and incubation machine, which will be a great access for future development of a commercially available technology in non-invasive chicken egg sexing of poultry industry, solving a huge ethical problem globally, where billions of one-day old male chick are being culled annually. The main contributions of this research are as follows:

1. Design and implementation of a simple and effective electrode for measuring egg impedance in incubation environment.
2. Relationship development between chicken egg impedance and its gender
3. Integration of the electrode system and the chicken egg incubation machine, proving the possibility of future development for a fully integrated system to non-invasive sexing method.

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CHAPTER 2. LITERATURE REVIEW

2.1. Electrical bioimpedance technology

Electrical bioimpedance (EBI) technique is to measures the electrical properties of biological materials (live animals, animal carcasses, or products include tissues, cells, blood, and many more) by applying alternative electrical current/voltage to the whole material or regions of interest, sensing corresponding voltage/current and calculating the impedance by using Ohm's law. It is possible to monitor not only material composition but also volume by monitoring EBI value, due to the fact that this value is related to both the resistive and dielectric properties and the geometrical dimensions of the material. Over the years, more and more studies on EBI technique has been made to develop methods for either healthcare monitoring or diagnosis. Current bioimpedance research applications include body composition assessment (BCA), impedance cardiography, cerebral monitoring, cancer cells/antibody detection, and microorganism viability.

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2.1.1 Electrical Impedance and dielectric properties of materials

Electrical impedance (Z) is a measure of opposition to the current flow in a circuit. It can be written as a complex in the Cartesian form or the polar form:

$$Z = R + jX = |Z|e^{j\theta} \qquad \text{(Equation 2.1)}$$

R, the real part, is the resistance, whereas the imagery part **X** stand for the inductive reactance, the magnitude of impedance $|\boldsymbol{Z}|$ and the phase difference $\boldsymbol{\theta}$ between the applied voltage (V) and corresponding current (I). Resistance **R** causes energy loss in the circuit, this won't cause phase shift between voltage and current. On the other hand, reactance **X** stores energy and causes phase difference between current and voltage. Usually reactance is introduced by capacitors or inductors in the circuit.

Electrical impedance can be calculated from Ohm's law:

$$Z = \frac{V}{I} \qquad \text{(Equation 2.2)}$$

Figure 2.1 shows the Impedance vector. The amplitude of the resistance (**R**) component is shown by a vector along the x-axis and the amplitude of the inductive reactance (**X**) is shown by a vector along the y-axis. The amplitude of the the impedance is shown by a vector that stretches from zero to a point that represents both the resistance value in the x-direction and the inductive reactance in the y-direction.

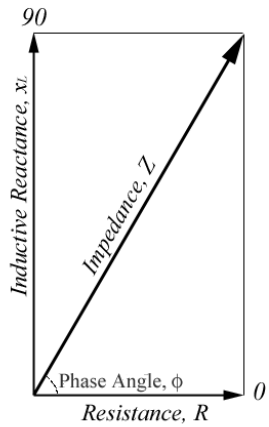

Figure 2.1. Impedance vector

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The phase angle θ of a circuit can be calculated as the ratio between the inductance and the resistance in the circuit or in other word, the ratio between the real and the imaginary part of impedance. And is demonstrated in the following equation :

$$tan\theta = \frac{X}{R} \quad or \quad \theta = \arctan\frac{X}{R} \qquad \text{(Equation 2.3)}$$

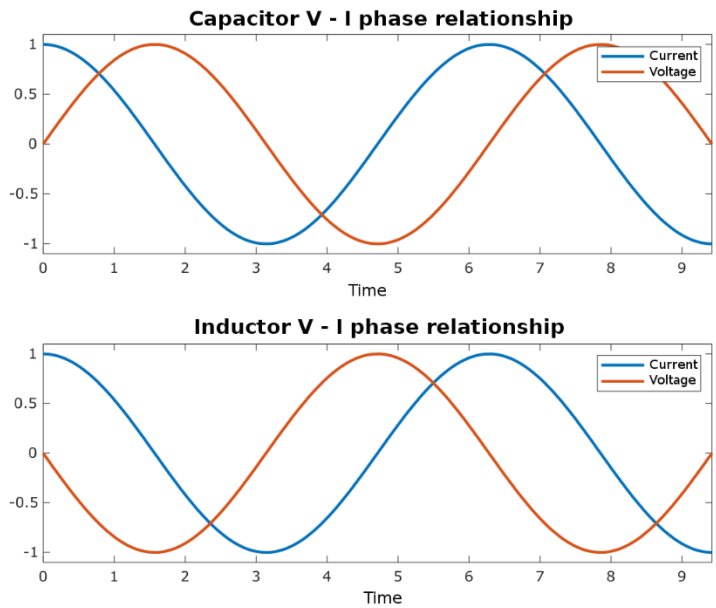

Figure 2.2. Illustration of the V-I phase relationships for a capacitor and an inductor.

The impedance phase angle for any component is the phase shift between the voltage across that component and current through that component. For a resistor, the voltage drop and current are always in phase with each other, and so the impedance angle of a resistor is said to be 0°. For a capacitor voltage drop always lags current by 90°, and so a capacitor's impedance phase angle is said to be -90°, and this phase angle is +90° for inductor (Figure 2.2).

Admittance Y, the inverse of impedance, can also be used to express the electrical properties of materials. Admittance can be written in formula as:

$$Y = \frac{1}{Z} = G + jB = G + j\omega C = K(\sigma + j\omega\varepsilon) = K(\sigma + j\omega\varepsilon_r\varepsilon_0) \qquad \text{(Equation 2.4)}$$

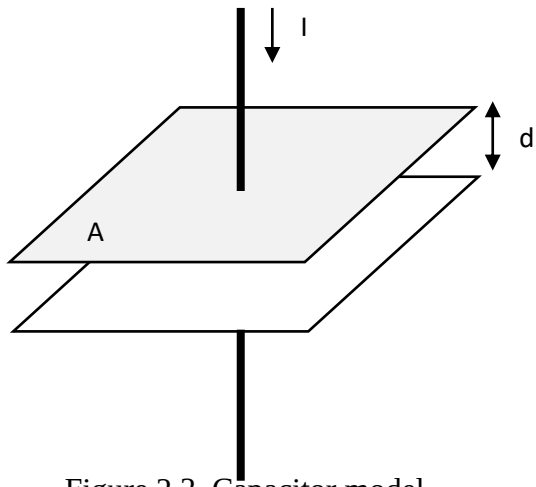

Figure 2.3. Capacitor model

The real part G is conductance while imaginary part B is the susceptance which is equal to ωC, where ω is the frequency of the applied electric field. A common capacitor model would include two electrodes with a same area A, on two parralel planes with distance d between the two planes, as illustrated in Figure 2.3. K is defined as the ratio between area A and distance d: K = A/d. σ is the conductivity of the material showing the material ability to conduct current. ε is the permittivity of the medium, indicating the transmittance cabability of electric field through the medium, between two planes which was further extended to $\varepsilon_r \varepsilon_0$, where ε_r is the relative permitivity of the material and ε_0 is the dielectric permittivity of vacuum. This parameter is related to the polarization of medium in response to the electric field.

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2.1.2 Electrical properties of living tissue

The cell is the most basic structural, functional, and biological unit of all known organisms. The electrical properties of biological tissue are determined by its constituents. Any tissue is formed by extracellular fluid and cells. This extracellular fluid denotes all body fluids outside of cells, surrounding the cell and contains proteins, ions, and can be divided into plasma (20%) and interstittial fluid (80%). Contrastingly, the cell is constituted by cell membrane which is a lipid bi-layer plasma. Inside this lipid bi-layer plasma, the cell contains intercellular fluid so called cytosol, organelles and the nucleus of the cell. Any biological tissue can be considered an electrolyte due to free ions contained into extracellular and intracellular fluid. Thus biological tissue is also considered an ionic conductor where K^+, Na^+ and Ca^{2+} are the most important ions contributing to the ionic current.

Table 2.1. Approximate concentration of ions in living tissue [20]

Concentration	Na^{+}	K^{+}	CA^{2+}	Mg^{2+}	Cl^{-}	HCO^{3-}	Protein	$HPO4^{2-}$
Intracellular	10	100	-	123	2	8	55	149
Extracellula	142	5	5	2	105	24	16	2

Biological tissue having both conductance properties and dielectric properties. A dielectric could be described as any material that can store capacitive energy. Although this ability is owned by living tissue at any level due to its constituents, the cellular structure with the major contribution to the dielectric behaviour, as mentioned above, is the plasma membrane. This membrane is a lipid bi-layer structure with a very poor intrinsic electrical conductance.

Figure 2.4 shows the structure of plasma membrane and its structure. As the cell membrane can be considered as a dielectric with low conductance, the extracellular fluid membrane and intracellular fluid form a conductor-dielectric-conductor structure, and can be seen as an analogy to a capacitor model.

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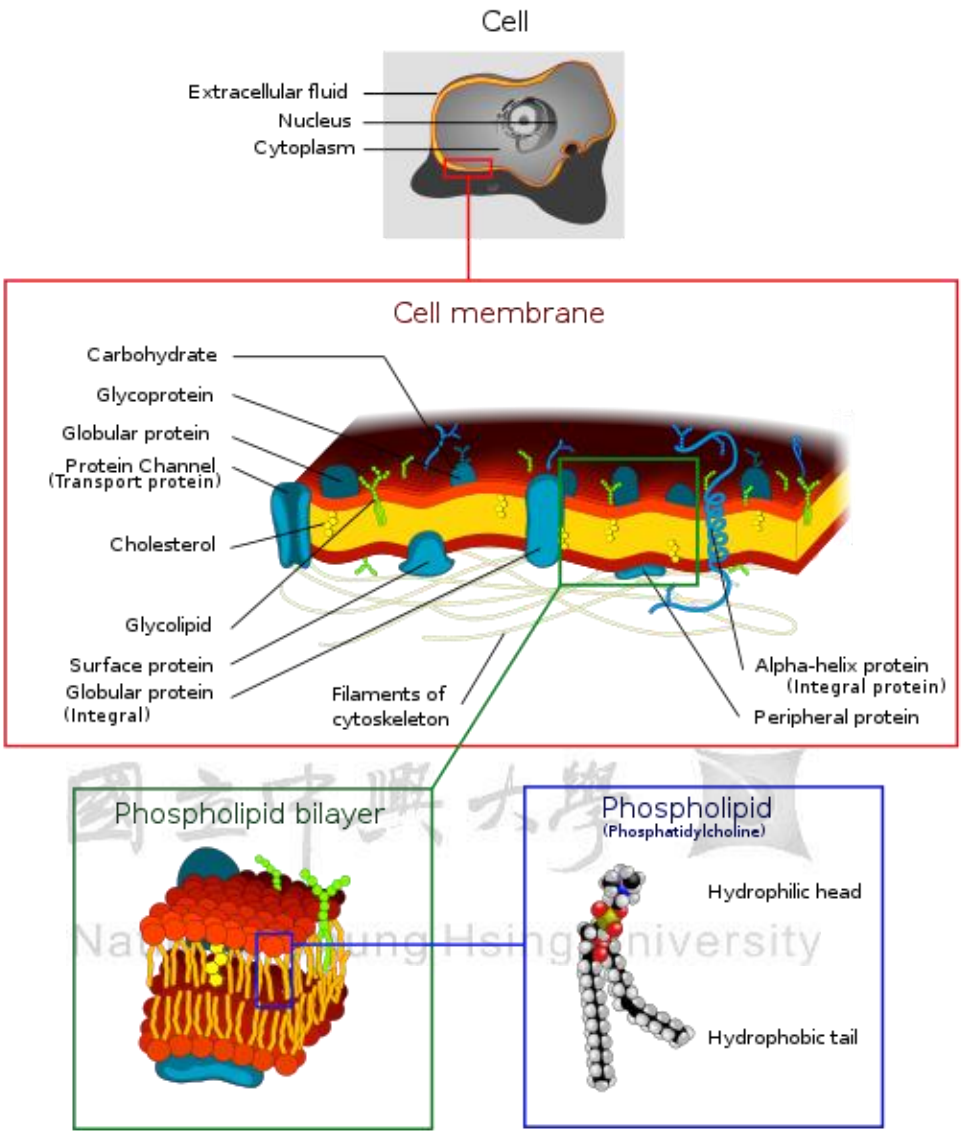

Figure 2.4. Plasma membrane and its structure. [21]

2.1.3. Living tissue as dispersive medium

Living tissue is considered as a dispersive medium because permittivity and conductivity are both functions of frequency (Figure 2.5). According to the changes in permittivity and conductivity, there are four major dispersions that were identified and named by Schwan as: α-, β-, γ-, δ-dispersion.

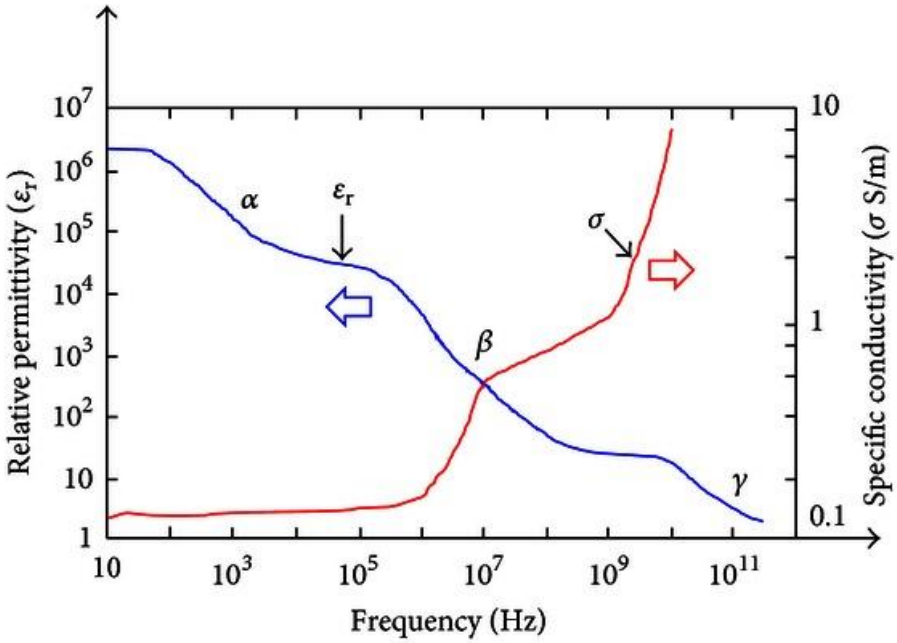

Figure 2.5. Frequency dependence of permittivity and conductivity of living tissue [22]

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2.1.3.1. α-dispersion

Happens at low frequencies from 10Hz upto 10kHz. Three main causes contribute to this frequency dependence are: the effect of the endoplasmic reticulum contributes, the channel proteins present in the plasma membrane, and the relaxation of counter-ions on the charged cellular surface.

2.1.3.2. β-dispersion

β-dispersion includes the frequency range from 1kHz to several Mhz. This dispersion is considered to be related to the behavior of membrane, caused by the low conductivity and capacitivity properties of the cell membrane and other internal membrane structures and their interactions with the extra and intra-cellular electrolytes.

2.1.3.3. γ-dispersion

This dispersion is cause by the high content of water in cell and tissue. The relaxation frequency of this dispersion is nearly about 20 GHz, almost like normal water. The range of dispersion is from hundreds of MHz to some GHz.

2.1.3.4. δ-dispersion

Was found nearly 40 years after others dispersions, this dispersion is a weak relaxation between β-, and γ-dispersion. Its dispersion range is from around 300 MHz to around 2 GHz. This dispersion is caused by rotation of amino acids, partial rotation of charge side groups of proteins, and proteins bonded to water.

For medical applications, α- and β-dispersion has been the main focus believing in these range is where most changes between pathological and normal tissue occur.

2.1.4. Electrical model of cells

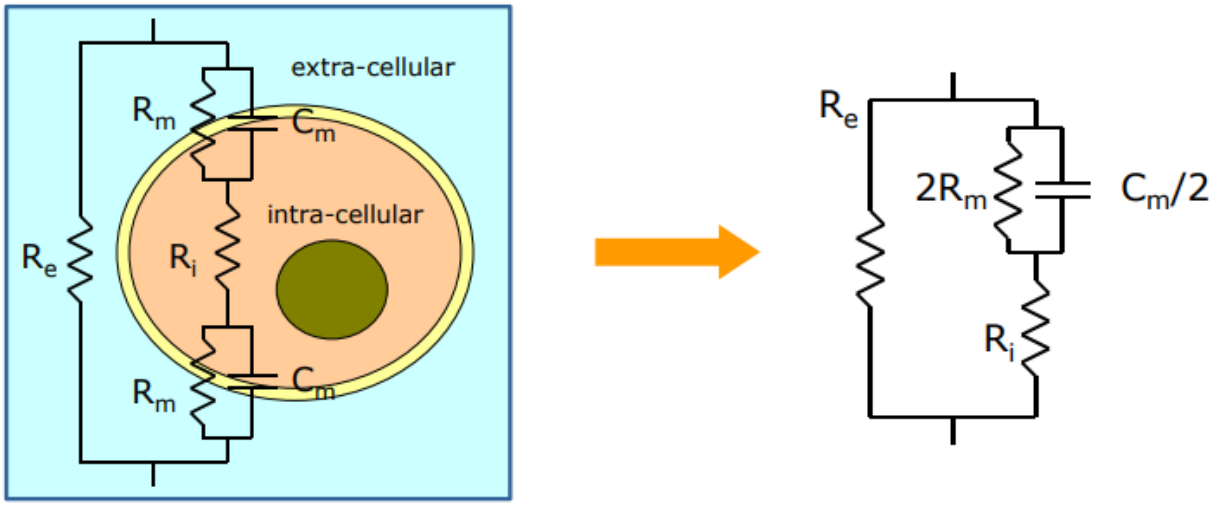

Figure 2.6. Equivalent electrical model of a cell. Re represents resistance of extra cellular fluid. Ri represents resistance of intra cellular fluid. Rm represents resistance of membrane. And Cm represents capacitance of membrane [23]

As mentioned above, electrical properties of living tissue are caused by their constituents. Applying theory of electrical circuit to these constituent, [Fricke 1924] has deduced an explanatory and descriptive electrical model for the cell, as shown in Figure 2.6. Rm is commonly neglected as its conductance is very low and the equivalent circuit can be simplified as Figure 2.7.

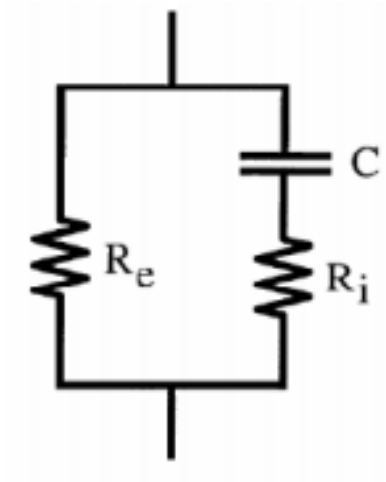

Figure 2.7. Simplified equivalent model of a cell

According to this simplified model, a frequency dependence introduced by the capacitor C exists in biological tissues. At low frequencies, the electrical current is impeded by the membrane. So the extra cellular fluid plays a major role in conducting electrical current. . It can be observed as at near DC, the current is not able to penetrate the cell because of the plasma membrane acts as an insulator. Therefore, the measured impedance is only the extracellular fluid resistance, Re. At high frequencies, the electrical current passes through the membrane and the measurement is dependent on both extra cellular fluid and intra cellular fluid.

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2.1.5. Bioimpedance and body composition measurement

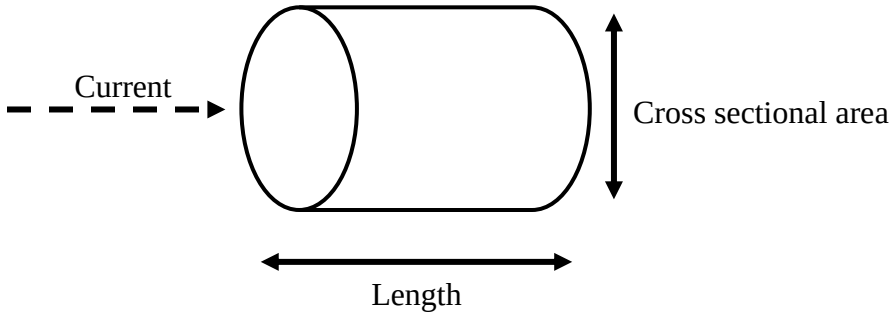

Figure 2.8. Cylinder model for the relationship between the impedance and geometry

Bioelectrical impedance analysis (BIA) is a method which measures the impedance or opposition to the flow of an electric current through the body fluids contained mainly in the lean and fat tissue. Due to the impedance in lean tissue, where intracellular fluid and electrolytes are

primarily contained, is lower than in fat tissue and because of the measured body resistance differs depending on the amount of body fluids, it is possible to estimate the body composition.

The impedance of a volume conductor is obtained depends on the electrical properties of the material and the shape and orientation respect the Electrical Field. Assuming the body as a cylindrical volume, the resistance of a length of homogeneous conductive material of uniform cross sectional area is proportional to its length and inversely proportional to its cross sectional area (Figure 2.8).

$$R = \rho \frac{L}{A} \qquad \text{(Equation 2.5)}$$

Where R is the resistance, ρ is the resistivity, L is the length of the cylinder, and A is the cylider cross sectional area. Although in reality, the body is not a uniform cylinder and its conductivity/resistivity is not constant, a relationship between the volume of body water and the ratio length^2 to resistance (impedance) can be established as:

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$$V = \rho \frac{L^2}{R} \qquad \text{(Equation 2.6)}$$

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Where R stand for the resistance of the body, ρ is the resistivity, L is the conductive length, and V is the volume of water, which contains electrolytes that conduct electrical current though the body. Instead of using the conductive length which is harder to measure, equation 2.6 can be modified using height measurement:

$$TBW = k \frac{Ht^2}{R} \qquad \text{(Equation 2.7)}$$

Where TBW is total body water, Ht is height, R is the resistance of the body, or segment, and k is a coefficiant which makes the ideal cylinder to the real geometry. The value of this coefficient, k, depends on several factors such as sex, age, weight or other anthropometric parameters.

There are many methods for BIA measurement. Firstly, single frequency BIA (Figure 2.9) where a single current at, most commonly, 50kHz is passed between surface electrodes placed on hand and foot and is permit to estimate Fat Free Mass (FFM) and Total Body Water (TBW). Secondly, the Multi-Frequency BIA, measuring over a range between 0 to 500kHz to

evaluate FFM, TBW, Intra Cellular Water (ICW), and Extra Cellular Water (ECW). Then there is Bioelectrical Spectroscopy (BIS) which is a more mathematical model with mixtrure equations to generate relationships between impedance and body fluid compartsment. And even more with Segmental-BIA, Localized BIA which focuses on well-defined body segments, and Bioelectrical Impedance Vector Analysis (BIVA) which is a new method for monitoring body fluid variation and body analysis.

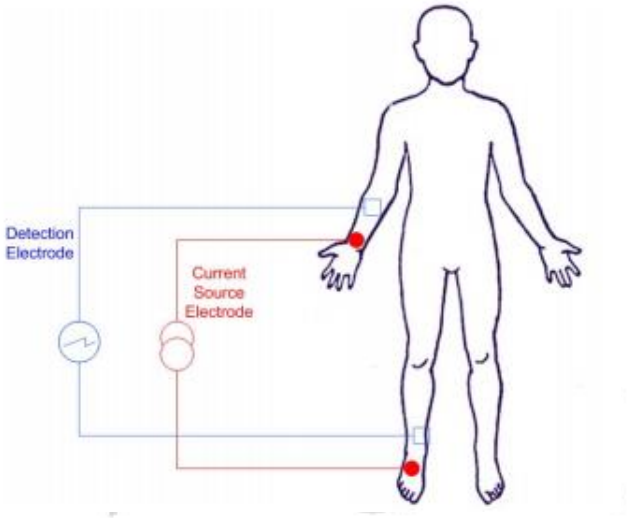

Figure 2.9. Standard placement of electrodes on hand and wrist and foot and ankle for tetrapolar SF-BIA and MF-BIA. [24]

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2.2. Fertilized chicken eggs development and bioimpedance potential

2.2.1. Fertilized chicken eggs development

A chicken embryo develops and hatches in 20 to 21 days and has been extensively used in embryology studies. Fertilized eggs can be easily nurtured in humidified incubators with a steady accurate temperature.

Soon after incubation begins, a pointed thickened layer of cells becomes visible in the caudal or tail end of the embryo. This pointed area is the primitive streak and is considered the longitudinal axis of the embryo. From here, the head and backbone of the embryo will develop, blood islands appear and later turns into blood or vascular system, digestive tract starts forming, as well as the eyes.

On the second day, the fore mentioned blood islands start linking and form a vascular system, with the heart also forming some where else. This newly formed heart and vascular system will join together and start bumping by the 44th hour of incubations. By this time two distinct circulatory systems have been established, an embryonic system for the embryo and a vitelline system extending into the egg.

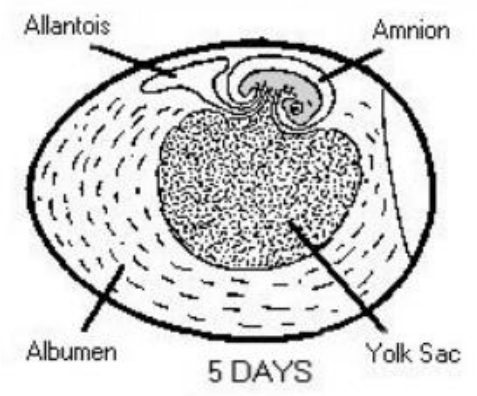

Figure 2.10. Embryo development after 5 days of incubation [25]

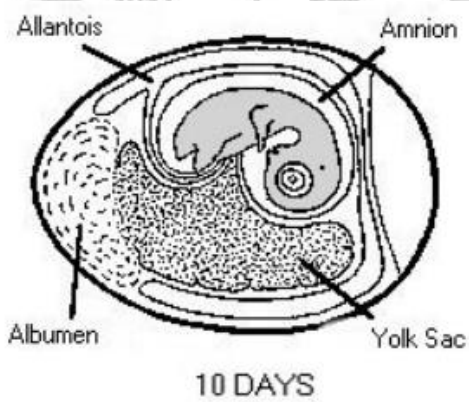

Figure 2.11. Embryo development after 10 days of incubation [25]

During the third and fourth day, most of the chicken organs are formed or being formed including the beak, limb buds for wings and legs, torsion and flexion, parts of the digestive and respiratory systems such as mouth, tongue, and nasal pits. The heart also continue to enlarge even though it has not yet to enclosed within the body. By the end of the fourth day of incubation, the embryo has all organs needed to sustain life after hatching (Figure 2.10).

The embryo rapidly grows and develops and by the seventh day, the heart is enclosed in the thoracic cavity, and the embryo looks more like a bird. During the tenth day of incubation

(Figure 2.11), feathers forms and the beak harden, readily to pierce through the air cell after the twentieth day (Figure 2.12).

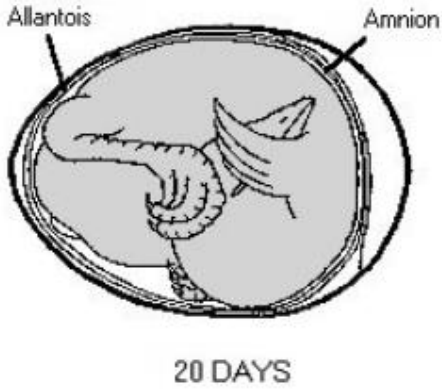

Figure 2.12. Embryo development after 20 days of incubation [25]

Details of events in fertilized chicken egg embryonic development were first described by A. L. Romanoff - Cornell Rural School Leaflet, September 1939. These events was marked as important events in embryonic development which included the before laying step, where the egg will go through fertilization, division and growth of living cells, and segregation of cells into groups of special function. After it has been laid, the egg stops growing, and is completely inactive until the start of incubation. Then the incubation process begins, with the embryo fully developing until hatch (Figure 2.13), as described in [25], is listed as follows:

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- First day:

  16 hours - first sign of resemblance to a chick embryo

  18 hours - appearance of alimentary tract

  20 hours - appearance of vertebral column

  21 hours - beginning of nervous system

  22 hours - beginning of head

  24 hours - beginning of eye
- Second day:

  25 hours - beginning of heart

  35 hours - beginning of ear

  42 hours - heart beats

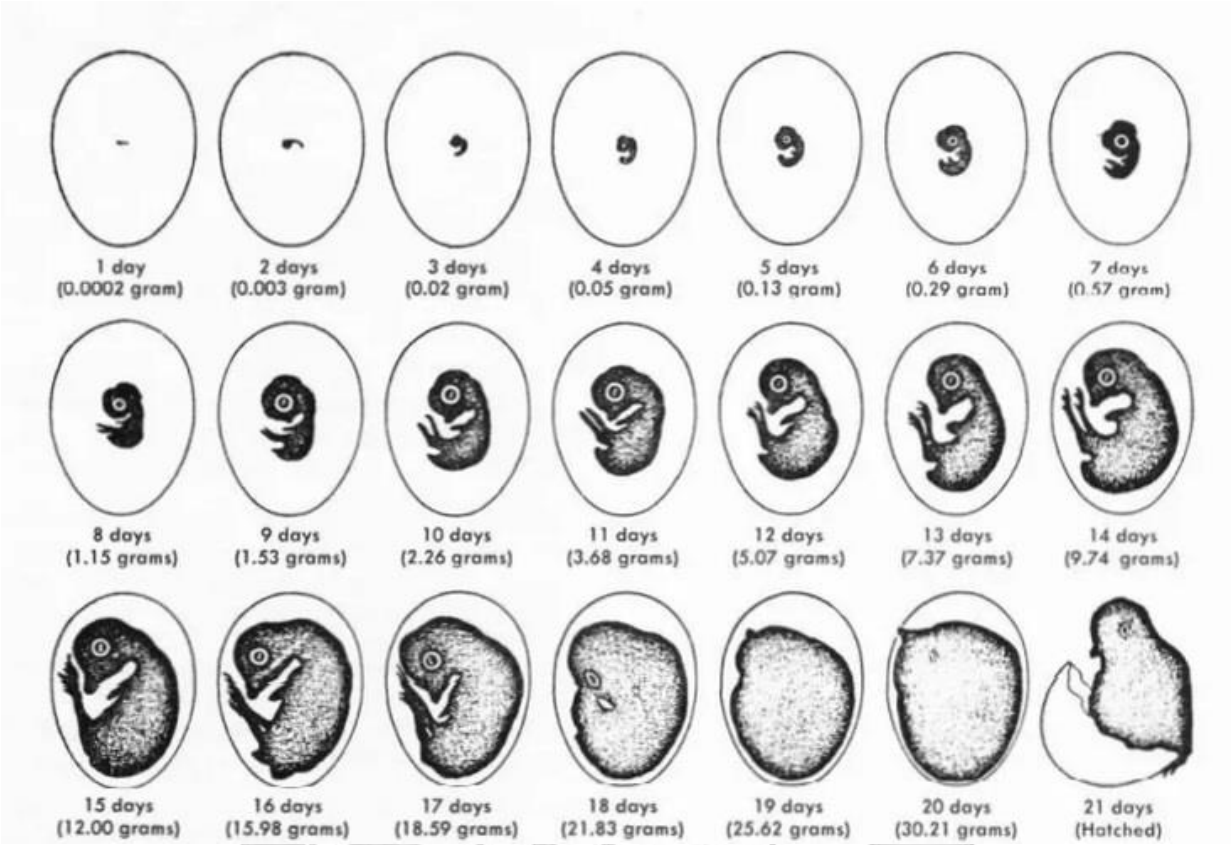

Figure 2.13. Chicken egg embryo development [25]

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- Third day:

  60 hours - beginning of nose

  62 hours - beginning of legs

  64 hours - beginning of wings
- Fourth day - beginning of tongue
- Fifth day - formation of reproductive organs and differentiation of sex
- Sixth day - beginning of beak
- Eighth day - beginning of feathers
- Tenth day - beginning of hardening of beak
- Thirteenth day - appearance of scales and claws
- Fourteenth day - embryo gets into position suitable for breaking shell
- Sixteenth day - scales, claws and beak becoming firm and horny
- Seventeenth day - beak turns toward air cell

- Nineteenth day - yolk sac begins to enter body cavity
- Twentieth day - yolk sac completely drawn into body cavity; embryo occupies practically all the space within the egg except the air cell
- Twenty-first day - hatching of chick

2.2.2. Fertilized chicken eggs bioimpedance potential

In 1989, William H. Burke and Peter J. Sharp was lead by previous studies that at day 1 of age (the day of removal from the incubator) male chicks is already heavier than females and was not convinced by the cause, neither by hatching time nor even different degrees of dehydration in the incubator. They also recognise that the body weight of male chick are heavier and this differences increase day by day and eventually reach statistical significance. Ultimately they developed a study to know when the differences in body weight occur before attempting to elucidate the mechanism(s) responsible.

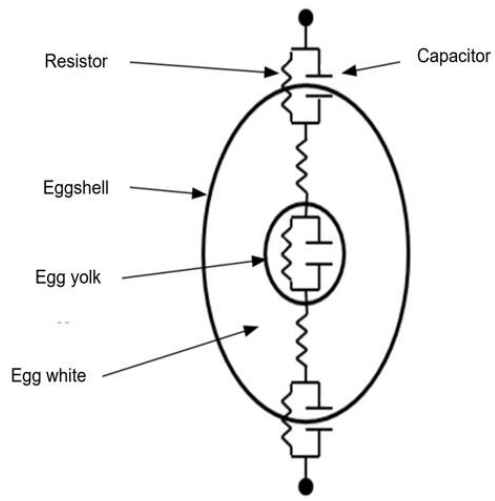

Figure 2.14. The proposed equivalent circuit for an egg as a cell.

By weighing the eggs at the same day, but different between experiments, about 60 minutes after removed from the incubator. The eggs were opened, the embryos were removed, quickly killed by cervical dislocation, briefly rinsed in water, blotted dry on a paper towel, placed in a preweighed pan, and weighed. The gender was determined by visual examination of

gonadal region using magnification. Then the embroys were let dried for 48h at 70-80 Celcius degree and then weighed again.

In conclusion, Burke and Sharp has found that the male and female egg embryo has a significant different in weight by at least 11 days of incubation and also. With male eggs have a heavier embryo than female egg. Also the eggs that hatched into roosters have larger yolks, the Opacity Value of their eggs is larger than 10.

Imagine the eggshell as the cell membrane, the protein as the cytoplasm, and the egg yolk as the nucleus, the egg is now considered as a cell and its equivalent circuit is shown in figure 2.14.

By measuring the egg bioimpedance non-invasively using a standard BIA electrode configuration it is believed that these differences can be pointed out thus distinguished the chicken egg gender.

2.3. Bioimpedance measurement for chicken egg system design

There are two methods to obtain the impedance. Either to inject an electrical current and measure the corresponding voltage or a voltage to the object of interest and measure the output current. Comparing these two methods, the one that injects current is considered more safety as it is easier to manage current within medical safety limits. The downside of current source is that it gives poor performance at higher frequencies due to output impedance decreases.

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2.3.1. Electrodes and connection

2.3.1.1. Two-electrode method

The two-electrode method or bipolar method (Figure 2.15) uses two electrodes to inject an alternating constant current to the biological material and measures the output voltage, drop over this material using the same electrodes. Apart from the impedance of biological material Zm, the impedances at two electrode contacts Zc are also included in the measured impedance.

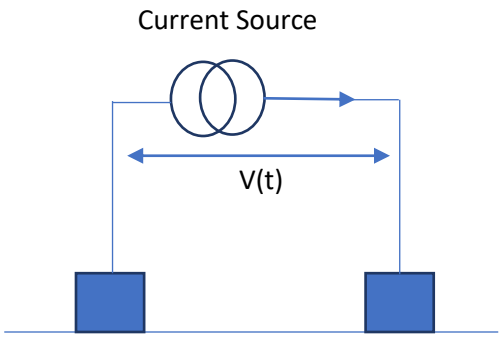

Figure 2.15. Bipolar bioimpedance measurement system

When the impedance of the tested material is higher than the impedance of electrode contacts at the BIA working frequency, this electrode configuration can work properly.

2.3.1.2. Four-electrode method

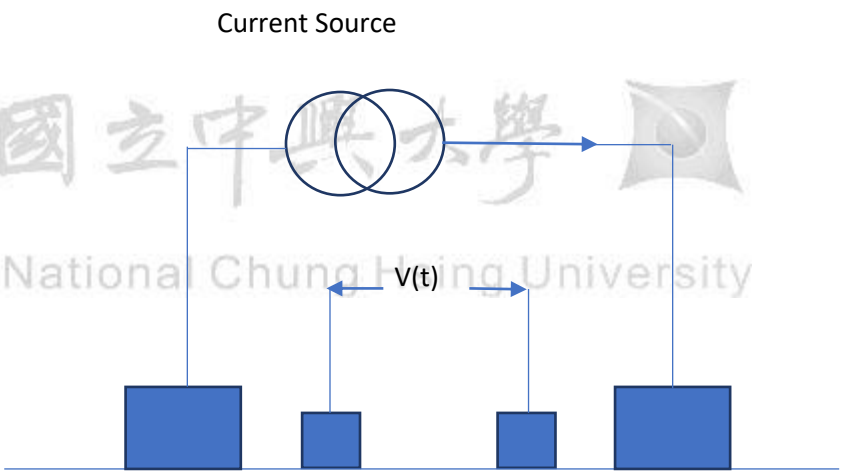

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Figure 2.16. Tetrapolar bioimpedance measurement system

The four-electrode system (Figure 2.16) or tetrapolar system uses separate electrode pairs, one to inject current and the other to measure voltage. The measurement result will involve only the impedance of the tested material, ideally. This configuration is the most common method in current bioimpedance measurement studies.

2.3.2. Current source

A stable sinusoidal voltage signal generator should be considered to reduce the noises. For current source, the high profile Howland's topology current source with high output impedance is considered. It is a desire to get the current source to work up until several MHz as

the chicken egg bioimpedance measurement has not been reportedly performed by any previous study so there is no preset frequency range, and it would be great if a handheld device can perform a spectroscopy in such a large range. However, the output impedance of a current source drops as the frequency increases. This is caused by the parasitic capacitances that come from many components in the circuit, for example between electrodes. The output impedance of the current source should be at least several Mega Ohm. More information will be shown in chapter 3, methodology.

2.3.3. Impedance analyzing

One of the easier method for analyzing the impedance of biological material is using a gain phase detector. The low-cost AD8302 has been used by many researchers to get the impedance and phase angle for developing bioimpedance measurements systems. This IC uses the principle where the inject voltage controlled current signal, and the output voltage signal is compared and gives out stable voltage signal determining the magnitude and phase different between the two signal. With a bandwidth of up to 2.7GHz it is sure to be a promising IC for this research.

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AD8302 comprises a phase detector and a closely matched pair of logarithmic amplifiers. It compares the two input signals A and B, and generates two output voltages: v_{mag} and v_{phs} which can be expressed as:

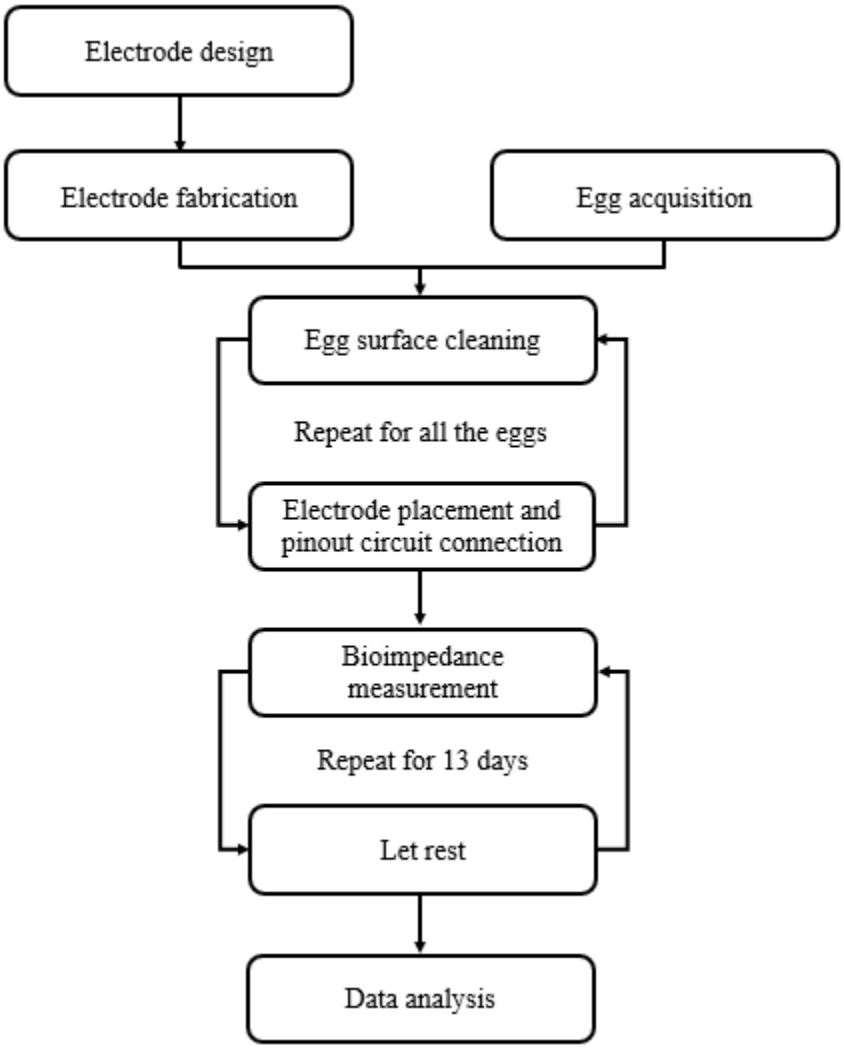
$$\begin{cases} v_{mag} = \alpha_1 . \log\left(\frac{|V_A|}{|V_B|}\right) + 900mV \\ v_{phs} = \alpha_2 . (|\theta_A - \theta_B| - 90^o) + 900mV \end{cases} \quad \text{(Equation 2.8)}$$

Where α_1 and α_2 are two coefficients: α_1 = 600mV/decade (30 mV dB^{-1}), α_2 = $-$10mV/degree. This equation can be used to solved the magnitudes ratio and the phase different between the input and output signals.

CHAPTER 3. METHODOLOGY

3.1. Chicken egg bioimpedance measurement

The parts under this section describe the procedure on how the chicken egg bioimpedance measurement experiment were conducted. As electrode performance are very important in impedance spectroscopy technique, requiring not only contain low noises, but also has to be accuracy and repeatable. This study evolve designing a simple but effective electrode that can be used inside a fix environment with high humidity of 55% and moderate temperature of 99.5^{o}F. The electrode positions on the chicken egg were also discussed in the section. The following diagram describe the experimental design for the bioimpedance measurement of chicken eggs.



3.1.1. Materials preparation

Kinmen county, Taiwan chicken strain were chosen for the experiment. The fertilized chicken eggs were obtained from Department of Animal Science, National Chung Hsing University, Taichung, Taiwan.

Potassium permanganante ($KMnO_4$=158.04) and Formalin (HCHO=30.02) extra pure reagent was purchased from Choneye Pure Chemical (Taiwan) for the incubation sterilization process.

Blood Genomic DNA purification kit (Biokit, Barcelona, Spain) was purchased for the embryos gender identification PCR process.

3.1.2. Electrode design, fabrication, and attachment

First step of designing the electrode, a medical disposable gel conductive electrode (Figure 3.1) (standard for ECG measurement) was considered. As it is commonly proposed as measuring probes for electrochemical impedance spectroscopy measurements. These electrode are cheap and easily available on the market, are composed of a Ag/AgCl electrode surrounded by a conductive gel, whose resistivity is about 100 Ω·m. The gel is surrounded by an adhesive material which is used to fix the electrodes. Although it is low-cost, accuracy, stable, and can be easily attached onto the egg surface, the conductive gel used to fabricate theses electrodes cannot be used for long time, especially at a high temperature of 99.5^{o}F, to continuously measuring the egg impedance day-by-day as it will dried and effect the impedance consistency.

Figure 3.1. A conductive gel ECG electrode

3.1.2.1. Electrode material

As the conductive gel of the standard ECG electrode cannot be used, others adhesive material was in consideration. Epoxy resin metal conductive adhesives was among these considerations. This adhesive contain conductive fillers such as silver, copper, and iron and can reach a volume resistivity as low as 0.0001 Ohms·cm. The curing temperature is at room temperature at around 25 Celcius degree, and will only took upto 5 minutes. Unfortunately, this conductive adhesive was not able to bond with the eggshell. Materials such as copper cannot be used due to it toxicity to giological tissue and may harm the chicken embryos.

Carbon epoxy (Figure 3.2) was chosen as the conductive adhesive. The volume resistivity is stable and as low as < 40 Ohms·cm. This epoxy is capable of adhere signal wire onto the eggshell without harming the embryos breathing process, as it takes up oxygen and discharge carbon dioxide by simple diffusion through microscopic pores in the eggshell. But with a high curing time of 24 hours at room temperature, another material was used to keep the carbon epoxy and the signal wire in place before cured.

Figure 3.2. Carbon Epoxy

Electrical conductive tapes (ECT) is an excellent option. Commonly used for grounding, electromagnetic interference (EMI) shielding, and attaching components in a wide range of electronic applications. These tapes have conductive filler particles that enable interconnection between substrates while also providing electrical insulation. ECTs are also very easy to use and integrate into the manufacturing process. Thay have a variety of surface substrate including gold, stainless steel, copper, and scrim. These ECTs come with different adhesion strength, bond line thickness, optimal temperature range, and even environmental condition. After consideration, conductive fabric tape with the width of 10mm and the thickness of 0.12mm was chosen (Figure

3.3). With a peel adhesion of 11N(gf)/W25mm, and low electrical resistance of 0.05Ohm/cm^2. This tape will be use to keep the carbon epoxy and the signal wire in place before the epoxy is cured, and shield the whole electrode system during the incubation and data measurement process.

Figure 3.3. Conductive fabric tape

The signal wire should be able to carry high-frequency electrical eignals with low losses. Coaxial cable would be able to fulfill this criteria, this type of cable consist of an inner conductor surrounded by a concentric conducting shield, seperated by a dielectric material. The cable is protected by an outer insulating jacket. Normally, the outside of the shield is kept at ground potential and a signal carrying voltage is applied to the center conductor. The advantage of coaxial design is that with differential mode, equal push-pull currents on the inner conductor, and inside of the outer conductor, the signal's electric and magnetic fields are restricted to the dielectric, with little leakage outside the shield. Further, electric and magnetic fields outside the cable are largely kept from interfering with signals inside the cable, if unequal currents are filtered out at the receiving end of the line. The chosen signal wire was the coaxial RG 1.13 (Figure 3.4). With an outer jacket diameter of 1.13mm, and the inner conductor consisted of 7 silver coated copper 0.24±0.02mm diameter wire.

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Figure 3.4. RG1.13 coaxial cable

3.1.2.2. Electrode shape

While designing the shape of the electrode, one thing to keep in mind is that the egg is breething through the pores. This is a limitation for the electrode because its size should not be so large that cover too much pore area suffocating the embryos. But it also needs the adhesive to be strong enough to withstand the peel adhesion equal to the weight of the signal wire. The shape of the electrode will effect its impedance, as the electrode impedance increases , the common mode rejection of the system decreases, and the S/N ratio of the recording decreases. This is primarily due to the fact that random differences between the impedances at the different electrode sites are typically magnified when the impedances are increased. Thus, it is important to keep a low electrode impedance but still meet the prior requirements.

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3.1.2.3. Electrode fabrication

Firstly, 10mm*10mm (±1mm) square pieces of conductive fabric cloth (Figure 3.5) was cut out and measured to make sure that the area of the pieces are almost identical.

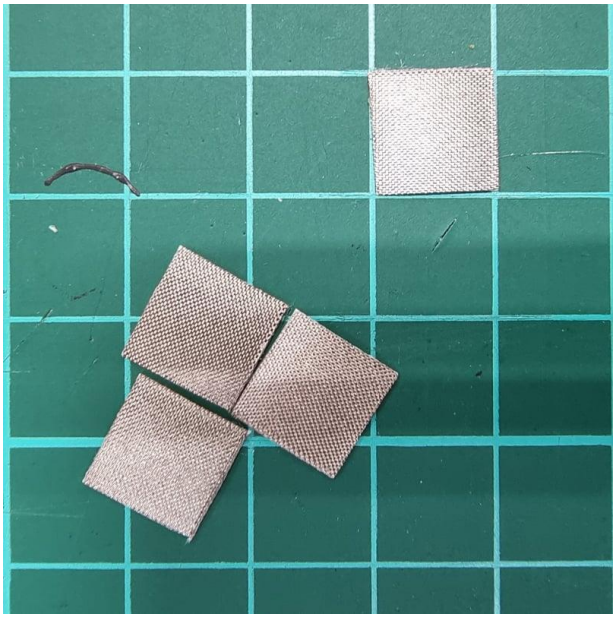

Figure 3.5. 10mm*10mm pieces of conductive fabric cloth

It is also notable that on the end of the coaxial cable where it is connected to the egg, the shielding layer should be wrapped together (Figure 3.4) and insulate using electrical tape or heat-shrink tubing (Figure 3.6).

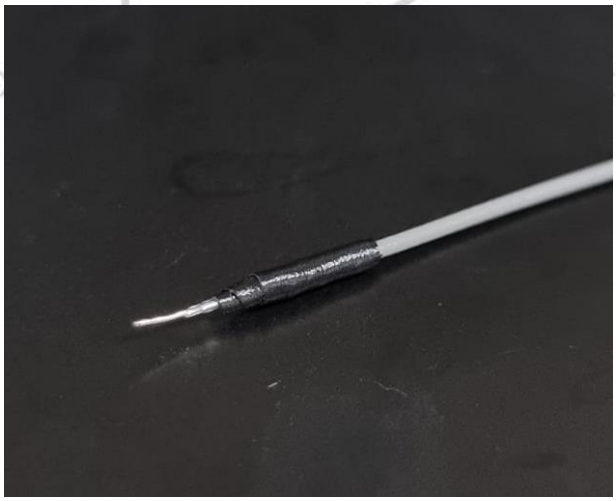

Figure 3.6. Shielding layer of coaxial cable insulated using electrical tape

A small amount of carbon electrode are dropped onto the adhesive side of the conductive fabric cloth piece using a 5mm diameter straw, as shown in Figure 3.7. This amount was chosen through test cases, calculated as a cylinder form with the base diameter of 5mm and vary heights, the height was test from 1 to 3mm and 2mm was found to be the optimized. The height of 1mm was too less to adhere the signal wire with the eggshell and the conductive fabric cloth after

curing, making the signal cable under the cloth piece unstable, thus effecting the electrode performance. while 3mm was too much that when put onto the eggshell, this amount would excess the conductive fabric cloth.

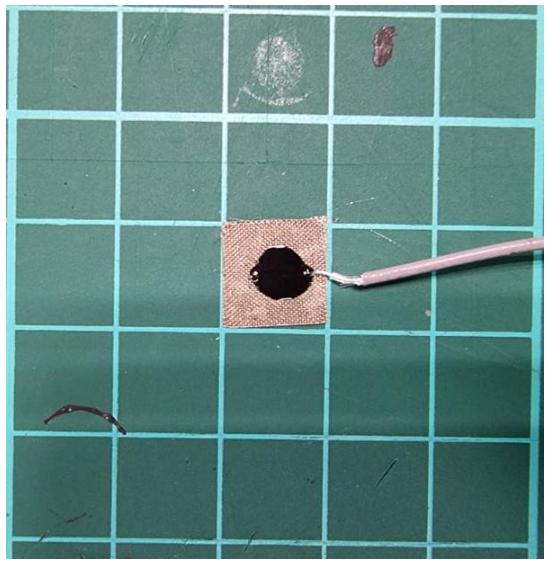

Figure 3.7 Carbon epoxy dropped onto 10mm x 10mm piece of adhesive conductive cloth and connected with the signal cable.

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Then this whole electrode will be taped onto the eggshell at designated points. Each egg has four electrode attached on it, with each signal cable was 1 meter long. After attachment, the electrodes are left in room temperature for 24 hours, allowing the carbon epoxy to be cured. After cured the electrodes are now stable and have similar impedance, ready for bioimpedance spectroscopy.

The resistance of the carbon epoxy can be calculated using the resistivity equation:

$$\rho = \mathrm{R}\frac{A}{l} \qquad \text{(Equation 3.1)}$$

Where R is the resistance, A is the cross-sectional area, and l is the length. With the volume resistivity of around 40 Ohms·cm, the resistance of the carbon epoxy drop was calculated to be around 2 Ohms. This resistance might be affected, lower or higher, by the amount of carbon ions in the droplet.

The contact resistance of the conductive cloth was very low at 0.05 Ohm/cm2, the conductive cloth pieces was cut out at 10mm x 10mm meaning the resistance of each piece is 0.05 Ohm. Making the electrode configuration resistance a total of 2.05Ohm by calculation.

3.1.2.4. Electrode placement

The placement of electrode on the egg is very important, by simulation Opitz et al (2018) proved that just a little more than 1cm displacement of electrode can lead to dramastically change in brain wave signal for EEG measurement. For bioelectrical impedance analysis of body composition, Yamada (2009) stated that the electrode placement can help to improves the accuracy in body composition measurement. Many more electrode placement studies have been and are being made to improve the precision of measurement.

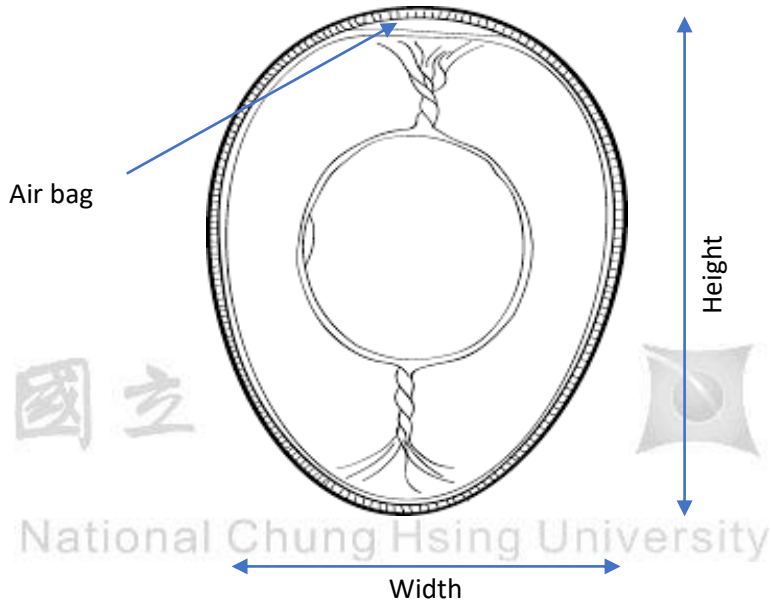

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Figure 3.8. Egg anatomy

As no previous reported study has been made for egg impedance analyzing, and electrode placement, a proposed position setting has been made to measure the chicken egg impedance.

Firstly, two electrodes were chose to be placed on either end of the egg. These are called the blunt end and the pointy end of the egg, because the air bag (Figure 3.8) inside the egg should always face up during the incubation process, these are considered the vertical ends of the egg and is the height of the egg. As two electrodes have already been placed on these ends of the egg, two more were also considered to be put on its width, creating a formation of 4 electrodes as illustrated in Figure 3.9.

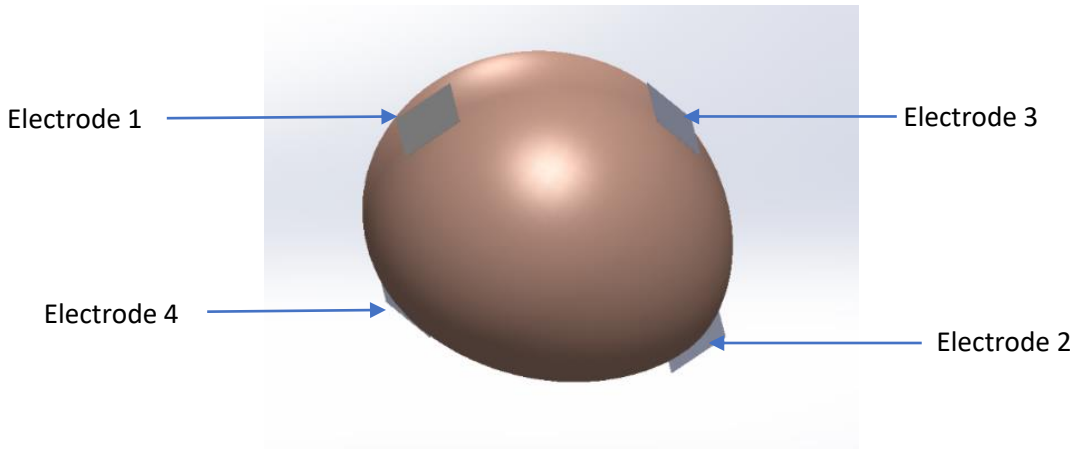

Figure 3.9. Electrode placement on experimental egg

Electrode placement determination device

As it is not easy to determine the width circumference, a device (Figure 3.10) was designed and fabricated using 3D printer to help with it. This device will determine where the eggs will in tangent with the poles, then by using a pencil to mark the tangent point the width circumference of an egg can be acquired.

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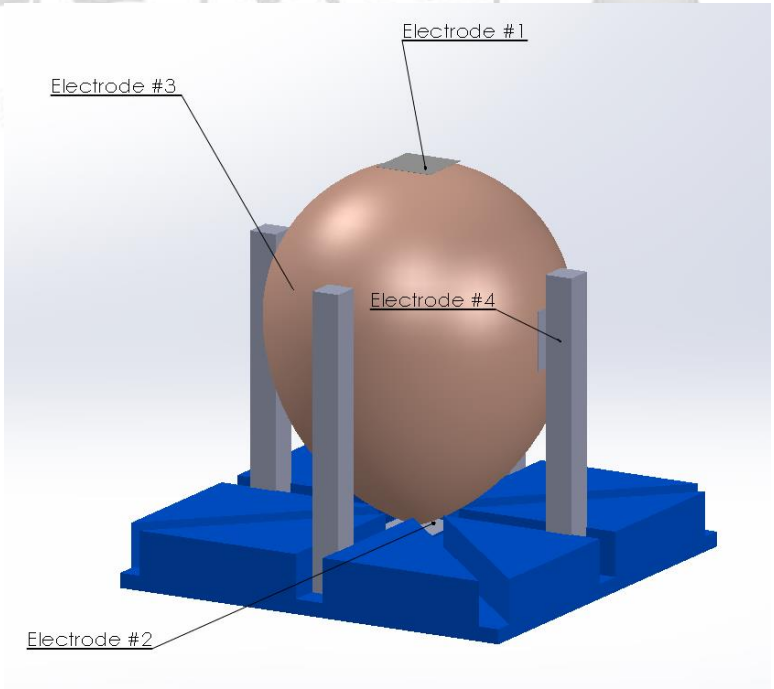

Figure 3.10. Electrode placement determination device

The device consisted of a square base of 50mm*50mm, this base has trenches to fit tangent poles into. The poles are 5mm width, 5mm height and 30mm length. The average

chicken egg size was found to be 34.4mm in length, 24.7 in width, so this device would be fitted for all the acquired eggs.

3.1.2.5. Signal cable connector and pinout circuit

Standard SMA headers for RG1.13 were used as a minimal connector interface, designed for use from DC (0Hz) to 12GHz. The SMA male headers were connected to the other end of the RG1.13 coaxial cables, heat-shrunk tubing was used to shield the exposed part. A pinout circuit was designed so that it would be easier to connect the impedance analyzer clip leads. The schematic is shown in Figure 3.11.

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Figure 3.11. SMA male header for the free end of the coaxial cable.

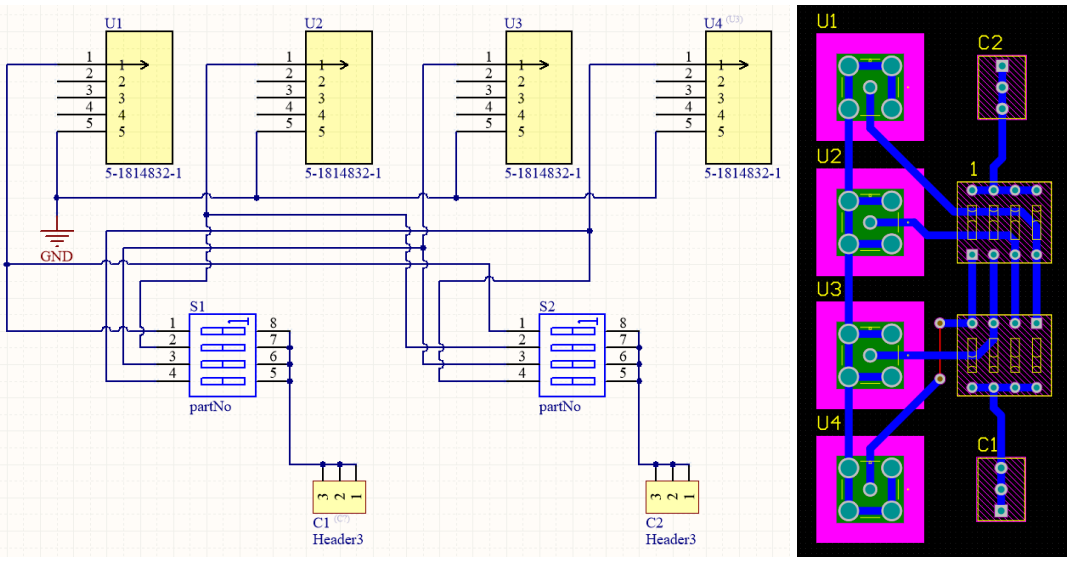

Figure 3.12. Schematic of chicken egg pinout circuit

This pinout circuit not only makes it easier to connect the impedance analyzer clip leads, switching between electrodes during measurement of each eggs, but also grounding shielding layer of the coaxial cables reduce the noise causes by electrostatic of a large number of wires.

Figure 3.12 shows the schematic of the pinout circuit. U1, 2, 3, and 4 are female SMA connectors which will connect to the SMA male connecters of the signal wires. S1 and S2 are dipswitches used for switching between electrodes. C1 and C2 are headers where the impedance analyzer clip lead will be clipped on.

During the impedance spectroscopy data gathering process, the pinout circuit was used as electrode switchs. As shown in figure 3.12, the first dipswitch (S1) is connect to electrode 1 and 3, the second (S2) connects to the others. By controlling the dipswitch, impedance spectrums of difference pairs of electrode connected to the eggs were measured. Based on standard BIA measirement, pair combinations are collected as pair 1 and 2, 3 and 4, 2 and 3, and 2 and 4.

3.1.3. Incubator preparation

A standard incubation was purchased and was set to operates at 55% humidity and 99.5 Fahrenheit degree.

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3.1.3.1. Incubator modification

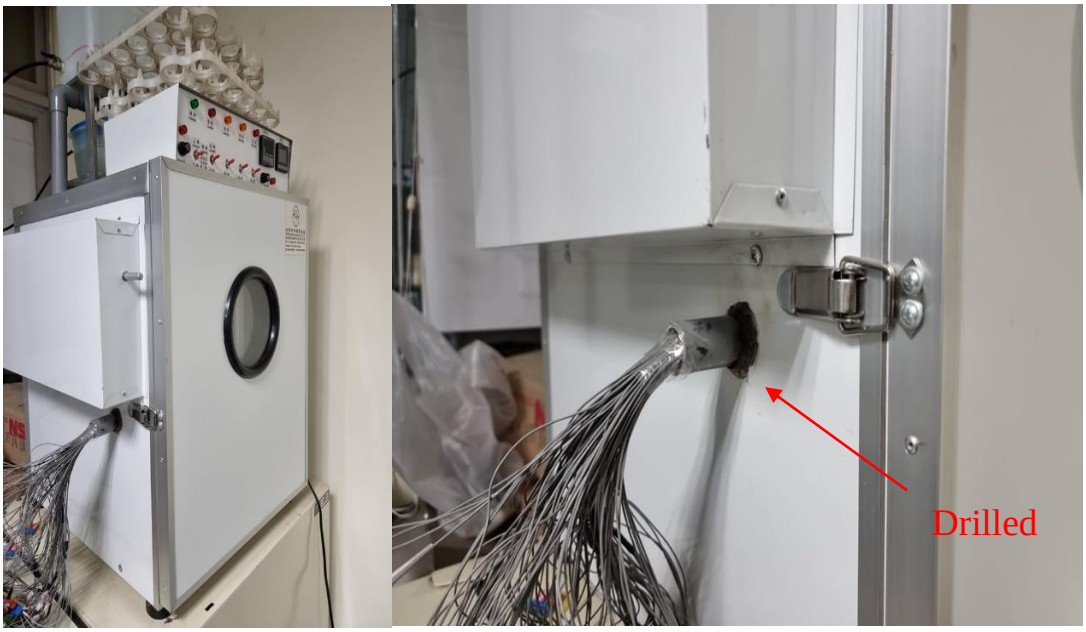

Figure 3.13. A circle hole with diameter of 2cm was drilled out of the incubator

A standard incubator was used in this research, but a little modification was needed in order to help with the bioimpedance spectroscopy process. As mentioned above, each fertilized egg will be attached with an electrode system including four one meter long signal cables. These wires must be led outside through a drilled hole. With a diameter of 20mm, a PVC tube was inserted into the hole and used to tunneled out the cable (Figure 3.13). The PVC tube was tightly fixed onto the incubator using silicone sealant.

3.1.3.2. Incubator sterilization

Standard fumigation process was applied to sterilize the incubator before each incubation process. Fumigation is a hazardous process so was performed carefully in solitary. The fumigation process was perfromed as follow:

a. Clean the inside of the incubator thoroughly using clean towel
b. Seal all of the air holes on the incubator, making sure no air leakage can happen
c. Exact 3mg of potassium permangannate was spooned into a ceramic container
d. Put the ceramic container inside the incubator through the main door
e. Exact 6ml of formalin was pipetted into the container
g. Immediately closed the door, and tightly sealed it as chemical reaction happen inside, where a poisonous brown-purple formaldehyde gas being produced.
h. After 20 minutes, the vent and other air holes of the incubator can be opened for air circulation.
i. Remove the ceramic container and the incubator is sanitized.

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3.1.4. Experimental procedure design

The eggs impedance are collected from 1st to 13th day of incubation process. The measurement at day 1 will be used as the base line as gender different was yet to appear inside the egg. From day 4, accordingly to Romanoff (1939), this is when the gender differences starts to appear, and on day 11, where Burke and Sharp found that was earliest for the size and weight differences to be significant. These will be the center point for analyzing in this research.

Bioimpedance data were measured using the Wayne Kerr 6420C Impedance Analyzer & LCR Meter (Wayne Kerr Electronics). The parameters are set as:

- 100 mV rms voltage
- Frequency range of 20Hz to 10MHz (100 points)
- Measurement speeds of 80ms per frequency measurement

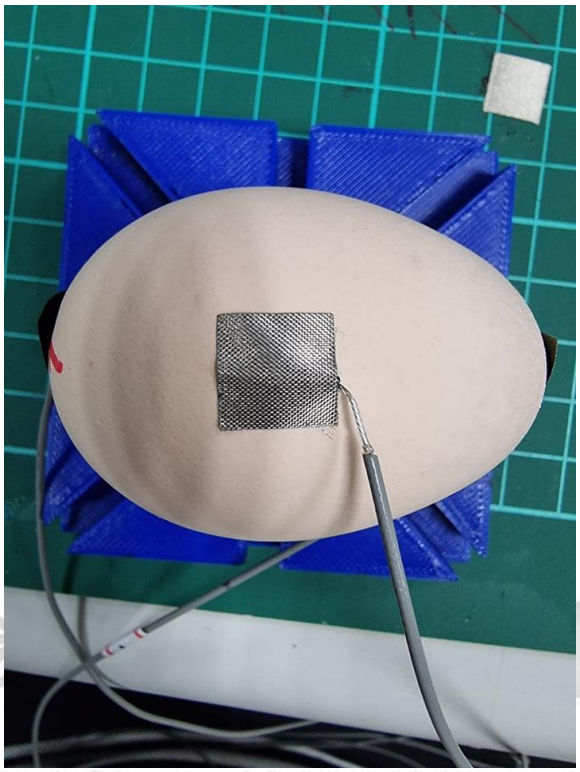

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Figure 3.14. Taping the electrode configuration on preset position of the egg

The experiment was performed step-by-step as described below:

a. Unsoiled the surface of the egg using a dry and clean towel, and marked using pencil. The eggs are kept at room temperature.

b. 10mm*10mm pieces of conductive fabric cloth were cut out and measured.

c. 1 meter long coaxial cables was cut, striped and shield layer insulated with heat-shrunk tubing.

d. A small amount of carbon epoxy as mentioned above was dropped onto the conductive fabric cloth.

e. Put the stripped end of the coaxial cable into the carbon epoxy dropped conductive fabric cloth.

f. Tape the whole configuration onto desired positions of each egg (Figure 3.14).

g. Let cured at room temperature for 24 hours.

h. Put the eggs inside the incubator, tunneled the wires out and connect them to SMA heads and pinout circuit.

i. Then every another 24 hours, make the impedance spectrum measurement.

3.2. Chicken egg genomic DNA extraction and gender identification using PCR

A standard PCR process described by Romanova et al (2019) [26](Figure 3.15), where genimoc DNA of the chick was collected from its blood and applied to the PCR, was used to determined the gender of the chicken egg after day 14 of incubation.

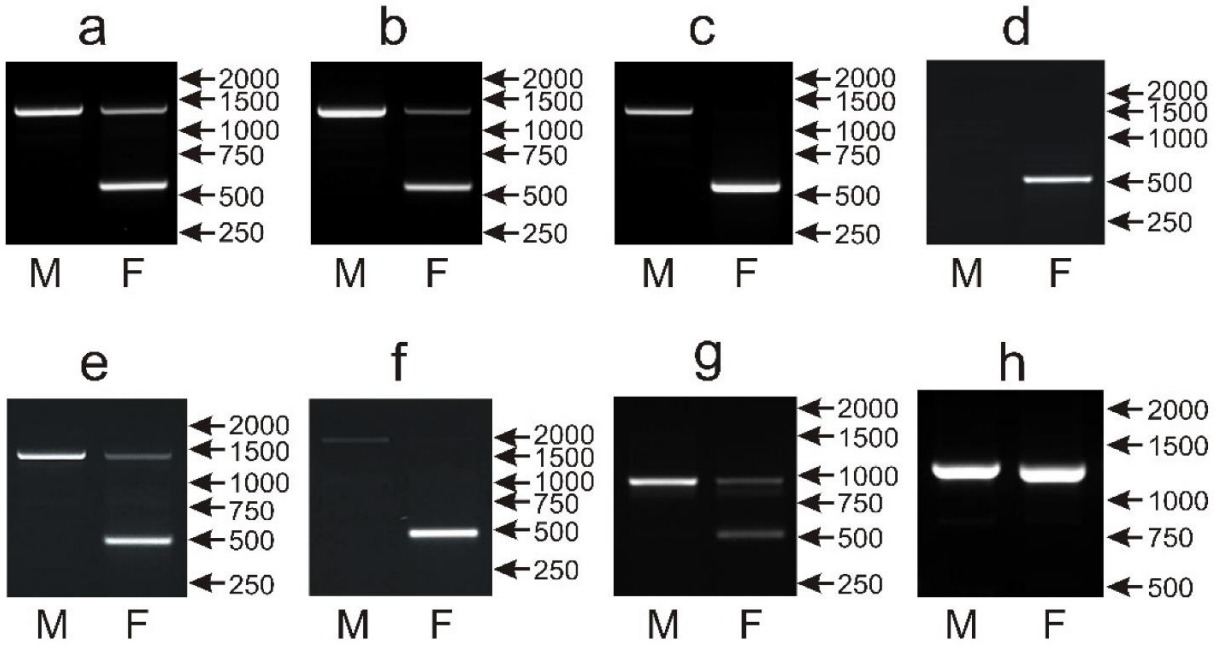

Figure 3.15. Sex identification for embryos using PCR [25].

The genomic DNA collection process is as followed:

a. A 10 µL whole blood sample was collected from each chicken fetus after 14 days incubation. The whole blood sample contained le.ukocytes and provided genomic DNA for gender identification. Genomic DNA was extracted using Blood Genomic DNA purification kit (Biokit, Barcelona, Spain).

b. The whole blood sample was added with 20 µL proteinase K, 170 µL Phosphate buffered solution and 200 µL binding buffer together.

c. The mixture was then vortexed and incubated in 70°C oven overnight.

d. After overnight incubation, each mixed sample was added with 200 µL 100% ethanol and transferred into filter column provided by Biokit genomic DNA purification kit.

e. This filter column was then centrifuged at 15,000 ×g for 1 min in room temperature.

f. The suspend solution was discarded, and pellet was washed with 600 µL washing buffer followed 15,000 ×g centrifuge for 1 min in room temperature, twice.

g. Centrifuged again at 15,000 ×g for 3 min in room temperature to remove remained water.

h. A 1.5 mL microtube was inserted with the filter column and 25 µL elution buffer was added into filter column to collect isolated genomic DNA.

Table 3.1. PCR program for chicken gender identification.

	Temperature	**Time**	
Initial	95ºc	30s	
Denaturation	95ºc	30s	30 cycle
Annealing	58ºc	1min	
Extension	72ºc	1min	
Finish	72ºc	5min	

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PCR was performed as Table 1. with each sample containing a total 25 µL mixture of 2 µL eluted genomic DNA, 16µL deionized water, 5µL master mix, 1 µL 2250 forward primer (5'-GTTACTGATTCGTCTACGAGA-3'), and 1 µL 2718 reverse primer (5'-ATTGAAATGATCCAGTGCTTG-3') using Biometra T-Personal thermal cycler (Analytikjena GmBH, Jena, Germany). The primer design is based on a pilot study for gender identification on avian (Romanova et al., 2019). Agarose gel (2%) electrophoresis was performed to analyze PCR products. Single band of 552 bp indicated a male fetus, whereas double bands in 358 bp and 552 bp indicated a female fetus.

3.3. Impedance measurement system design

The following part describe the impedance measurement system

The principle of the impedance and phase readout instrument (Figure 3.16) is as follow, an function generator generates a stable peak to peak voltage sine wave at different frequencies. The sine wave will go through a system of filters which will filtered out unwanted DC signal and the power line noise (60Hz), leaving only the frequencies that are in the desired range going through. This signal will then be the input for a Voltage Controlled Current Source (VCCS) circuit which will transform the input signal into a fixed frequency sine wave with a constant current. The current will then be injected onto the subject to be measured connected in series with a reference resistor. The voltage different as well as the sine wave frequency that passes through both the sensor and the resistor will be compared with the voltage different and sine wave frequency of the reference resistor by the help of the AD8302 which is a gain and phase detector intergrate circuit (IC). The impedance and the phase different will be the output of the AD8302.

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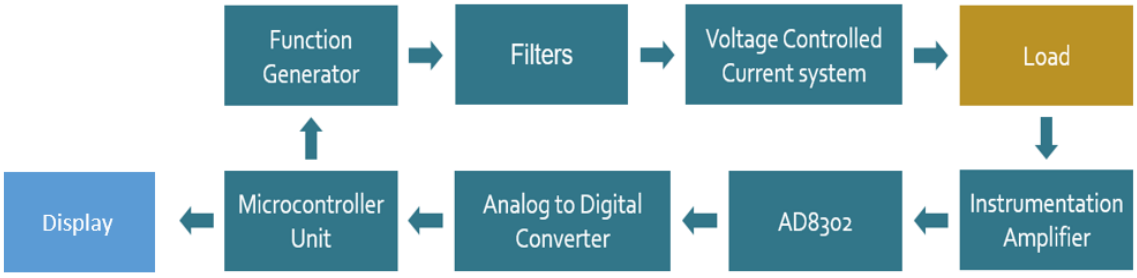

Figure 3.16. Impedance readout instrument system architecture.

The impedance readout circuit is mainly consisted of:

i. Sine wave function generator (Figure 3.17a): it is used for the generation of a fixed frequency sine wave voltage, which is then fed into the voltage controlled current source circuit. The integrated circuit (IC) MAX038 precision function generator was used to provide stable, precise, and adjustable frequency sinewave form with minimum external components required for the back-end system. The MAX038 operates with ±5V power supply and the output frequency can be controlled over a wide frequencies range of 0.1Hz to 20MHz. This is an excellent bandwidth and suitable for impedance measurement systems. The system will operate

at a total of 27 preset specific frequencies. The microcontroller will be used to control the analog multiplexer to select the required frequency signal to achieve various operating frequency ranges.

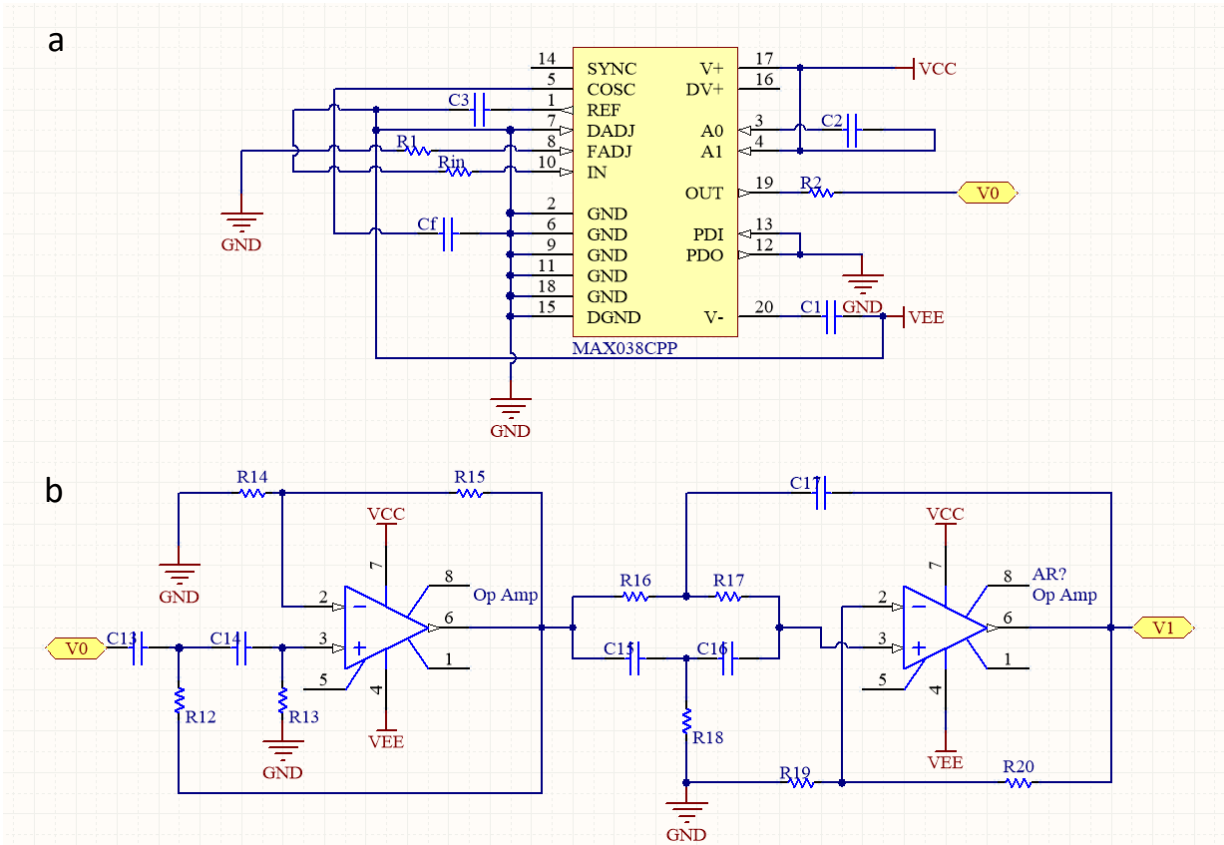

Figure 3.17. (a) Function generator with MAX038 and (b) the filter circuits.

ii. Voltage controlled current source (Figure 3.18a): its function is to generate a constant alternating current (AC) at a fixed frequency to inject into the target. A self-made enhanced Howland constant current source circuit was built, under different load conditions, generate a constant current through it.

The enhanced Howland's circuit needs only one Op-amp and five resistors to work. The output current is given by:

$$I = \frac{Vin}{R_5}\frac{R_6}{R_7} \qquad \text{(Equation 3.2)}$$

The resistance follows the ratio:

$$\frac{R_6}{R_7} = \frac{R_3}{R_4 + R_5} \qquad \text{(Equation 3.3)}$$

LM7171 operational amplifier was chose as the operating opamp. It has a gain bandwidth of 200MHz, very high slew rate of 4100V/µs, with a CMRR of 80dB.

iii. Instrumentation amplifier (Figure 3.18b): Use to amplify and read the potential different across the electrode. AD620 has a high Common-Mode Rejection Ratio (CMRR) value of 100dB, plus the low input bias current of 1nA can improve the signal-to-noise ratio and impedance measurement accuracy of the reading circuit.

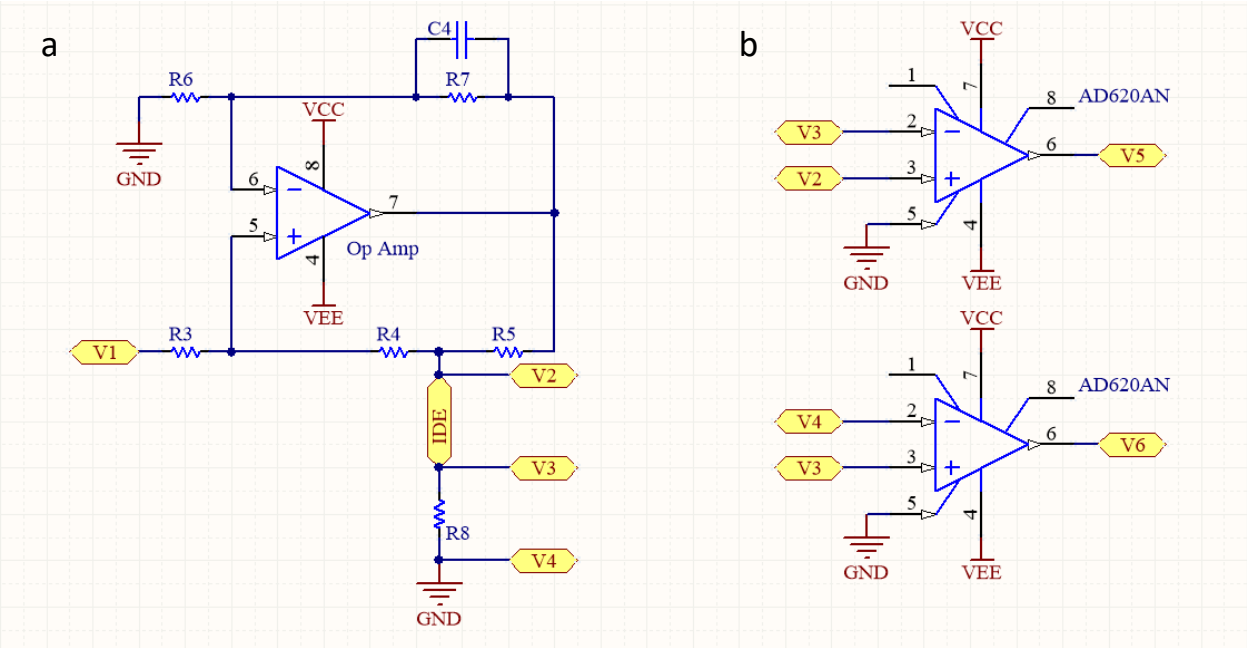

Figure 3.18. (a) Voltage controlled current source and (b) the instrumentation amplifiers.

iv. Filter circuits (Figure 3.17b): A combination of multi feedback active band pass filter, notch filter, and band reject filter to filter out the unwanted noise that is not within the targeted bandwidth of measurement. The operational amplifier chosen was the JFET op-amp AD8626.

v. Gain-Phase shift detector AD8302 (Figure 3.19): As the key component of the system for precise and complex impedance measurement, AD8302 is a fully integrated system for measuring gain/loss and phase shift. Its accurate phase measurement scale factor is 10mV / °, and the detection range can reach 180 °. With this chip, the magnitude of the impedance, and the phase difference between two signals can be measured.

The ac-coupled input signals can range from −60 dBm (316 µVpeak) to 0 dBm (316 mVpeak) in a 50 Ohm system, from low frequencies up to 2.7 GHz (AD8302 data sheet 2002).

Figure 3.19 shows the actual connections circuit of the AD8302. This also determine the impedance measurement range estimated by the equations:

$$\begin{cases} Z_{min} = \frac{316(\mu V)}{G*I} \\ Z_{max} = \min\left\{\frac{V_{omax}(mV)}{G*I}, \frac{316(mV)}{G*I}\right\} \end{cases} \quad \text{(Equation 3.4)}$$

Where G is the gain of the instrumentation amplifier, I is the desired output current and the V_{omax} is the output voltage amplitude of instrumentation amplifier.

vi. Analog-to-digital converter (ADC) circuit: its function is to convert an analog DC signal voltage into digital voltage, which is then fed into the microcontroller.

vii. Microcontroller unit (MCU): its function is to calculate the impedance, the real part of impedance, the imaginary part of impedance, and the phase shift between the signal before and after going through the target, and to control the ADC circuit and the display.

viii. Display: its function is to real-time display the testing procedures as well as impedance, real part of impedance, imaginary part of impedance, and phase changes.

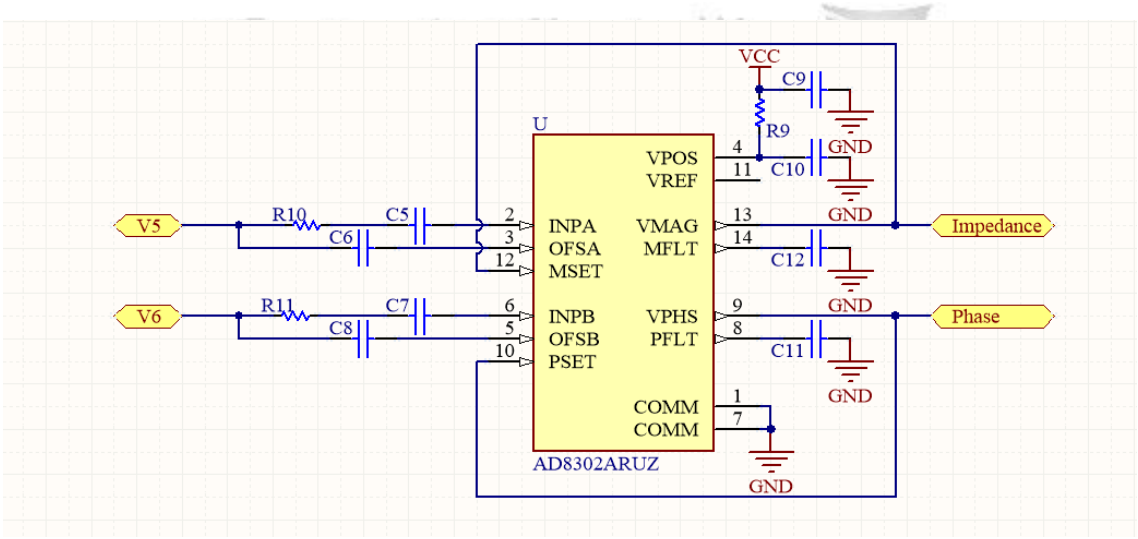

Figure 3.19. Gain and phase detector AD8302.

3.4. Statistical analysis method

Independent T-tests were performed to confirm the connection between the acquired measurement data and the chicken embryos PCR confirmed gender. The analysis were made using Microsoft Excel software.

CHAPTER 4. RESULTS AND DISCUSSION

4.1. Electrode validation tests

To validate the elctrode performance it was tested with two different test. The first one is to measure the resistance of the electrodes making sure that they are similar, reducing electrode impedance error. The second test is involve using an unfertilized, unhatchable egg. By measuring its impedance spectrum over a span of eleven days in and outside of the incubation, the performance of the electrode can be validated.

A total of 20 electrodes configuration was prepared without taping onto the egg. Their resistance was measured using a standard Fluke multimeter. As shown in Figure 4.1, the measured resistance was found varies from 1.78 to 3.12 with an average of 2.383. This value was found to be in agreement with the calculated resistance value of the electrodes meantioned in section 3.1.2.3.

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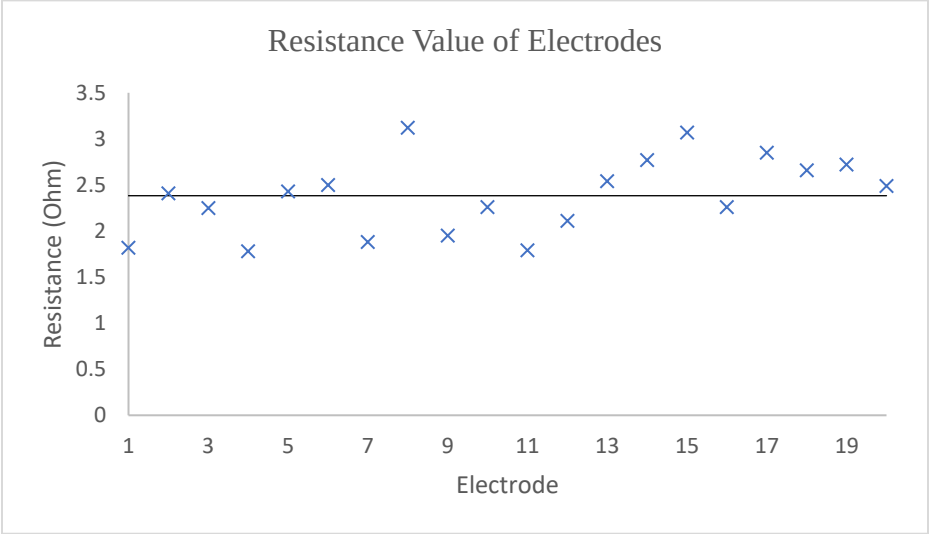

Figure 4.1. Electrode resistance validation

An unhatchable egg was attached with a pair of ECG electrode, an impedance measurement was made then remove the ECG electrode. After the ECG electrodes was removed, and its surface has been wiped clean with a dry towel, a pair of designed electrode was attached

to position 1 and 2 of the egg. The impedance spectrum was made after 24 hours of attachment at room temperature. Then again after 1 hour it has been put in the incubator at a temperature of 99.5 Fahrenheit degree, this was marked as Day 0 of incubation measurement. Then the impedance was measured again on Day 1, 4, and Day 10 of the incubation process, these measurement was made to test the electrode performance after a long time in a fixed humidity (55%) and temperature (99.5F) environment. The impedance spectrum is shows in figure 4.2, from here, it can be observed that there is a quite large variation in impedance value at very low frequencies (20Hz to 100Hz). But from the frequency of 100Hz and above, the agreement level is increasingly better.

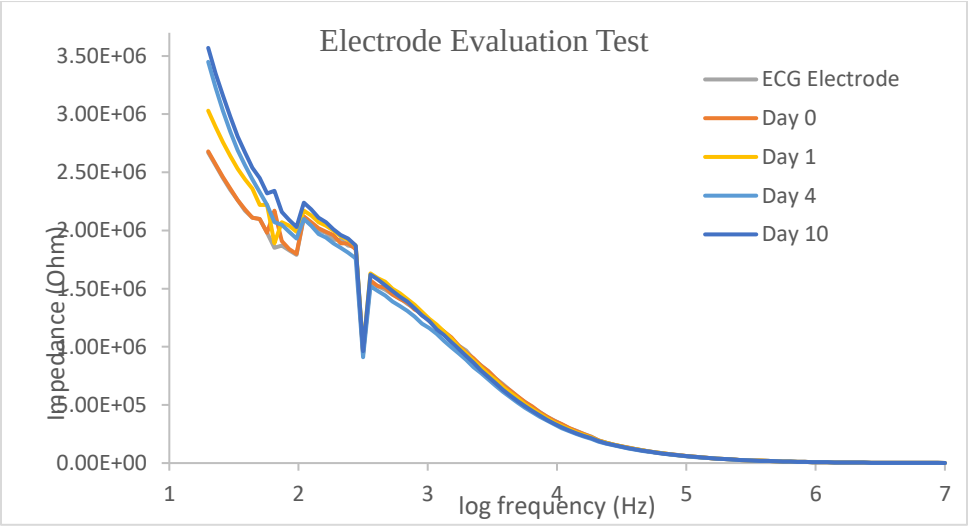

Figure 4.2. Impedance spectrum of unfertilized egg measured with ECG electrode, then with the designed electrode on different days of incubation process.

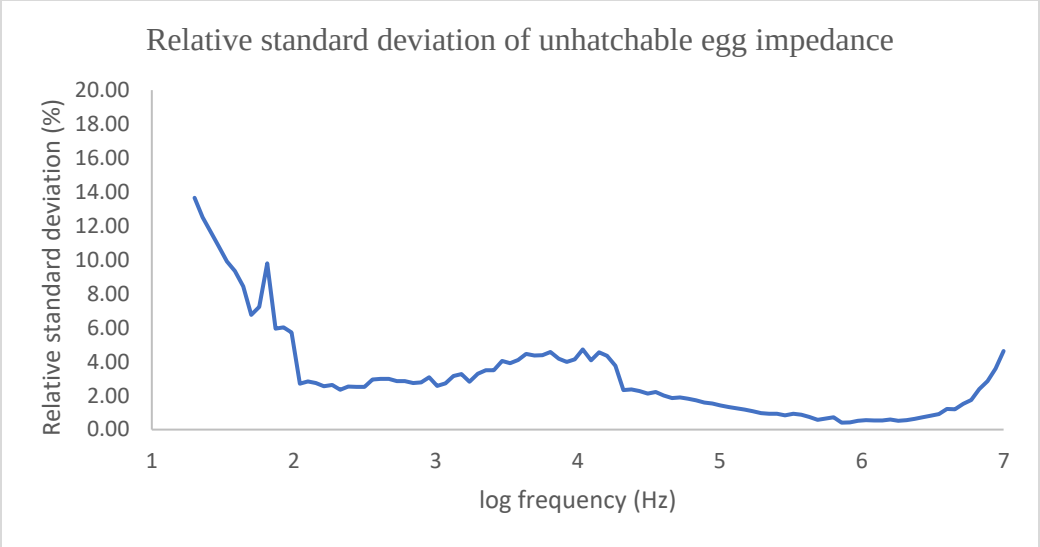

Figure 4.3. Relative standard deviation of impedance measurement made on an unhatchable egg

The relative standard deviation (RSD) (Figure 4.3) was used to determine the repeatability of the electrode. The highest RSD was 13% at 20Hz, from 100Hz to 10MHz, RSD value was found to be smaller than 5%. Even at high frequencies, the RSD remain below 5%. Proving that the electrode performance was acceptable for the remaining of the research. The variation from the ECG electrode maybe caused by the temperature change from inside of the incubator (at 99.5^{o}F) compare to 70°F of the room temperature where the original ECG measurement was made.

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4.2. Fertilized egg gender identification test

4.2.1. PCR chicken gender result

Table 4.1. Survival, gender records for 14 days incubated fetal Kinmen county chicken

No.	Status	Survival status	Gender
1	+	Good	M
2	+	Good	M
3	+	Good	M
4	+	Good	F
5	+	Good	M
6	+	Good	F
7	+	Good	F
8	+	Good	F
9	+	Good	M

10	+	Good	F
11	+	Good	F
12	+	Good	M
13	+	Good	M
14	+	Good	F
15	N/A	N/A	N/A
16	+	Good	M
17	+	Good	F

Remarks: N/A: none fertilized or dead with fetal condition, +: Survival, M: Male, F: Female

Among the 17 obtained eggs, 1 was found not abled to identified, 8 were male, and 8 were female. Table 4.3 summarize the result of the PCR chick egg sex determination. Figure 4.4. shows the obtained PCR result.

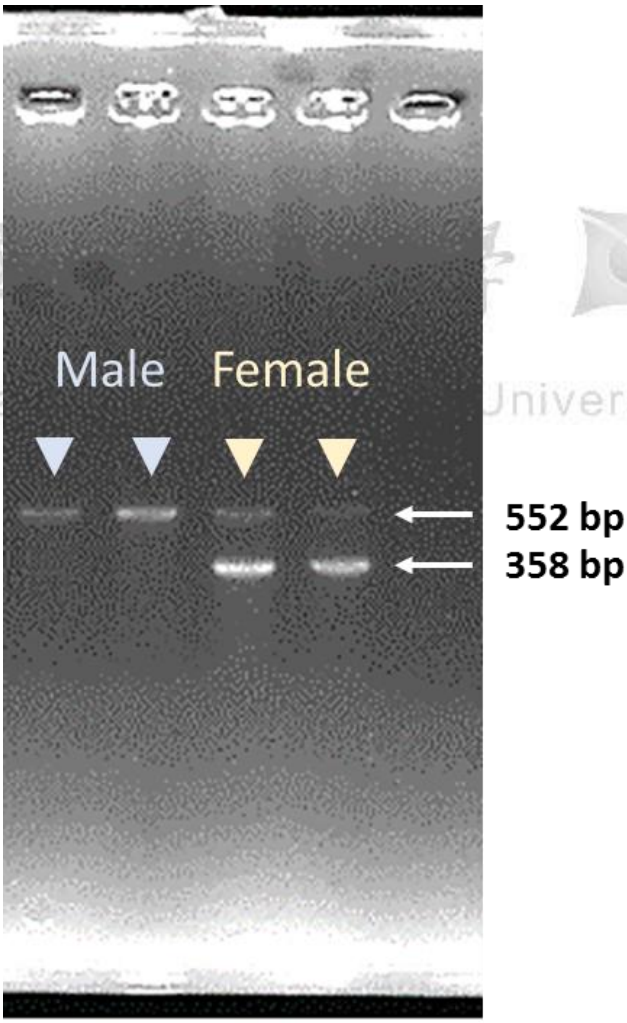

Figure 4.4. Sex identification for chick fetus using PCR. One single band at 552 bp indicated a male fetus and two bands at 552 bp and 358 bp indicated a female fetus.

4.2.2. Impedance measurement result

In this part, the impedance value of each eggs will be shown for each pair of measurement on difference days of the incubation process. The impedance data was found to contains quiet an amount of noises.

The impedance measurements were made for pair 1 and 2, 3 and 4, 2 and 3, and 2 and 4. The graphs are titled in the form of (Gender)_(Electrical Properties measured)_(Pair)_(day). As there are so many results, the figures are put in Appendix section of the thesis. In Apendix, Figure A.1 to Figure A.52 shows the impedance data recoreded from 8 male and 8 female eggs between electrode position 1-2, 2-3, 2-4, and 3-4.

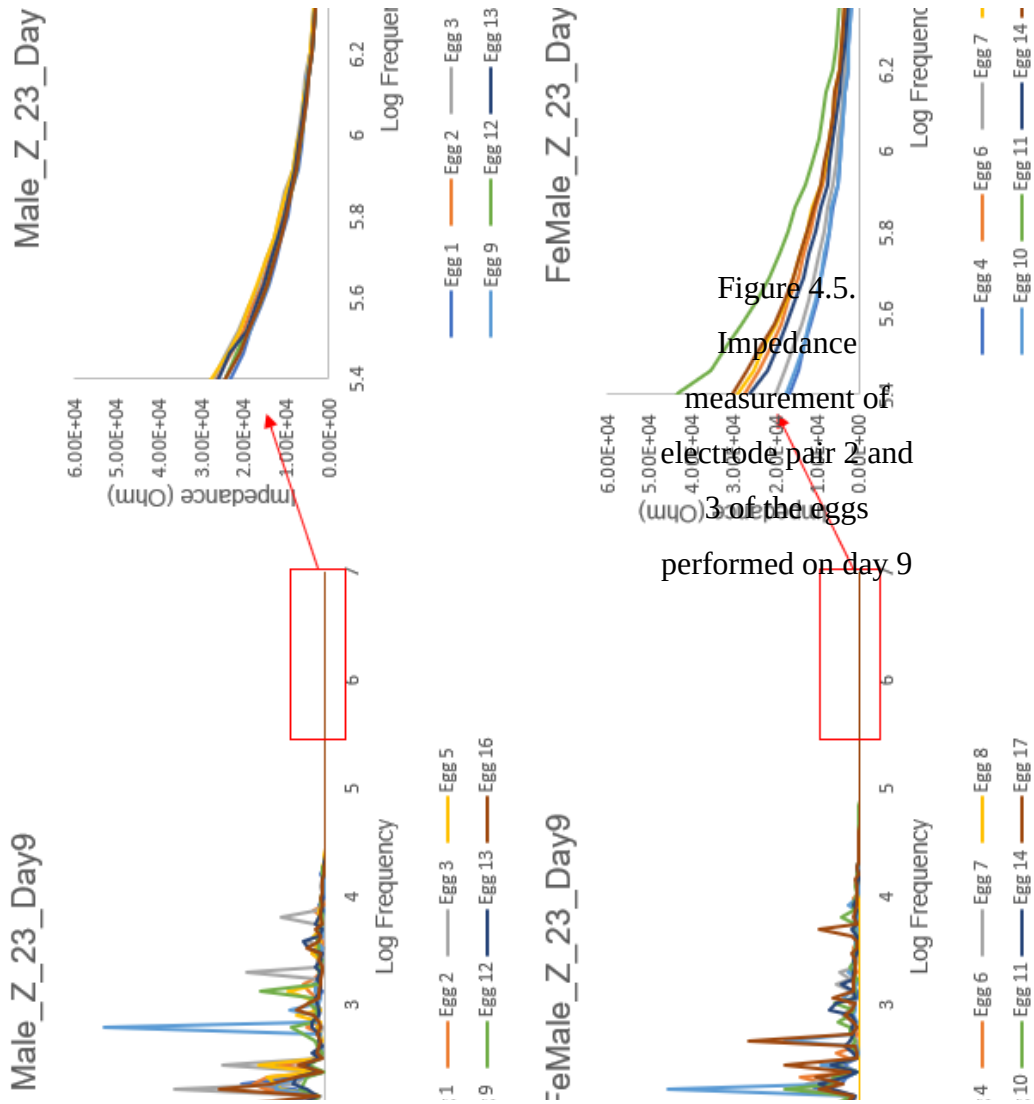

Figure 4.5. Impedance measurement of electrode pair 2 and 3 of the eggs performed on day 9

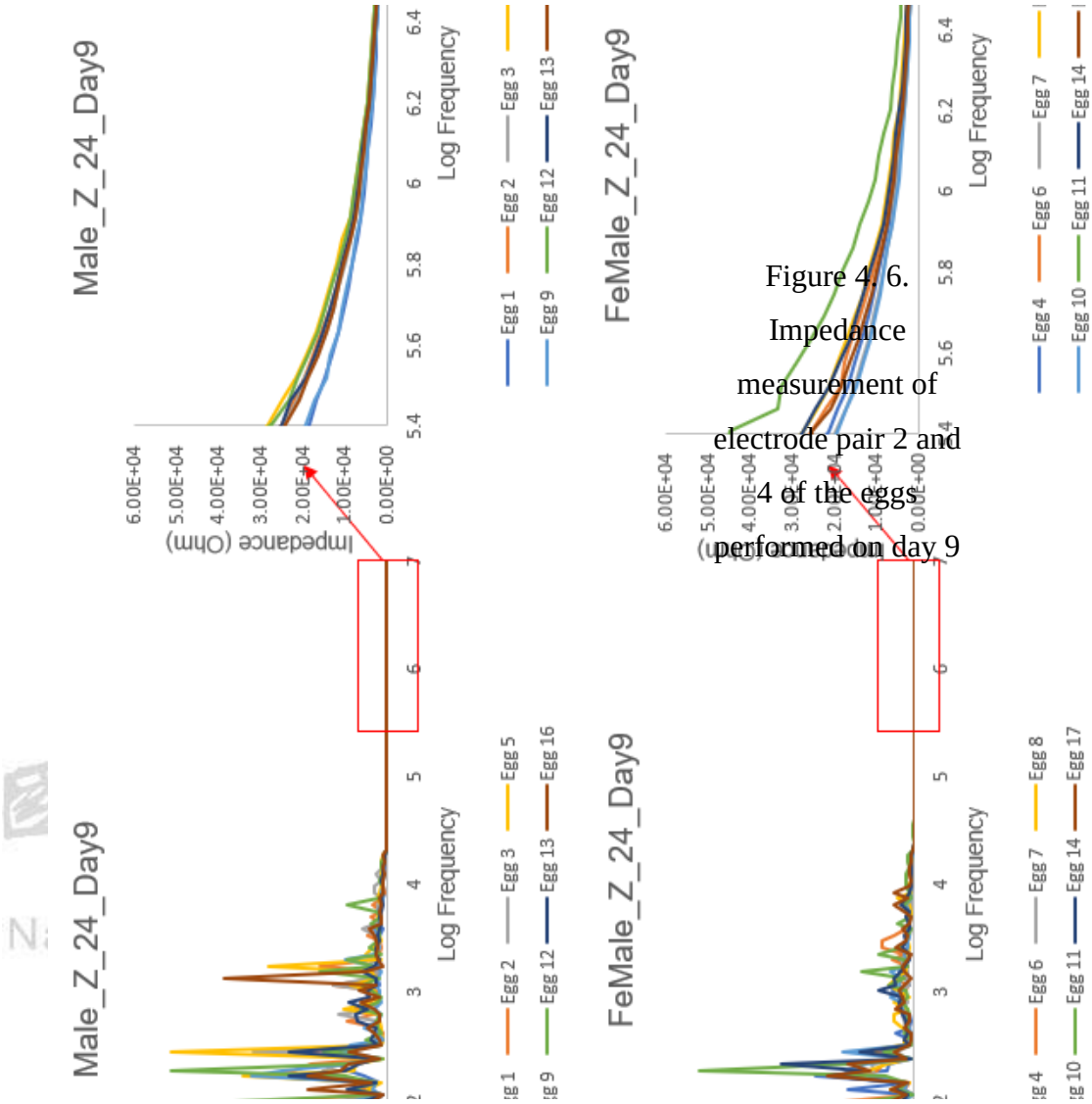

Figure 4. 6. Impedance measurement of electrode pair 2 and 4 of the eggs performed on day 9

Figure 4.5 and Figure 4.6 shows the male and female impedance spectrum measured for electrode pair 2-3 and 2-4. Even though having many noises, the impedance spectrums shows agreements at certain frequencies, having multiple peaks. These noises are believed to be caused by the rapid development of the embryos inside the eggshell. Nevertheless, data analysis was made in order to get the relation between chicken gender and collected data.

4.2.3. Self-normalization ratio

The data were measured between different pairs of electrode on the egg, these measurement at different pairs were then used to calculate a self-normalized ratio. Self-normalization idea has been utilized in studying the parameter, with the intention of keeping

away from the large embryos-to-embryos differences during data analysis. This ratio data would shows more independent and reliable on distinguished the gender differences.

4.2.3.1. Impedance self-normalization ratio

Normality test (Shapiro – Wilk) was conducted for all male and female of each pairs ratio. Data are normally distributed with all p-values greater than .05. The ratio data between pair 2-3 and 2-4 (Figure 4.7) shows an significant relationship between chicken egg gender and the impedance ratio at a range of frequencies as early as day 9 of incubation. The range was found to be from 376kHz to 2.69MHz. In this range of frequencies, the ratio between impedance measurement at pair 2-3 and pair 2-4 (Z_{2-3}/Z_{2-4}) can be used to distinguished the gender of the chicken egg. From day 9 to day 13 (Figure 4.8- 4.12), independant t-test was performed between male and female data set at each frequency among the range and all p value was found to be smaller than .05, meaning the impedance ratio (Z_{2-3}/Z_{2-4}) of male data set was significantly different from the impedance ratio (Z_{2-3}/Z_{2-4}) of female data set.

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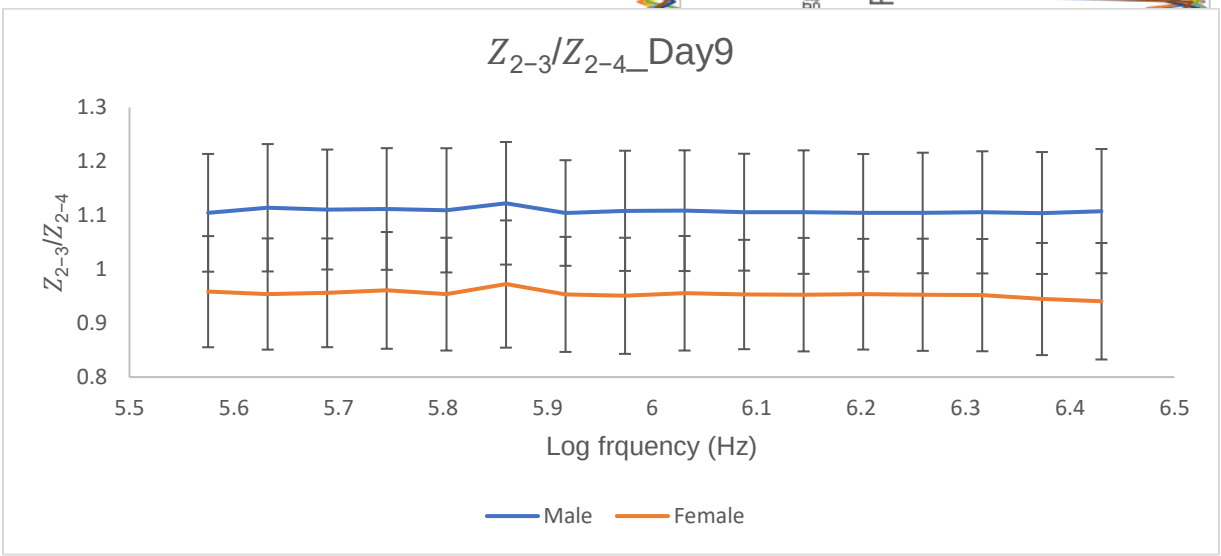

Figure 4.7.

Impedance ratio

Z_{2-3}/Z_{2-4} at day 9 of incubation process

Figure 4.8. Impedance ratio Z_{2-3}/Z_{2-4} at day 9 of incubation process between male and female data. Significant different (p value < 0.05) was found between frequency range (376kHz – 2.69MHz)

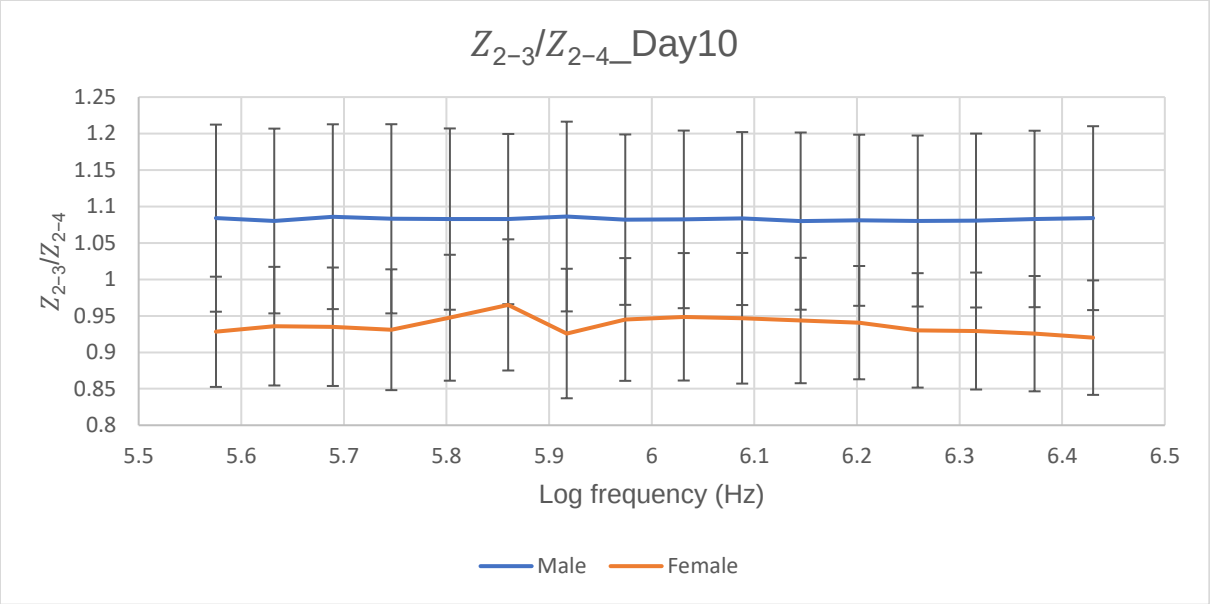

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Figure 4.9. Impedance ratio Z_{2-3}/Z_{2-4} at day 10 of incubation process between male and female data. Significant different (p value < 0.05) was found between frequency range (376kHz – 2.69MHz)

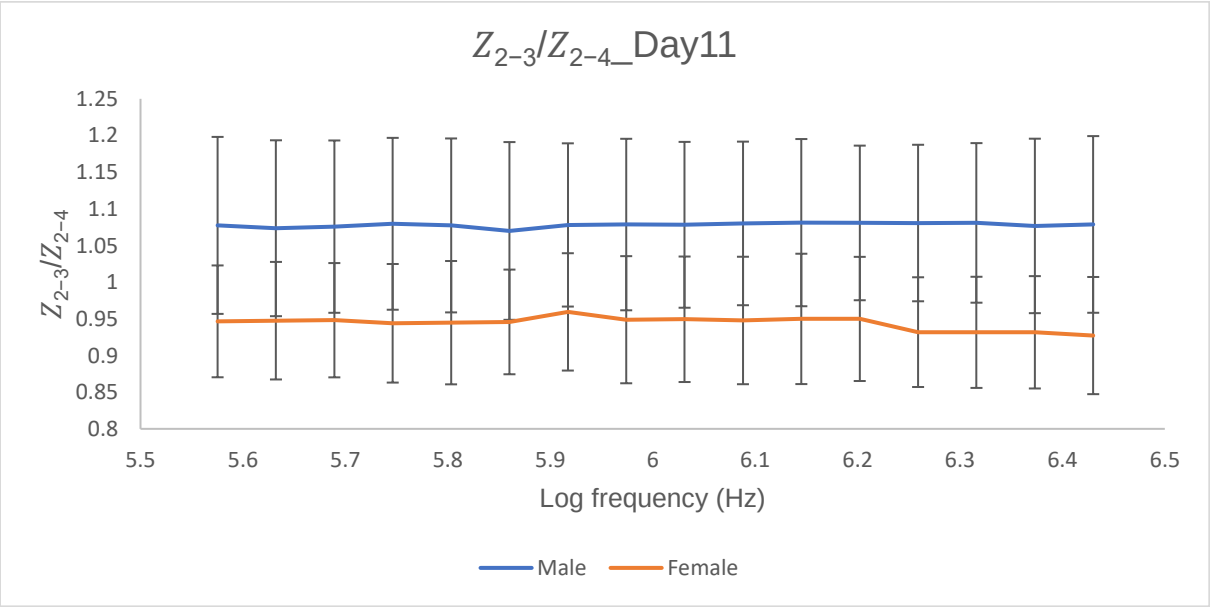

Figure 4.10. Impedance ratio Z_{2-3}/Z_{2-4} at day 11 of incubation process between male and female data. Significant different (p value < 0.05) was found between frequency range (376kHz – 2.69MHz)

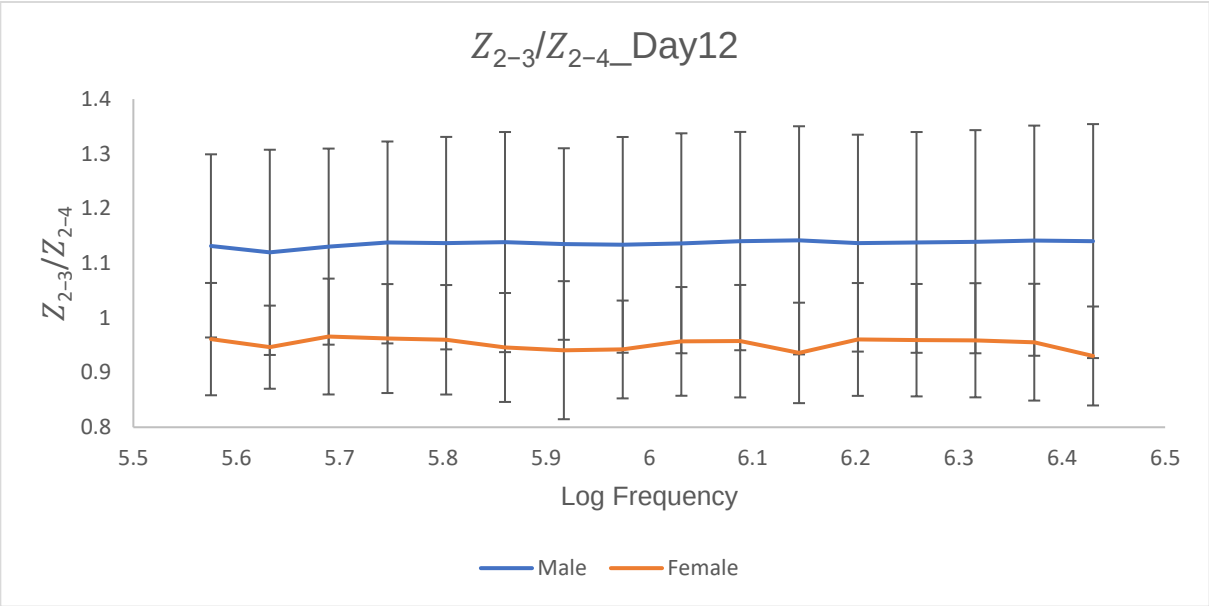

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Figure 4.11. Impedance ratio Z_{2-3}/Z_{2-4} at day 12 of incubation process between male and female data. Significant different (p value < 0.05) was found between frequency range (376kHz – 2.69MHz)

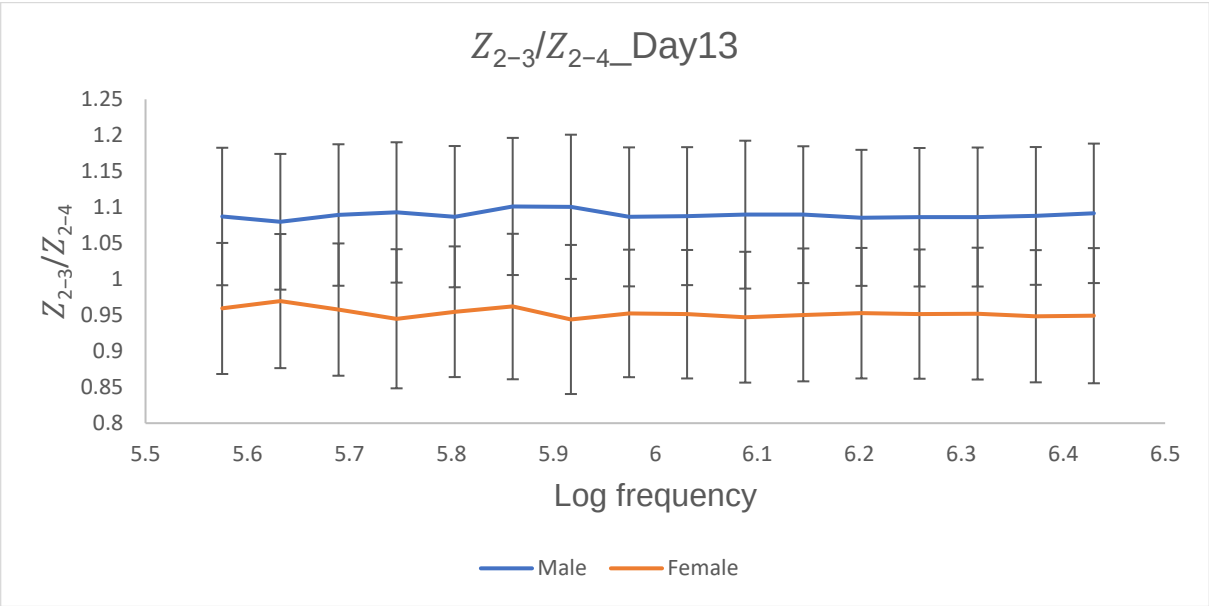

Figure 4.12. Impedance ratio Z_{2-3}/Z_{2-4} at day 13 of incubation process between male and female data. Significant different (p value < 0.05) was found between frequency range (376kHz – 2.69MHz)

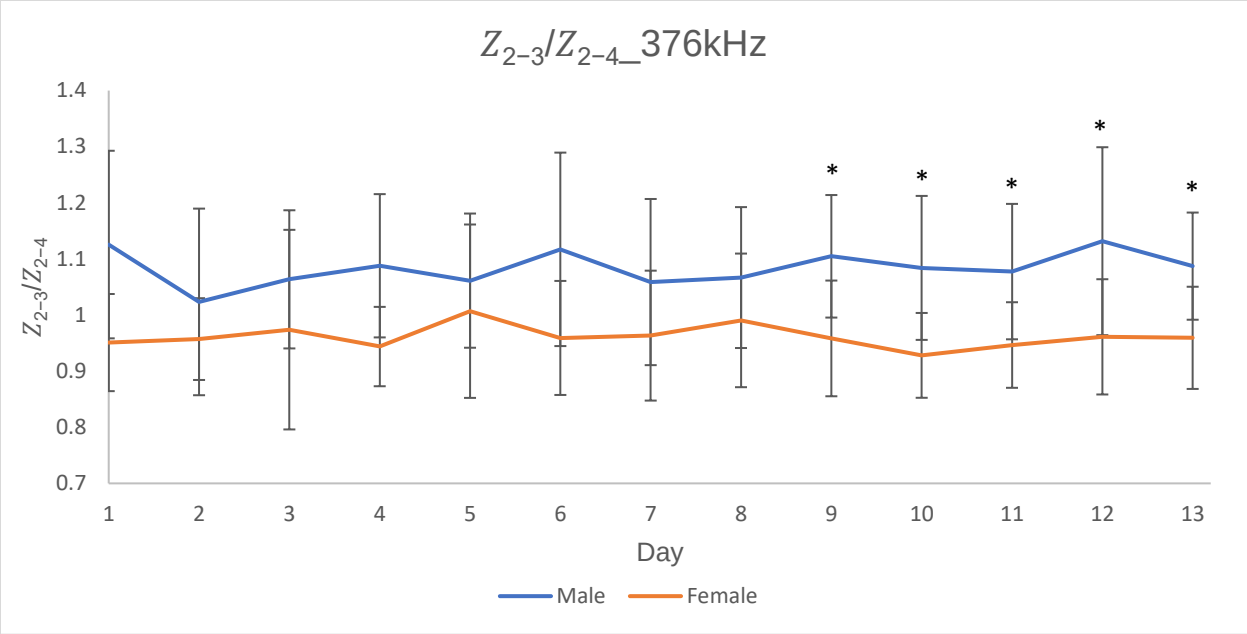

Figure 4.13. The impedance ratio Z_{2-3}/Z_{2-4} measured at optimum frequency 376kHz from day 1 to 13 of incubation process. *Remark: * p value < 0.05*

For development purpose the frequency of 376kHz was chosen as the optimum measurement frequency for the impedance ratio between pair 2-3 and 2-4. Figure 4.13 shows the impedance ratio of the pairs at different days of the incubation process with the significant can be found from Day 9.

A similar result with Burke and Sharp finding was found, where the different in the ratio equivalent to the gender can be found very early during the incubation days but it is not until day 9 that the differences become significant. From day 9 through day 13 of the incubation process, this differences is always significant proving that the applied method is consistent.

4.2.3.2. Capacitance self-normalization ratio

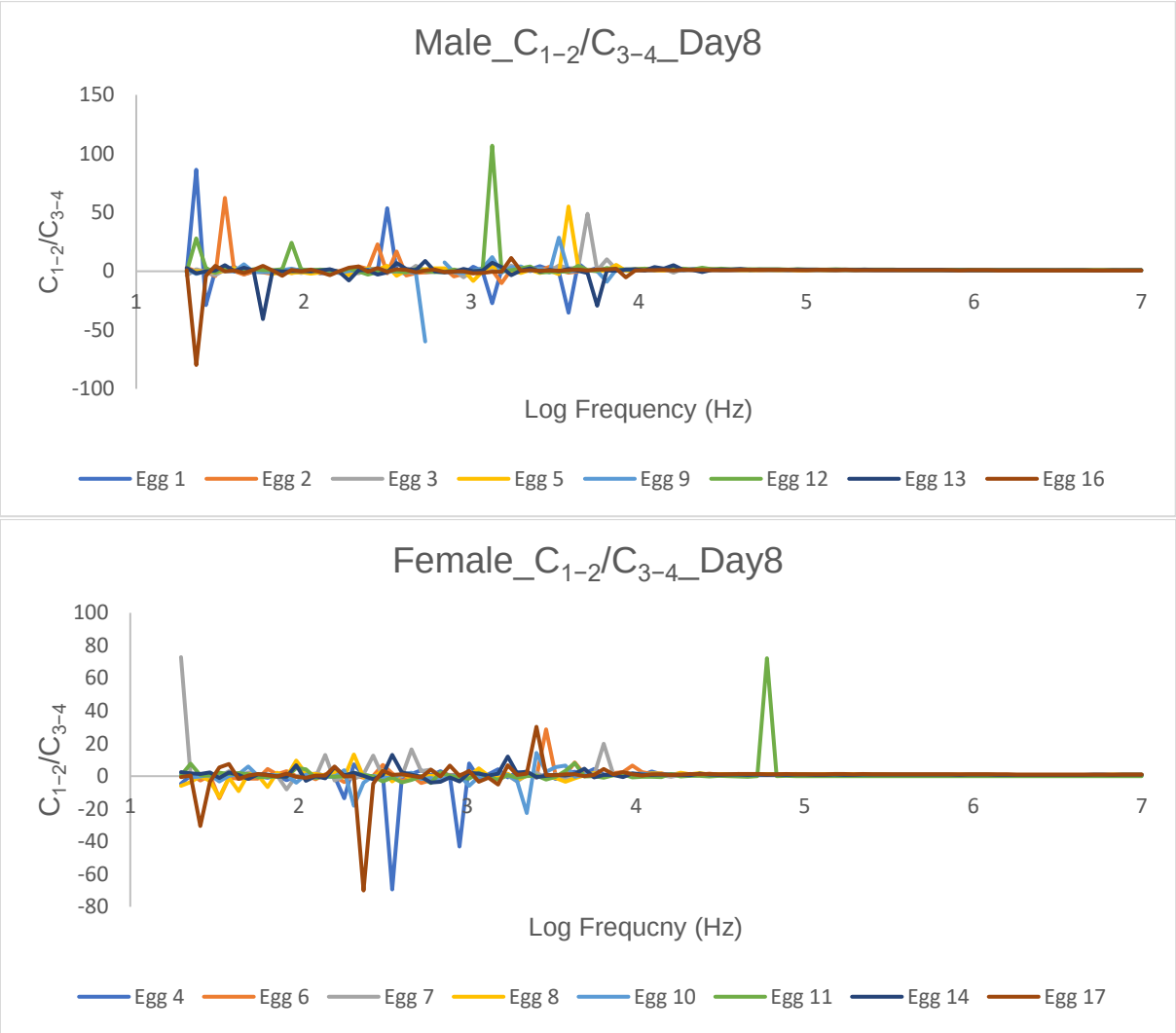

Figure 4.14. Capacitance ratio C_{1-2}/C_{3-4} at day 8 of incubation process

During experiment, capacatance data of different pair has also been collected for analysis consideration. After the impedance finding, capacitance was analyzed. The capacitance data are show in the Apendix.

The result shows that not only impedance data, but also capacitance data shows a similar finding. Figure 4.14 shows the capacitance ratio between pair 1-2 and pair 3-4 at day 8 of incubation process. For the capacitance, the optimum frequency was found to be 35kHz (Figure 4.15). The same phenomenon, the capacitance ratio between pair 1-2 and 3-4 can be used to distinguished the gender of the chicken egg. As can be seen from Figure 4.15, at day 1, the base

line data, the capacitance was found to be almost the identical, this difference increases and become significantly different from day 8 of incubation process. Meaning capacitance ratio value can be use to identify the gender of the chicken egg earlier than impedance ratio.

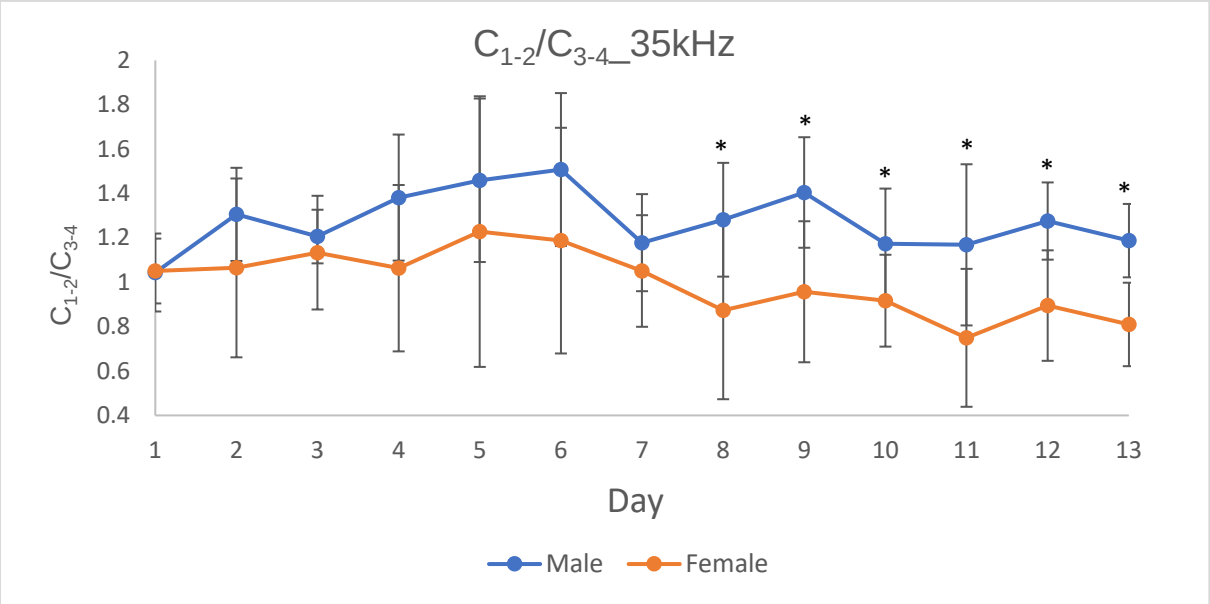

Figure 4.15. Capacitance ratio $C_{1\text{-}2}/C_{3\text{-}4}$ measured at optimum frequency 35kHz from day 1 to 13 of incubation process. *Remark: * p value < 0.05*

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4.3. Impedance measurement system

Bassed on the finding, the impedance measurement system specification was determined. The device should be able to measure the impedance value at the frequency of 376kHz. The highest impedance value found was approximately 40k Ohm.

In this study, the AD8302 was used, with the ac-coupled input signals can range from −60 dBm (316 μVpeak) to 0 dBm (316 mVpeak) in a 50 Ohm system. With a maximum measurement resistant of 40k Ohm, applying to equation 3.3

The gain of the instrumentation amplifiers are 1 (G = 1), the V_{omax} of the AD620 is 7.8V for a ± 9V supply system. So the desired inject current can be calculated as low as 225μA.

4.3.1. Function generator

The MAX038 function generator used an external resistor and capacitor to output a 2Vp-p sine-wave signal with a desired frequency.

This frequency is calculated as

$$F_0 = \frac{2\times 2.5V}{R_{IN}\times C_F} \qquad \text{(Equation 4.1)}$$

For an output of 376kHz sine wave signal, the R_{IN} was chosen to be 13.5kOhm, and the C_F was 10nF. The output sine wave generated from MAX038 waveform is shown in figure 4.16.

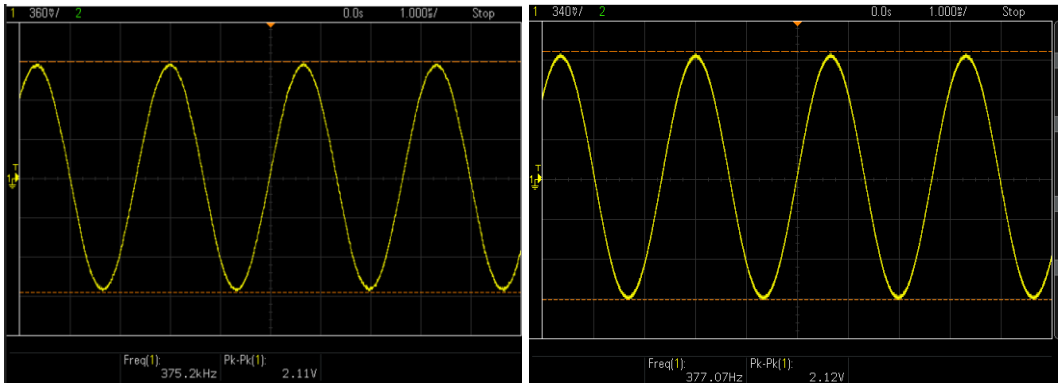

Figure 4.16. Output waveform of MAX038

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Due to component error, the ouput frequency varies from 375.3kHz to 377kHz. This error was found to be of only 0.2 percent making it acceptable.

4.3.2. Voltage-controlled current source

The current source was an enhanced Howland current pump (Figure 4.17)

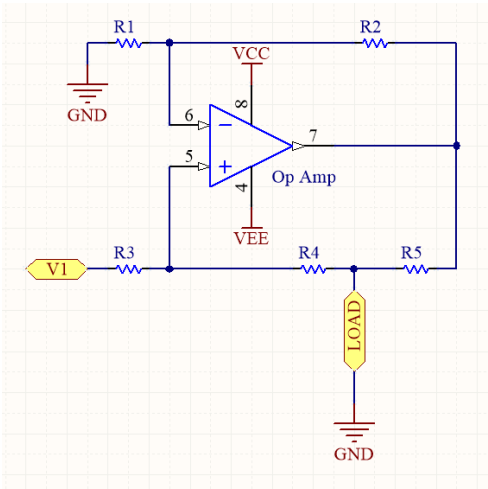

Figure 4.17. Enhanced Howland current pump

The chosen Opamp was the LM7171 with a gain bandwidth of 220MHz, the resistors are set to be $R_1 = R_2 = 47000$ Ohm, $R_3 = R_4 + R_5 = 47000$ Ohm.

The output current was calculated as $I = V_{IN}/R_5$. With V_{IN} is the output of the function generator MAX038, $V_{IN} = 1V$. Then R_5 should equal to 4400 Ohm, making R_4 value set to be 42600 Ohm.

Voltage controlled current source simulation result

Figure 4.18 shows the simulated current source frequency response of the designed circuit with the components error of 5%.

The simulation results with the Gaussian distribution for components error of 5% shows that the lowest gain bandwidth was at 167kHz and the best result falls at 18MHz using Monte Carlo method provided by PSpice® for TI simulation program. This result indicates that by building an exact circuit using the mentioned components, the voltage controlled current source has a high probability of capable of given out a stable current with a gain bandwidth of at least 10MHz, which is the median of the simulation result.

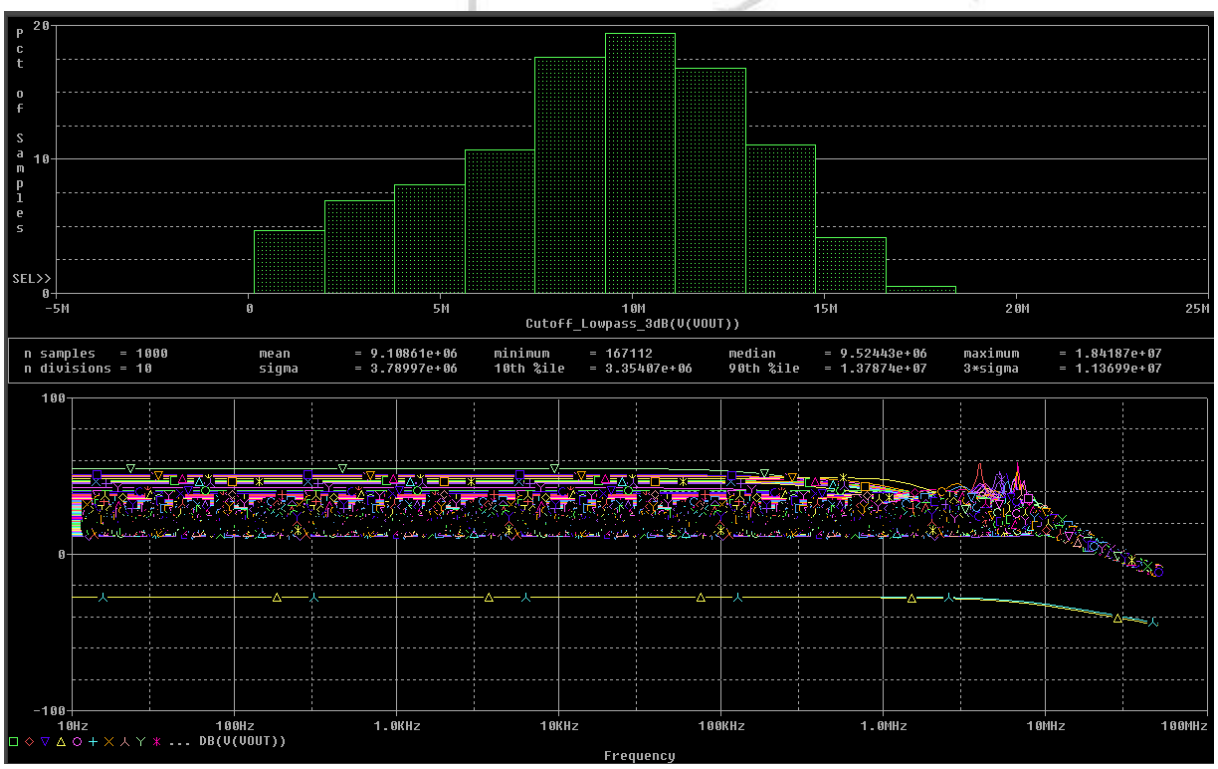

Figure 4.18. Simulated frequency response of current source injecting to a load of 40k Ohm

Voltage controlled current source PCB test result

Figure 4.19 shows the voltage controlled current source gain bandwidth of the designed PCB as obtained from the gain and phase detector AD8302 compare with the result from the precision impedance analyzer. The same load used was a pure resistor with the value of 43012 Ohm. The function generator MAX038 was set to give out 60 points of frequency in the range of 20Hz to 400kHz

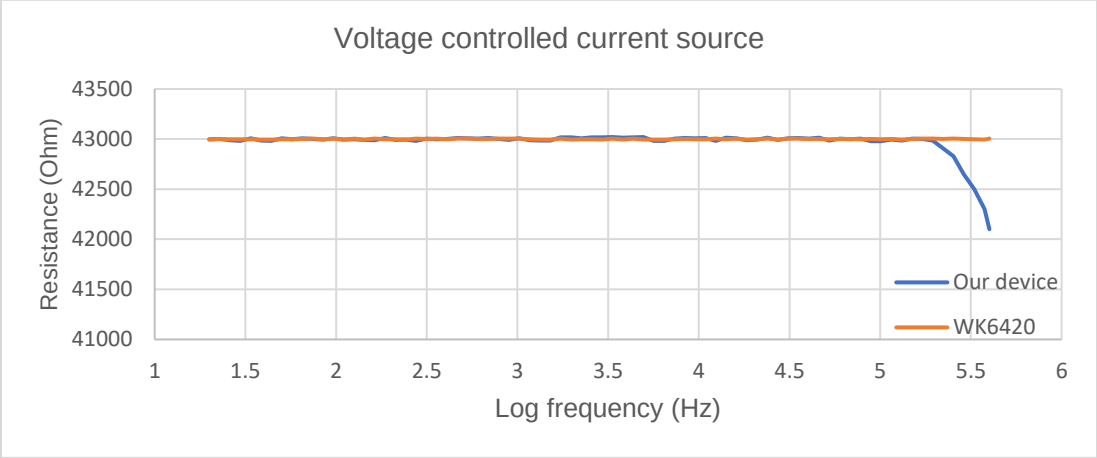

Figure 4.19. Fabricated VCCS gain bandwidth

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The results shows that the gain bandwidth is 220kHz, lower than the expected gain bandwidth and also lower than the desired frequency of 376kHz. This current source gain bandwidth was among the simulation results meaning that we have successfully fabricated a working voltage controlled current source. Unfortunately, the gain bandwidth was not enough for test on egg measurement.

4.4. Discussion

The designed electrode performance was found to be accurate and repeatable comparing to measurements made using a standard ECG electrode. The agreement level is very high for frequency range from 100Hz to 10MHz with a relative standard deviation of less than 5%.

Although having shown a good electrode performance, the impedance spectrums contain noises at very low frequency range of 20Hz to around 200kHz. These noises were believed to be caused purely by the egg and the components inside of the egg, including the evolution of the

embryo, and the proteins changes inside the eggshell. From frequency at 300kHz and above, the spectrum shows less noises, this is believed to be due to penetration of high frequency signal to the eggshell. As mentioned above, the egg can be imagine as a cell structure with the egg shell is the cell membrane. This cell membrane are commonly expressed by capacitor as equivalent circuit, thus higher frequencies would be more ideal to pass through this capacitor, leading the signal into and outside the eggshell.

Due to the noises found in the spectrum, self-normalization idea has been utilized in studying the parameter, with the intention of keeping away the large embryos-to-embryos differences during data analysis. This ratio data would shows more independent and reliable on distinguished the gender differences. And as can be observed, inspite of the large variations in the low frequencies zone (20Hz – 100kHz), at higher frequencies, characteristic curves was found. At this frequency range of 300kHz to 10MHz, for either male or female, the ratio between impedance value of the eggs shows a noticeable increase. To visualize this changes, mean value of male eggs and female eggs were calculated, and were scattered with days data at different frequency. From observing the spectrum, it is noticeable that for male eggs, the ratio data in the frequency range of 300kHz up to 10MHz has a value of larger than 1, while for female eggs, this ratio value is almost alway falls below 1.

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This different was confirmed by independent T-test. Among the frequency range of 376kHz to 2.69MHz , the ratio measurement of male and female eggs shown significant different with p value of independant T-test lowest at 0.006 at the frequency of 376kHz.

Burke and Sharp studies have proved a phenomenon, by cracking the eggshell and weigh the wet and dry embryos weight, the group stated in their studies that a different in embryos weight of male and female eggs at very early of incubation was found but the different only become significant from day 11 of incubation process. The results in our study are believed to have proved the same situation. By measuring the impedance /capacitance and compare the self-normalization ratio values, different between male and female eggs values are shown as early as day one of the incubation process, figure 4.13 and 4.15, and the differences was significant starts from day 9 for impedance ratio, and day 8 for capacitance ratio. With the bioimpedance measurement analysis knowingly used for human body mass composition, it is believed that the finding is reasonable.

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CHAPTER 5. CONCLUSION

5.1. General conclusion

This pioneer study aims to develop a non-invasive method for the detection of chicken egg gender. Kinmen county, Taiwan chicken strain eggs was used for the experiment. Evolving an electrode fabrication process, chicken egg impedance measurement process, and bioimpedance measurement device specific for chicken egg gender identification purpose.

The designed and fabricated electrode was accurate with the relative standard deviation at high frequencies only up to 5%. The resistance of the electrode was measure at 2.383 Ohm. Electrodes placement suggested by the study was sufficient, allowing impedance and capacitance measurement to find the relationship between collected data and chicken egg gender.

The incubator modification and incubation process applied in the study was found to be successful. 17 eggs were obtained from the Department of Animal Science, National Chung Hsing University, Taiwan, 16 alive and after PCR gender identification, 8 were identified to be males with the others 8 are females.

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For the impedance and capacitance measurement of fertilized chicken egg. The pure data include lots of noise, these are suspected to be due to the continuous development and constant changing of the chicken embryo But by applying self-normalization ratio, the embryos-to-embryos noise was reduced, making it easier to focus on the gender differences of each eggs. The self-normalization ratio between impedance data of pair 2-3 and pair 2-4 at frequency range from 376kHz up to 2.69MHz was found to be sufficient for chicken gender identification. This impedance ratio was able to recognize the gender differences from day 9 of the incubation process. For capacitance, the self-normalization ratio between capacitance value of pair 1-2 and pair 3-4 measured at 35kHz was found to be able to identified the difference between male and female egg as early as day 8 of the incubation process. Both the impedance and capacitance ratio value of day 9 at the optimum frequency can be used as a two step verification to point out the gender of the chicken egg.

The finding of this study has agreement with Burke and Sharp study on chicken embryos. Burke and Sharp stated that the differences found out that the weight and yolk size different between the two gender has appeared on the egg very early during the incubation process. But the difference only become significant at at least day 11. For our study these differences was found at day 8 using capacitance ratio, and day 9 using impedance ratio.

5.2. Limitations

The electrode fabrication and attachment process is very long, for at least 24 hours included the curing time of carbon epoxy. Meaning that even though the electrode shows acceptable performance, it should be optimized for future studies.

The study included a small pool of subject and the standard deviation value is large, this could be improve by using a much larger pool of subject.

This study failed to developed a measurement device specifically for chicken egg gender identification. As the impedance ratio optimum frequency was found to be 376kHz, the current source frequency response cannot keep a constant current to inject into the egg.

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5.3. Future works

5.3.1. Electrode optimization

A study on making a more efficient electrode should be made, reducing the curing time, or even better material choosing. Another study should be made on electrode placement on fertilized egg, proving the similarity or different between different electrode placement. Finding the optimum electrode placement for chicken egg gender identification.

Electrode optimization can also lead to lower noise upon the impedance and capacitance measurement. With lower noises, it is believe that pure impedance or capacitance data con exposed even more on chicken egg information, whether it is viability, quality, or gender information.

5.3.2. Measurement system

This study tries to approach impedance measurement using current injection. This approach was mainly for the safety of biological tissue on the egg. But it was prove to be non-

efficient as the optimum frequency was 376kHz and the current source frequency response cannot fulfill this criteria.

In future study, a voltage source which is usually have much better frequency response should be considered. A study by Qureshi [27] and his team has simulated a voltage source with a gain band width of 18MHz. This would be an optimization for the measurement system development in the future.

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Appendix

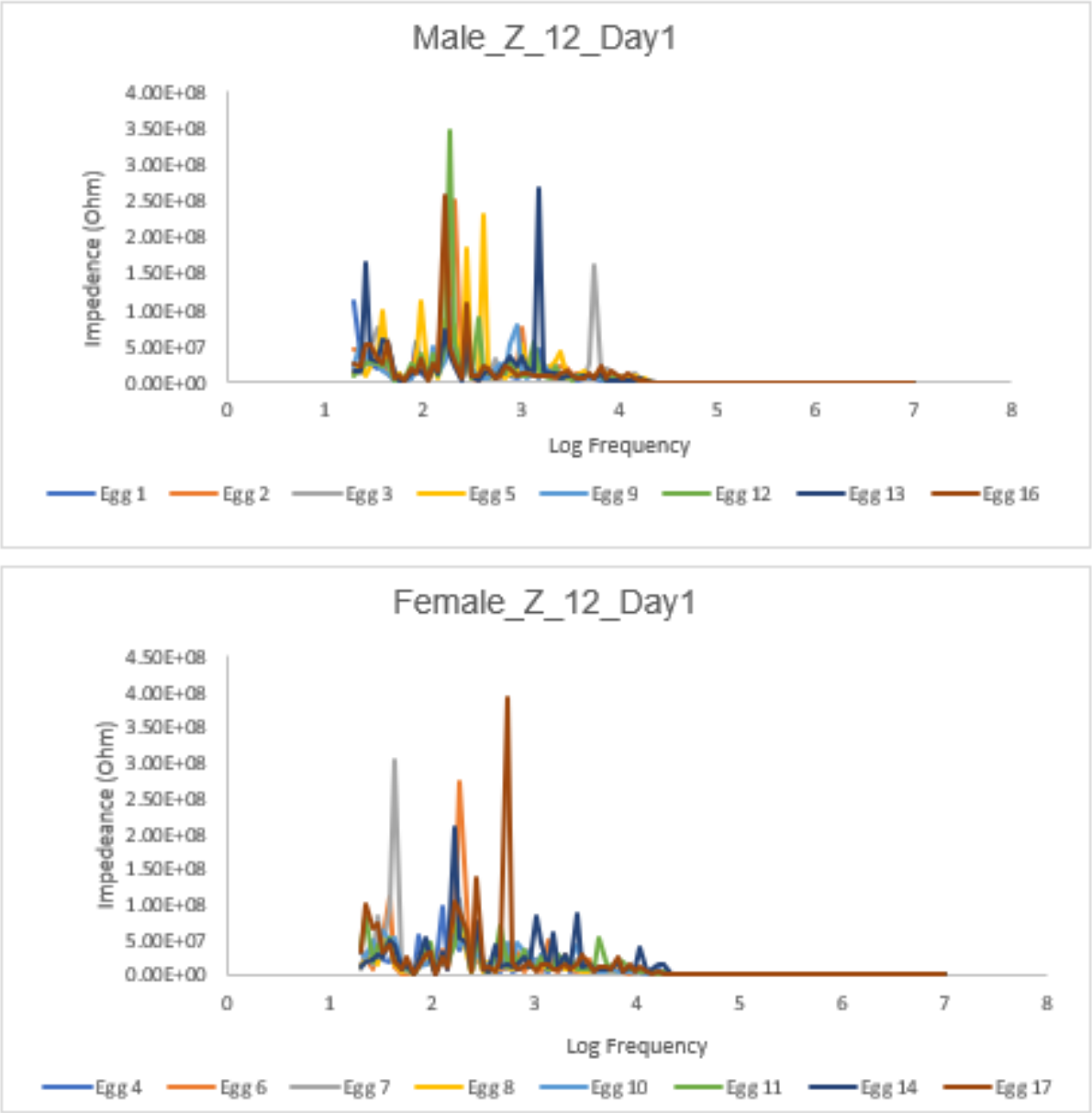
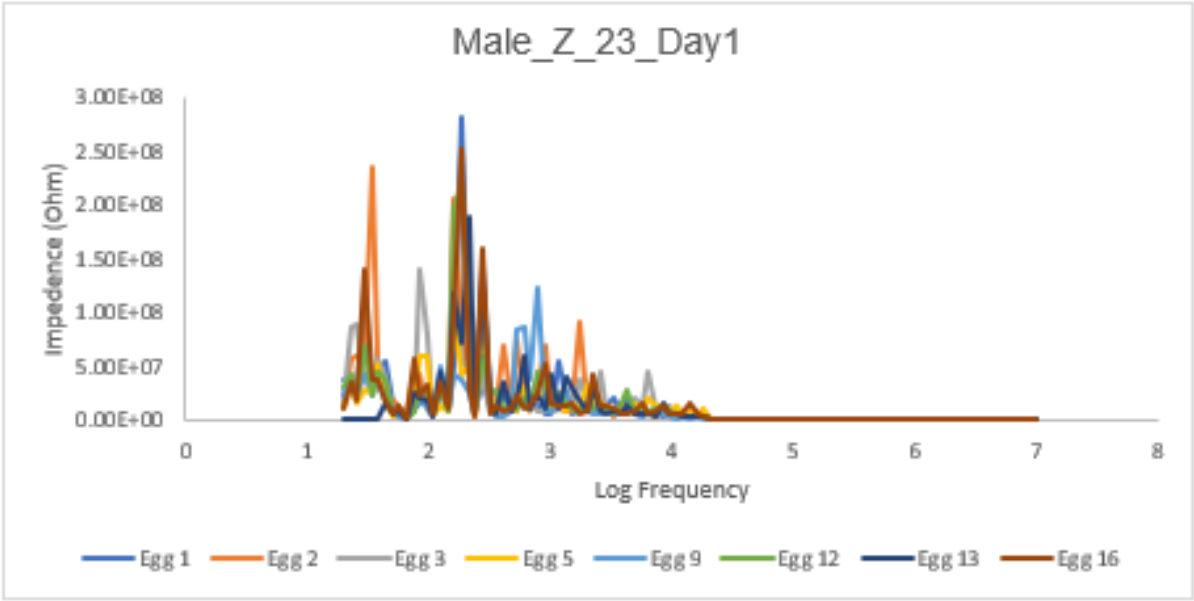


Figure A.1. Impedance measurement of electrode pair 1 and 2 of the eggs performed on day 1



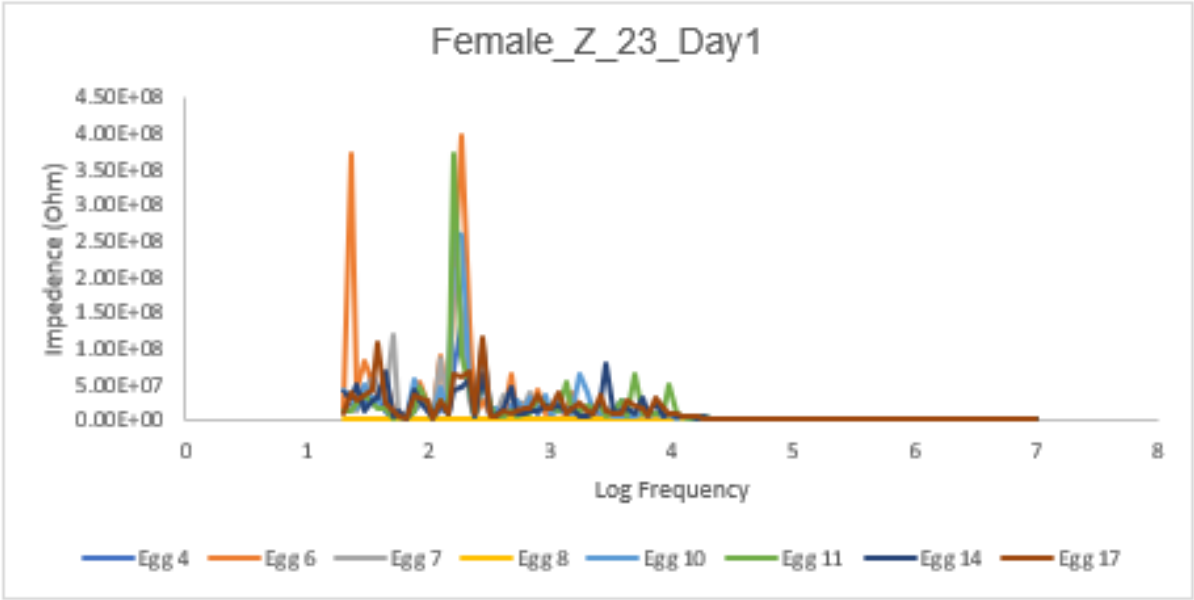
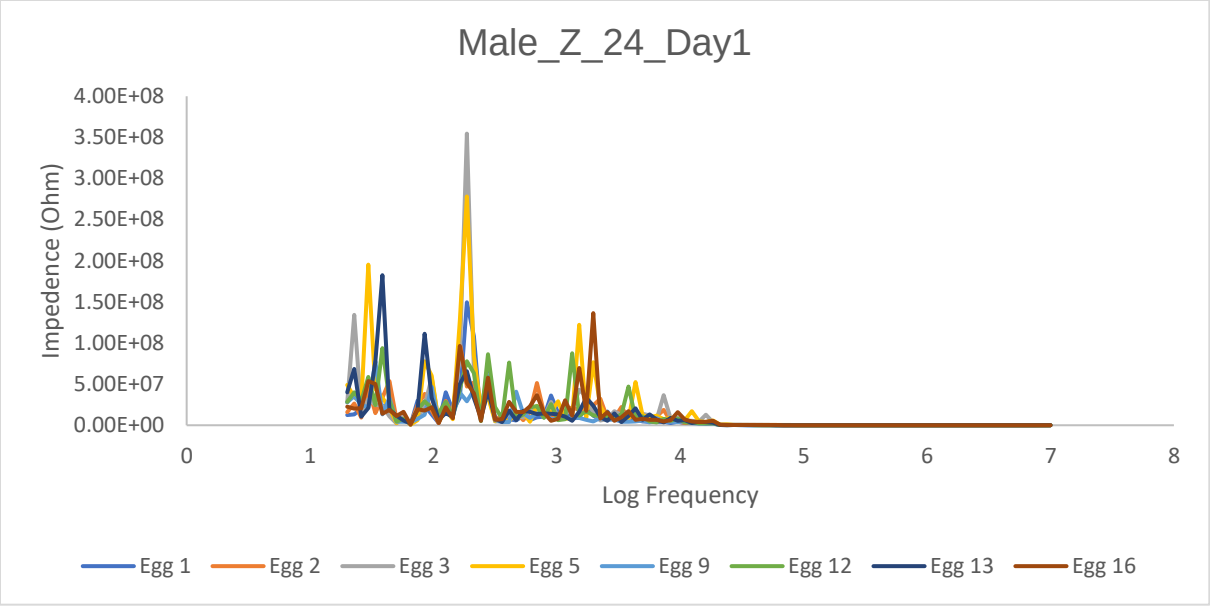


Figure A.2. Impedance measurement of electrode pair 2 and 3 of the eggs performed on day 1



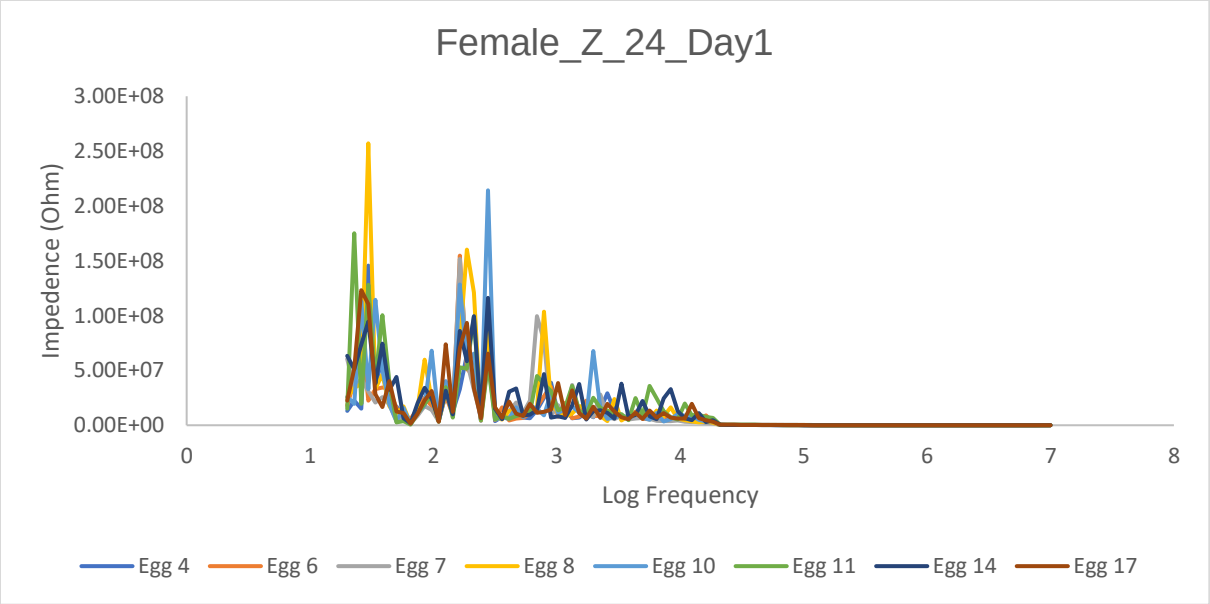
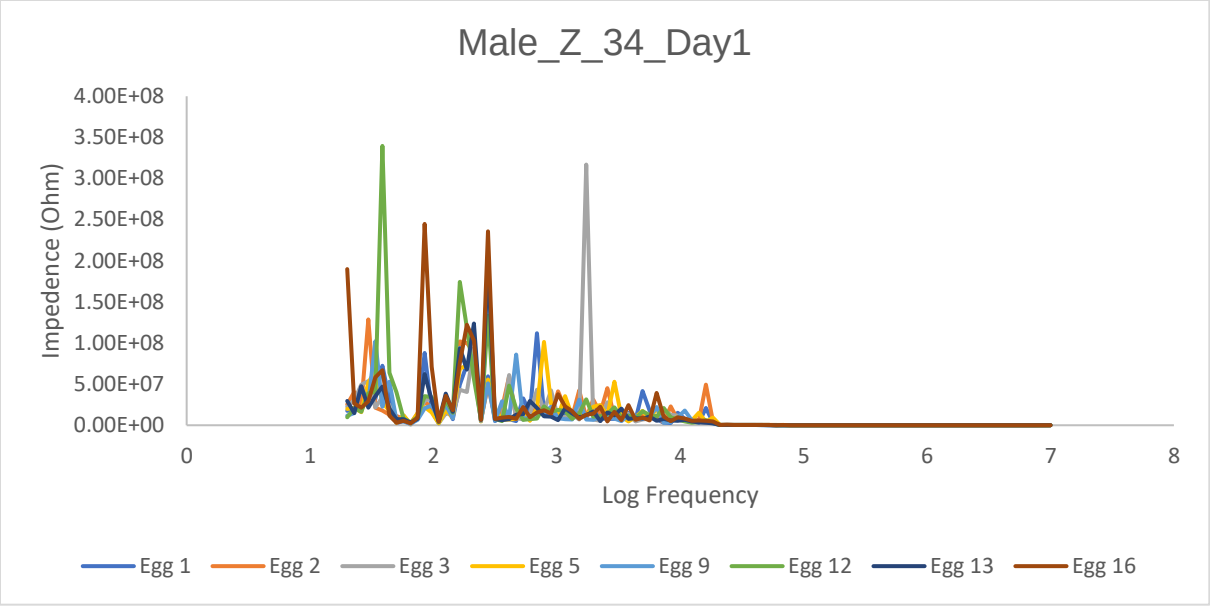


Figure A.3. Impedance measurement of electrode pair 2 and 4 of the eggs performed on day 1



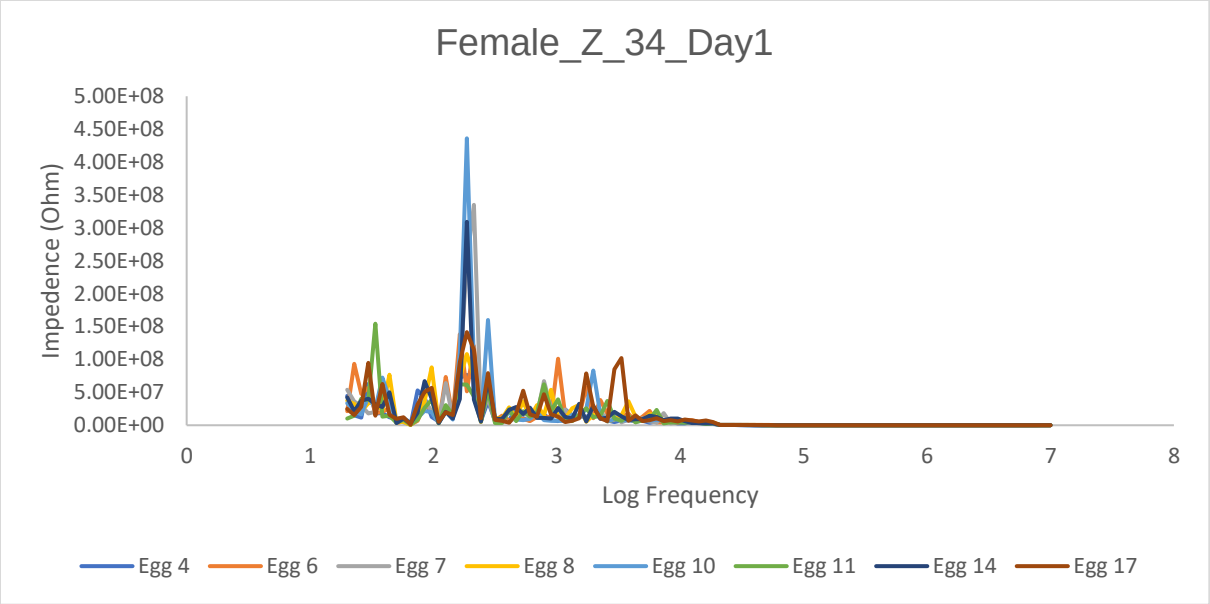


Figure A.4. Impedance measurement of electrode pair 3 and 4 of the eggs performed on day 1

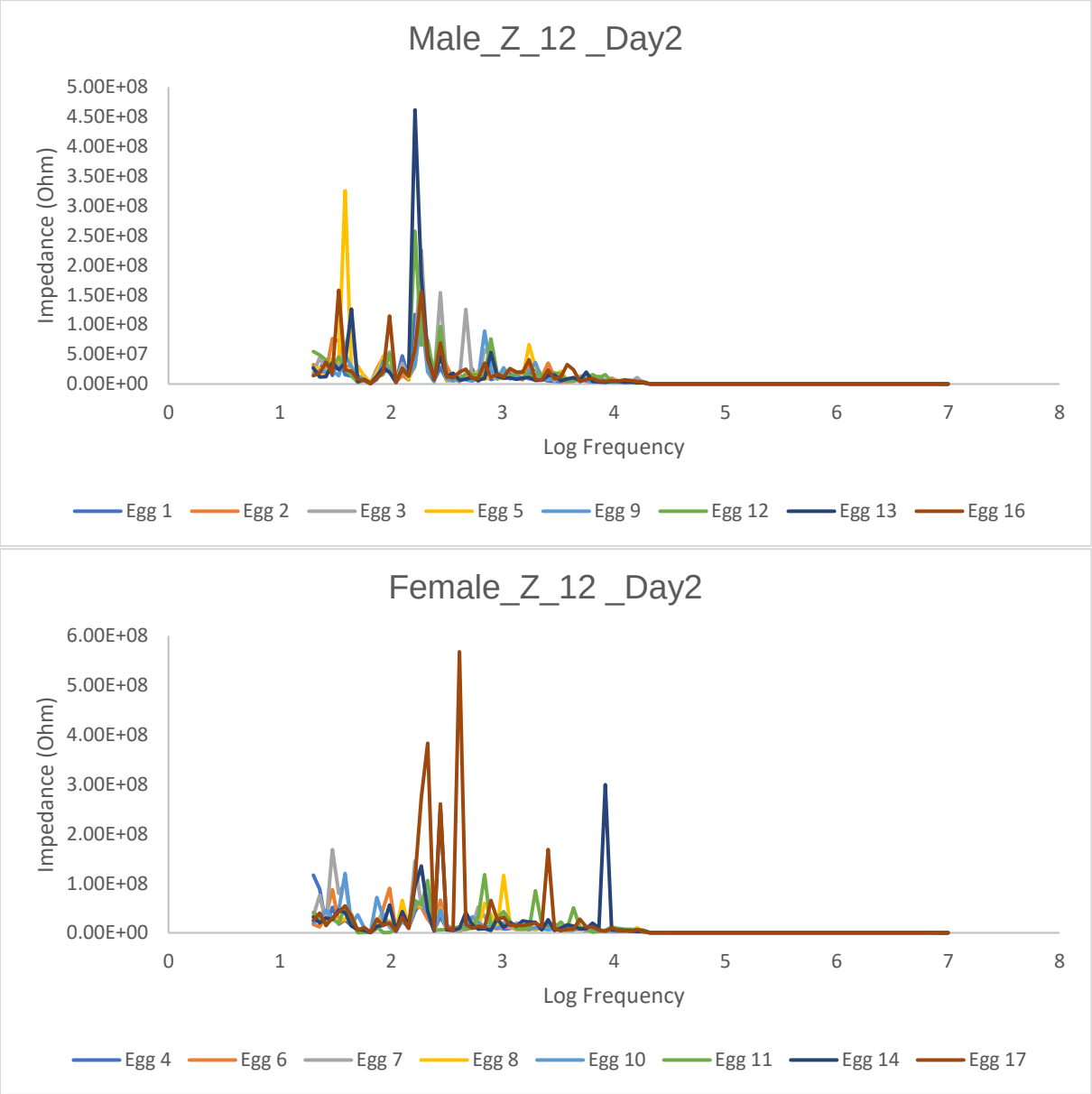
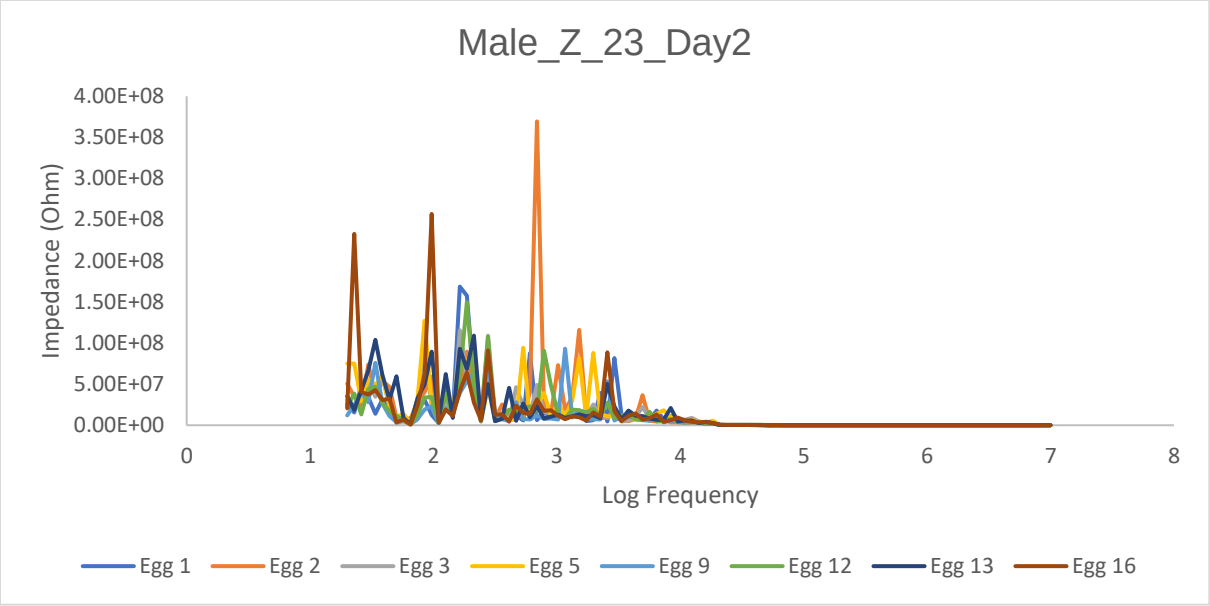


Figure A.5. Impedance measurement of electrode pair 1 and 2 of the eggs performed on day 2



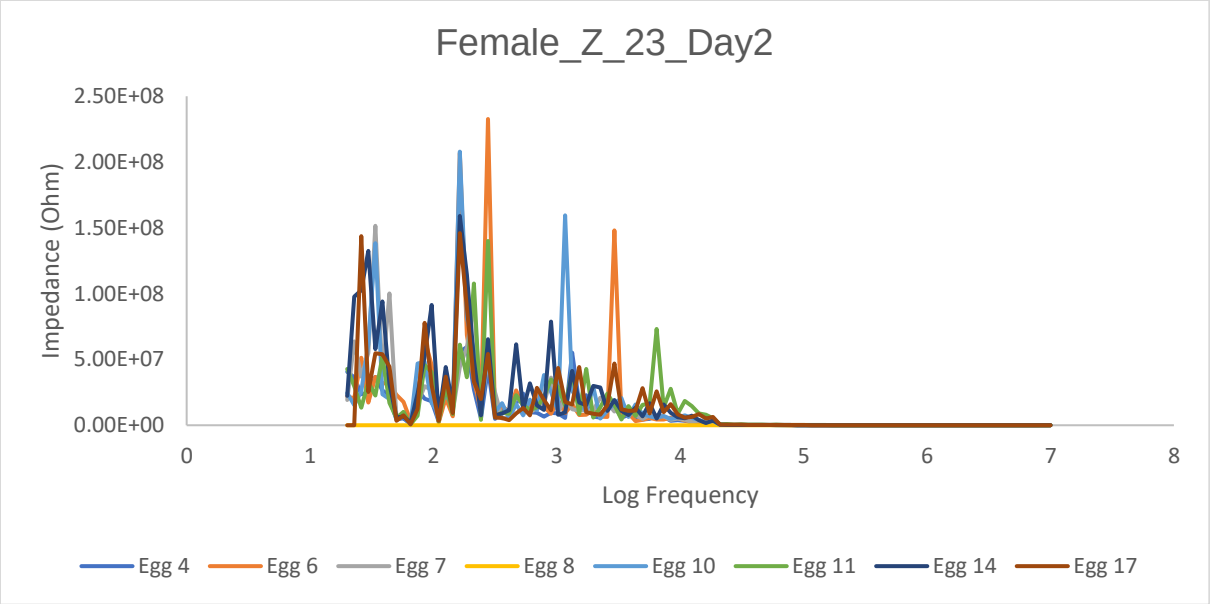
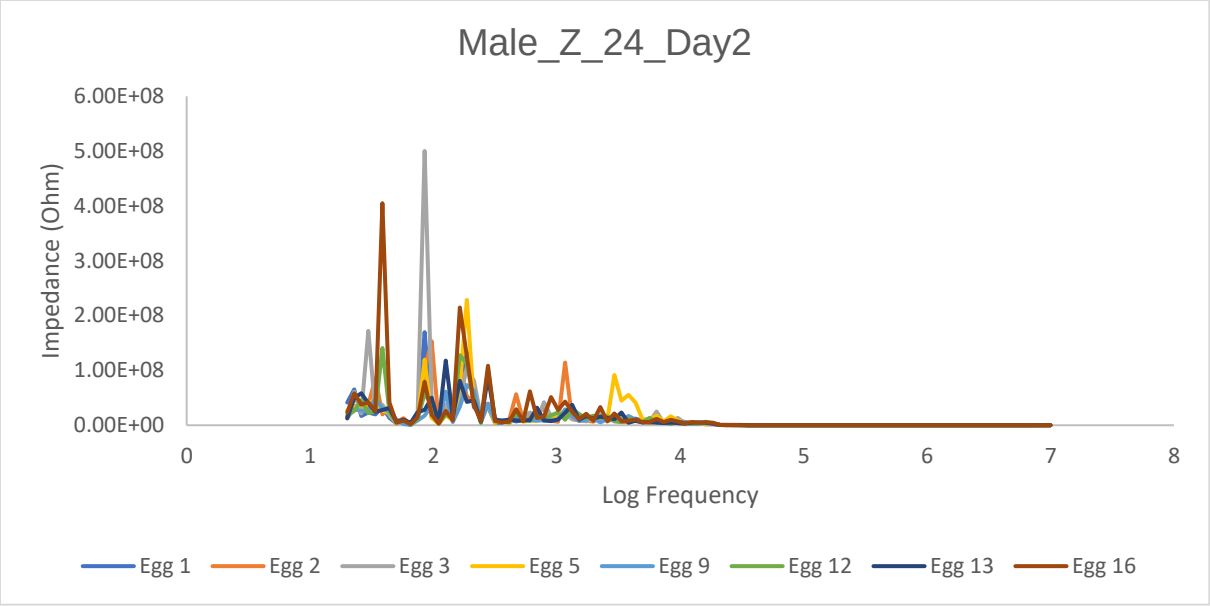


Figure A.6. Impedance measurement of electrode pair 2 and 3 of the eggs performed on day 2



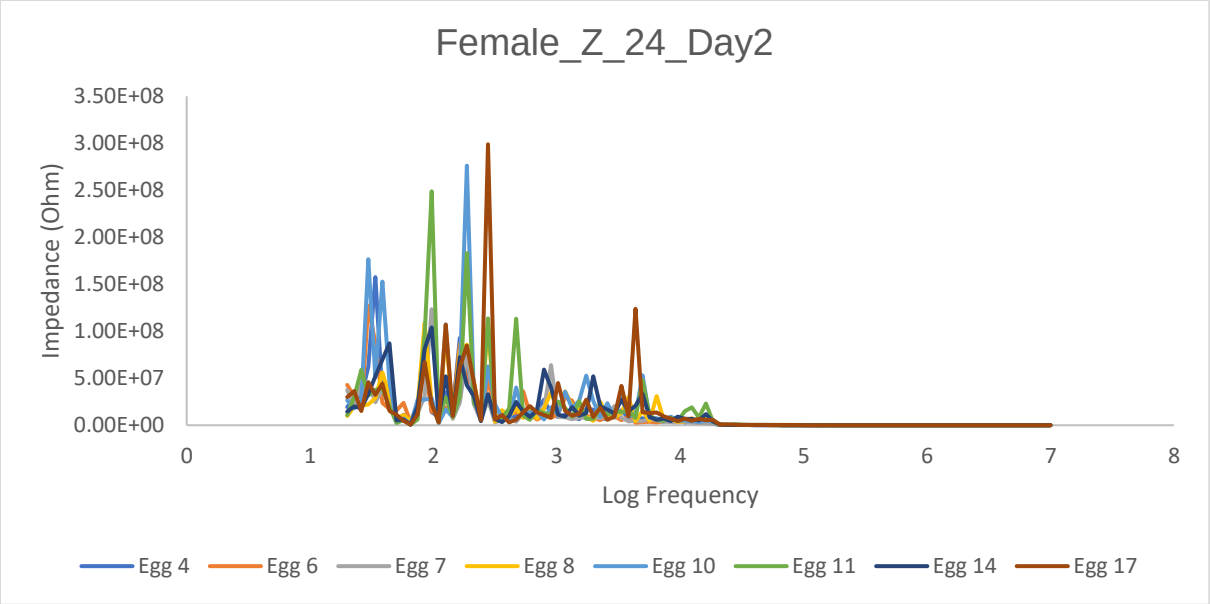
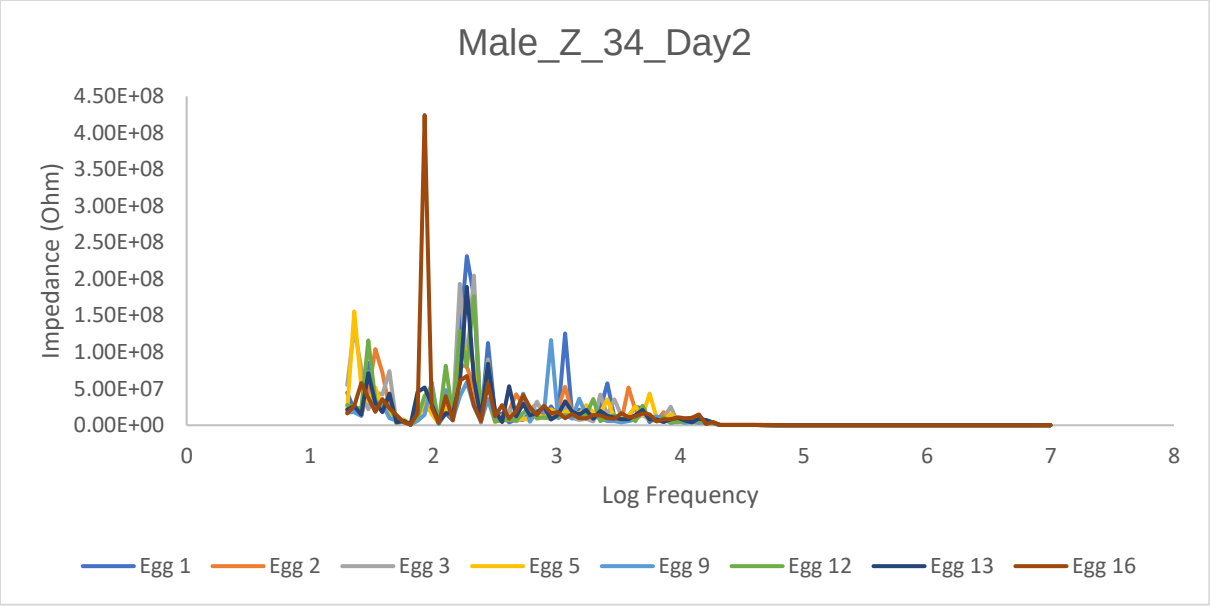


Figure A.7. Impedance measurement of electrode pair 2 and 4 of the eggs performed on day 2



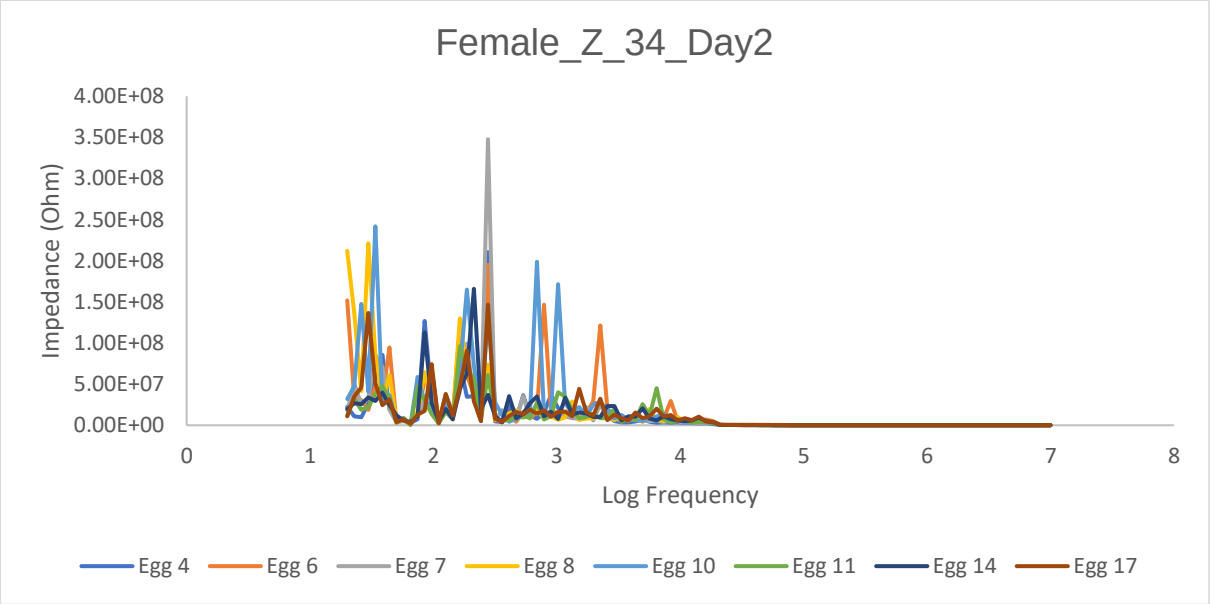
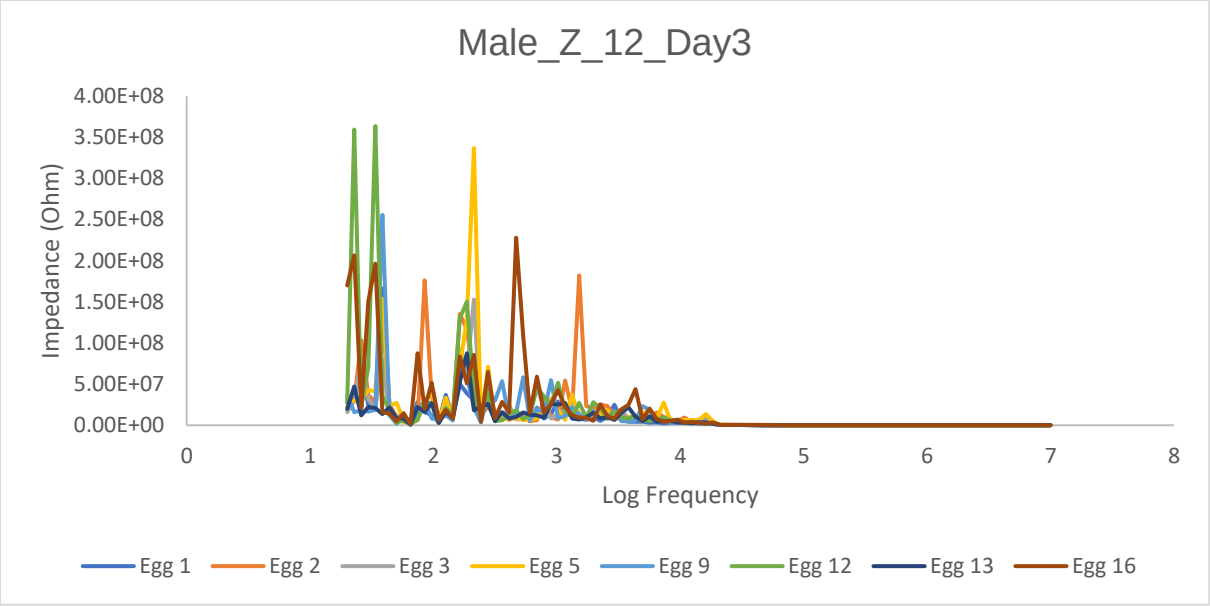


Figure A.8. Impedance measurement of electrode pair 2 and 3 of the eggs performed on day 2



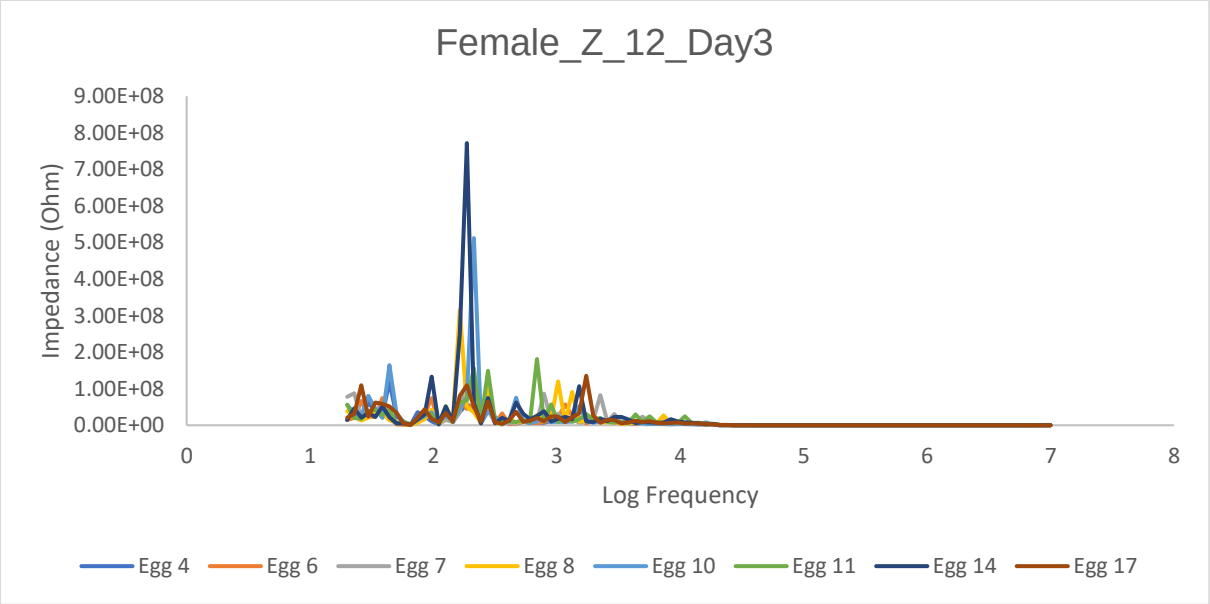
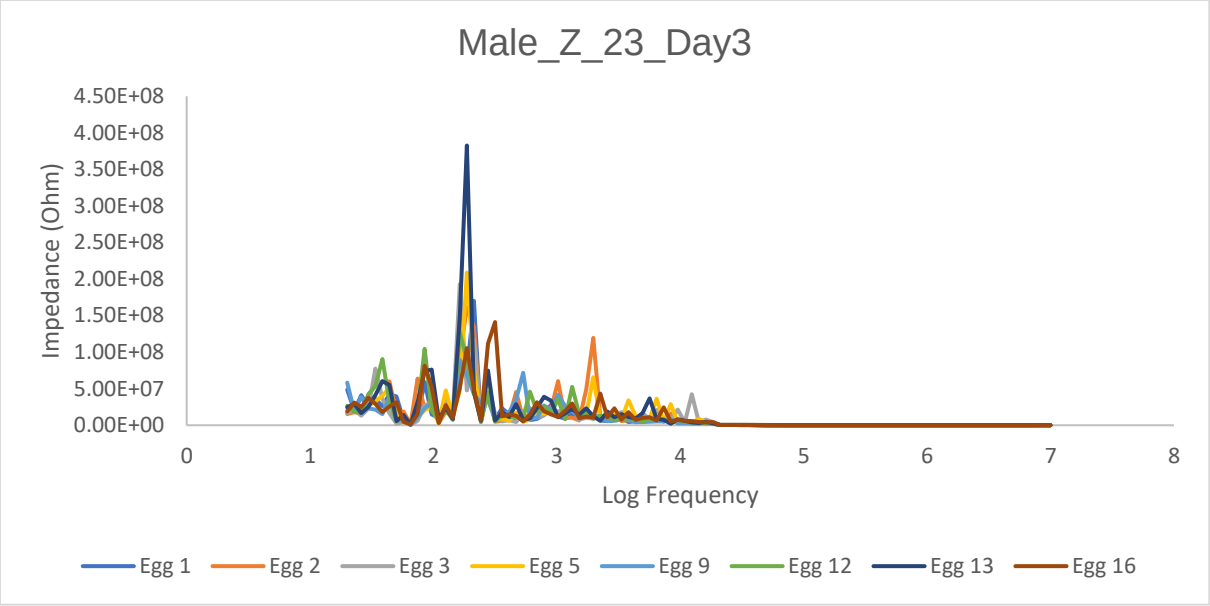


Figure A.9. Impedance measurement of electrode pair 1 and 2 of the eggs performed on day 3



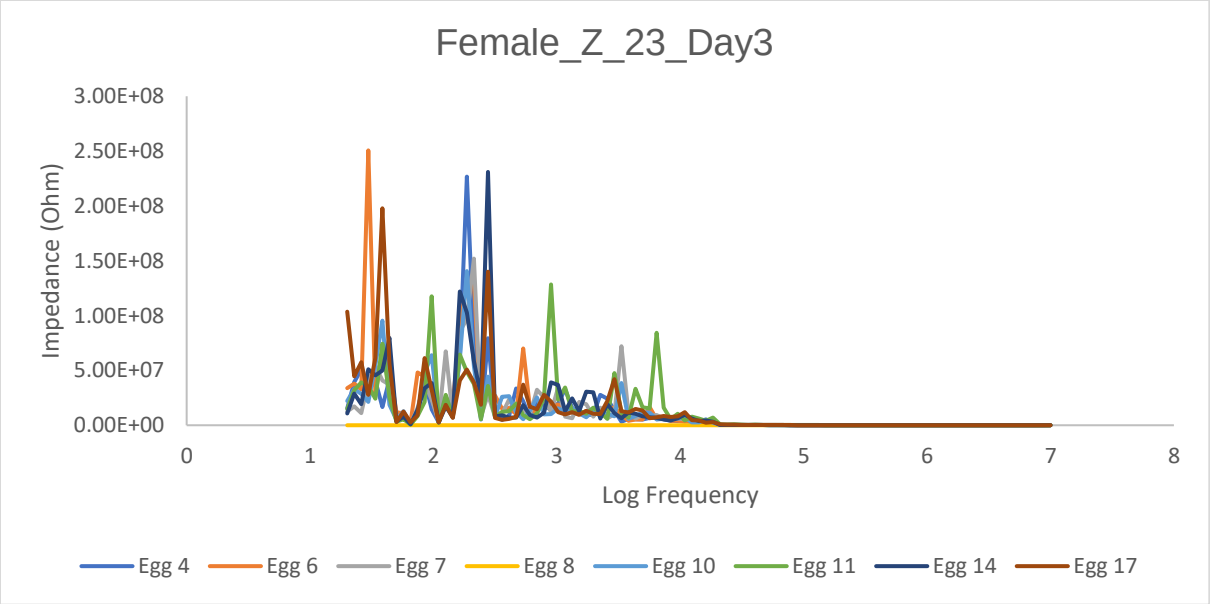
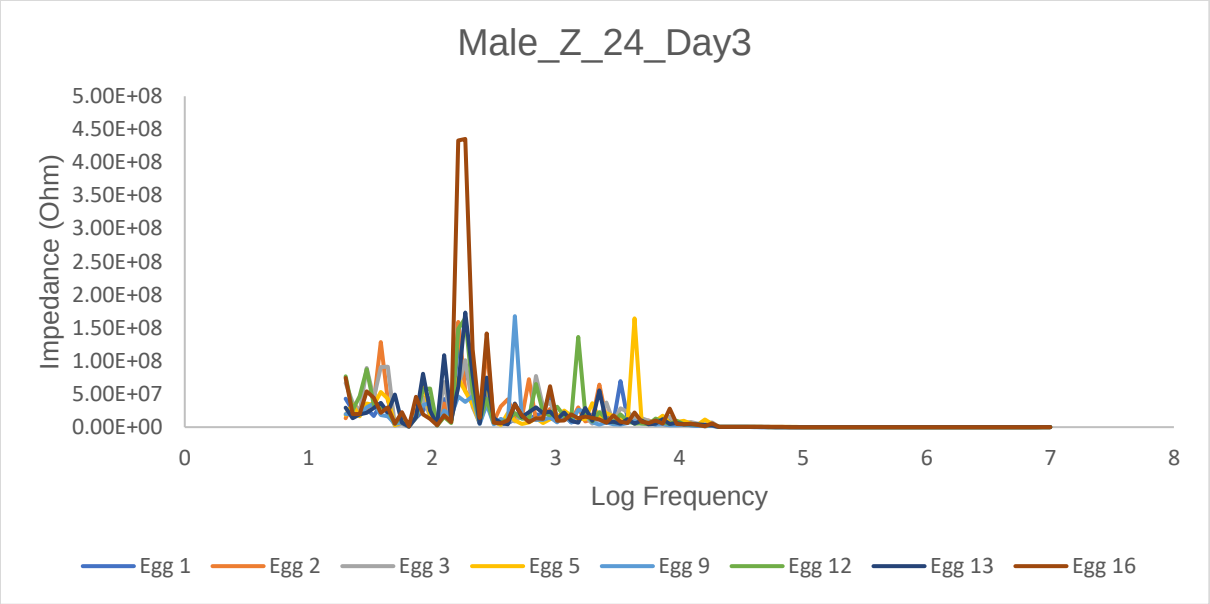


Figure A.10. Impedance measurement of electrode pair 2 and 3 of the eggs performed on day 3



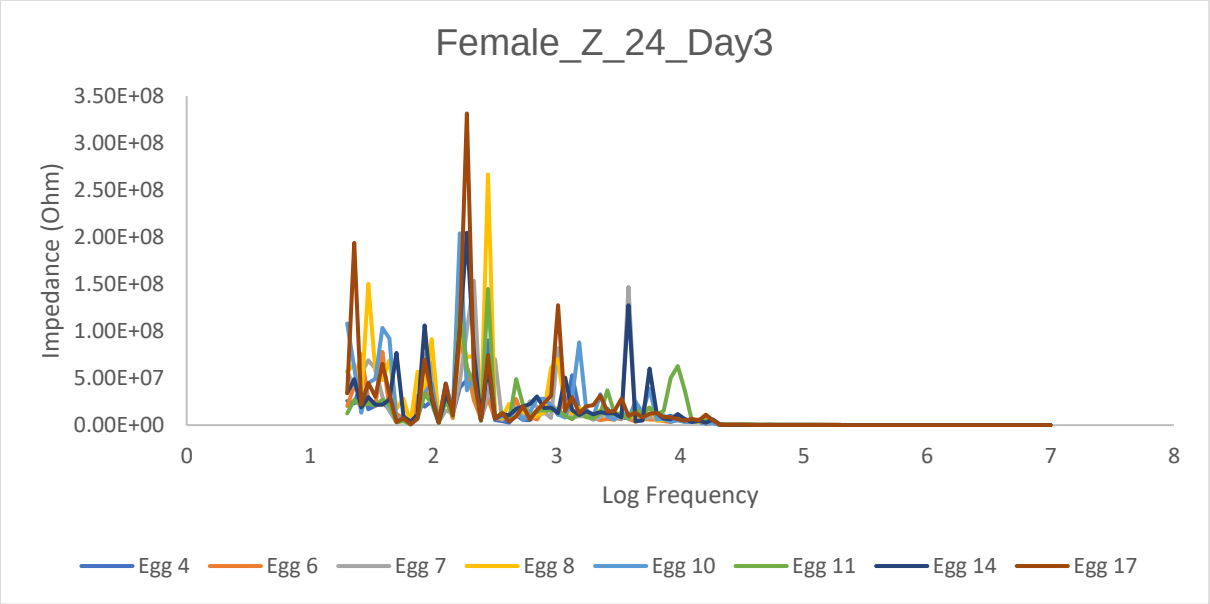
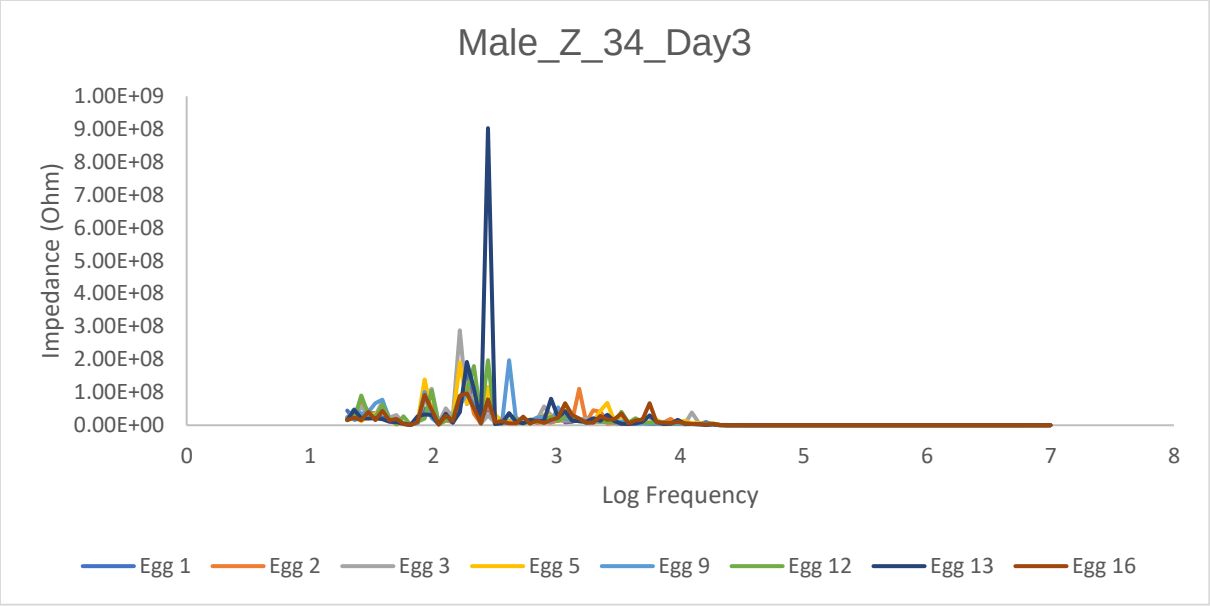


Figure A.11. Impedance measurement of electrode pair 2 and 4 of the eggs performed on day 3



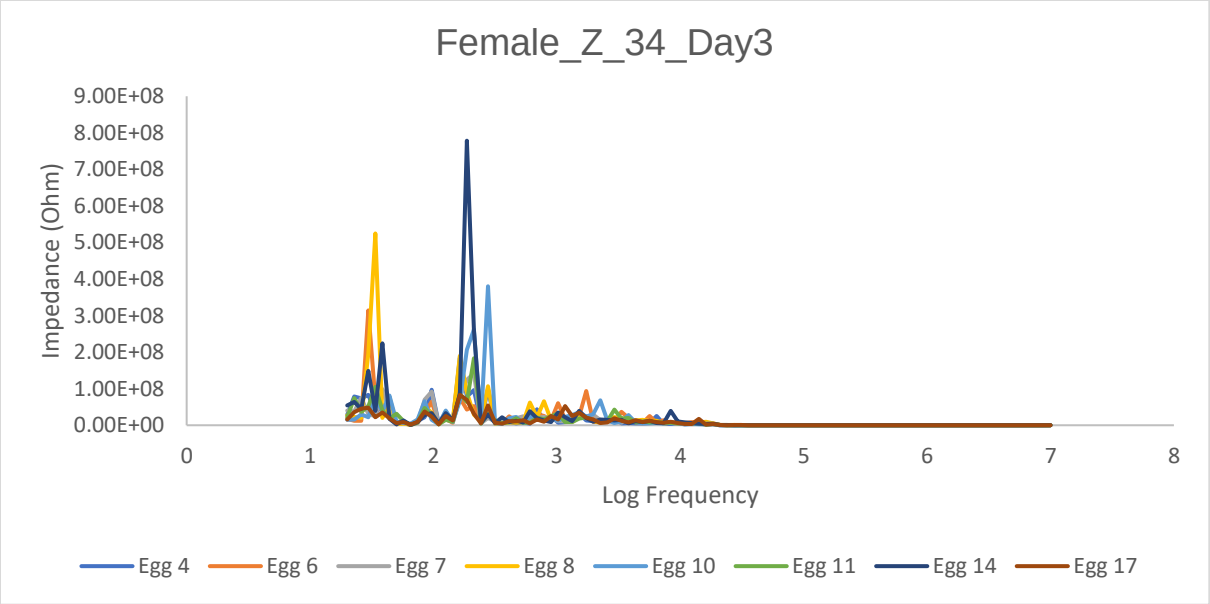
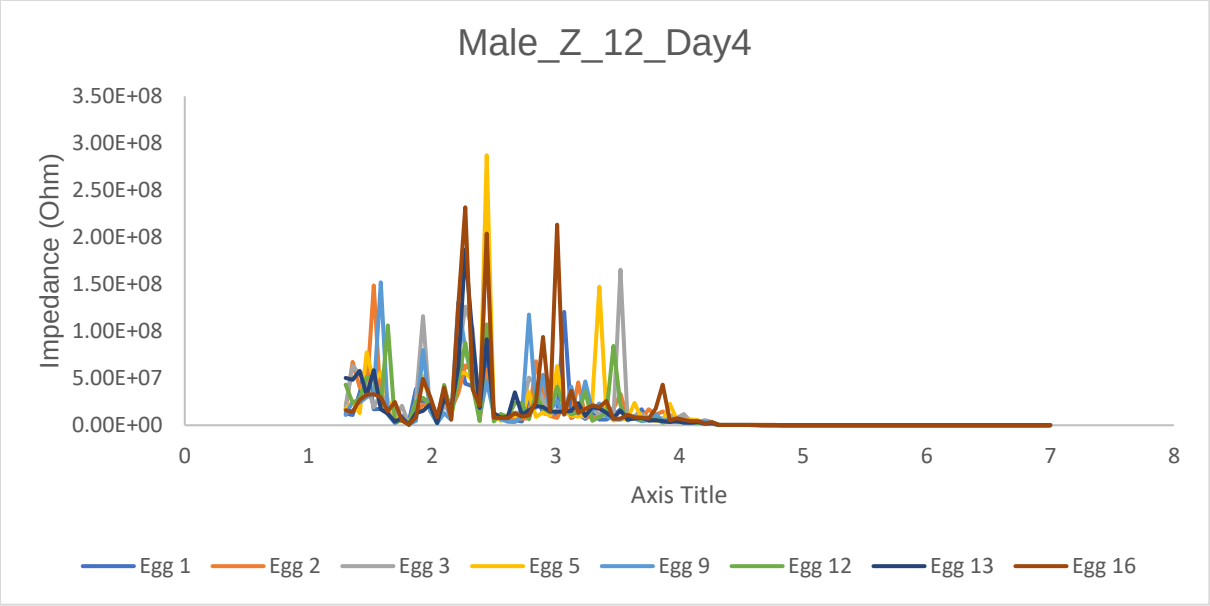


Figure A.12. Impedance measurement of electrode pair 3 and 4 of the eggs performed on day 3



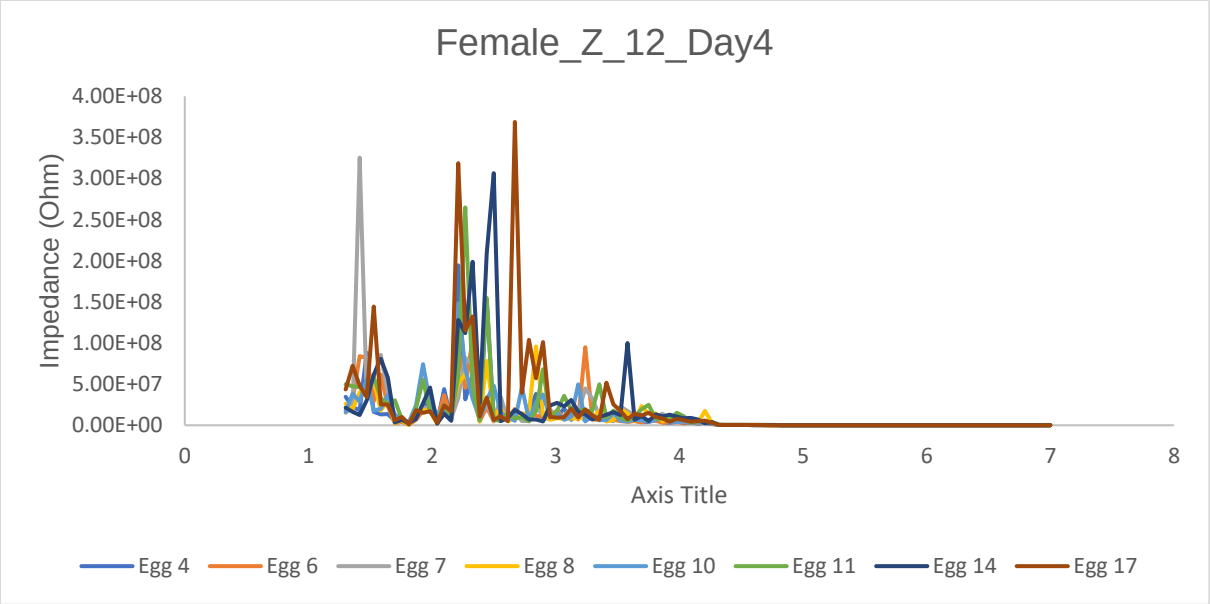


Figure A.13. Impedance measurement of electrode pair 1 and 2 of the eggs performed on day 4

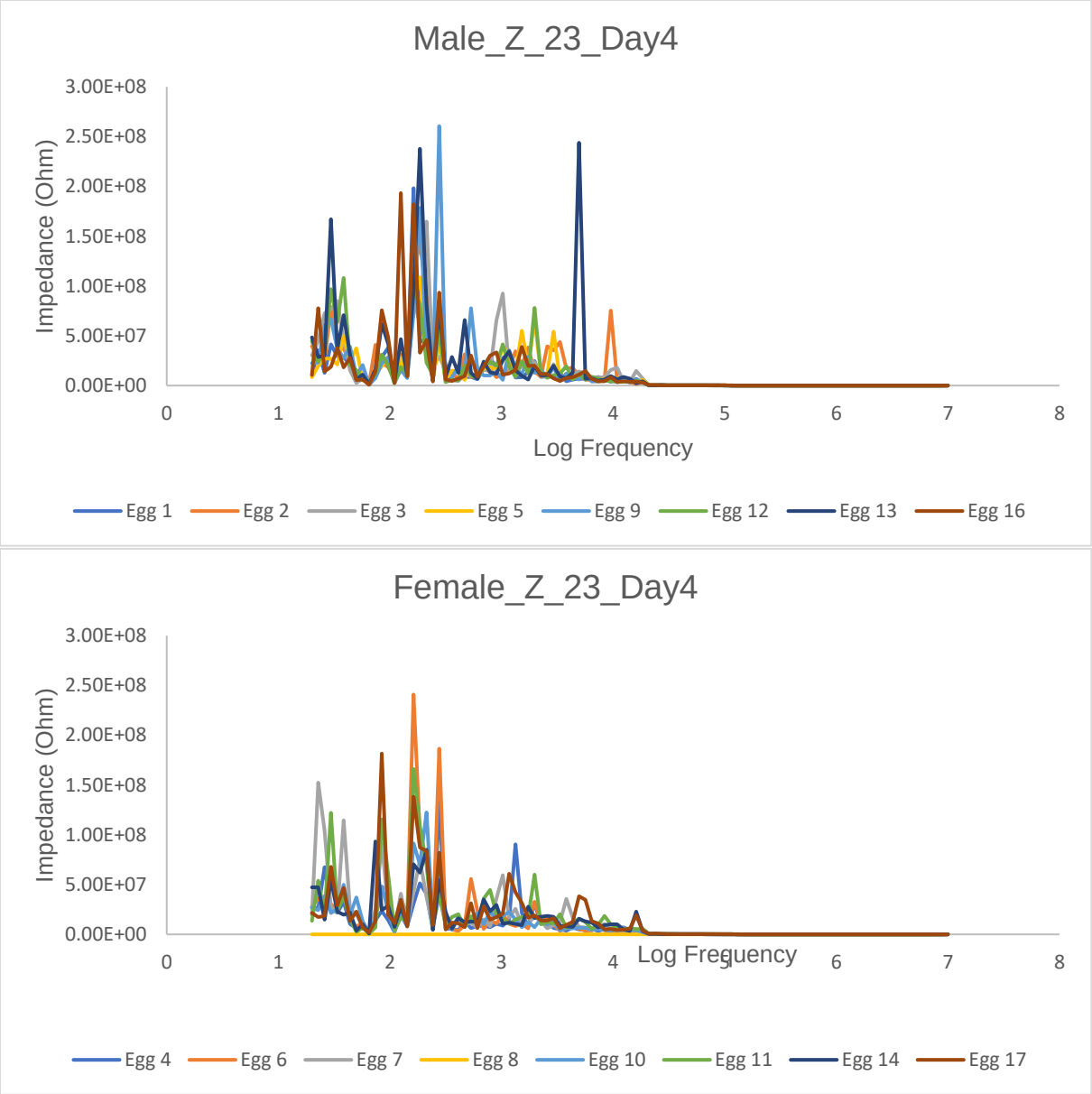
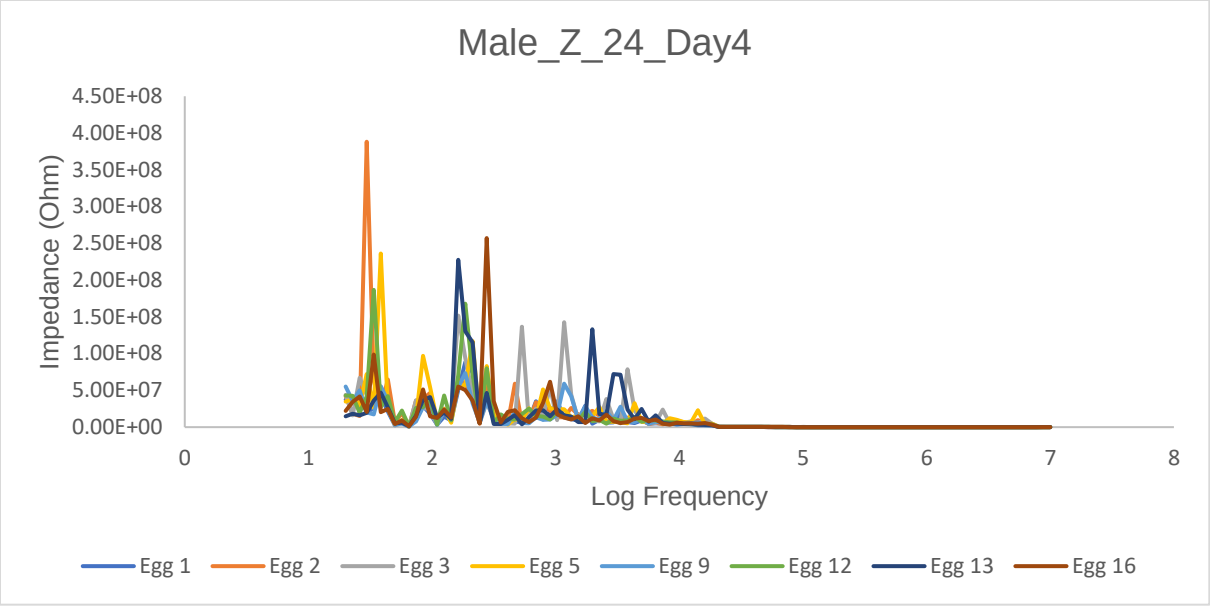


Figure A.14. Impedance measurement of electrode pair 2 and 3 of the eggs performed on day 4



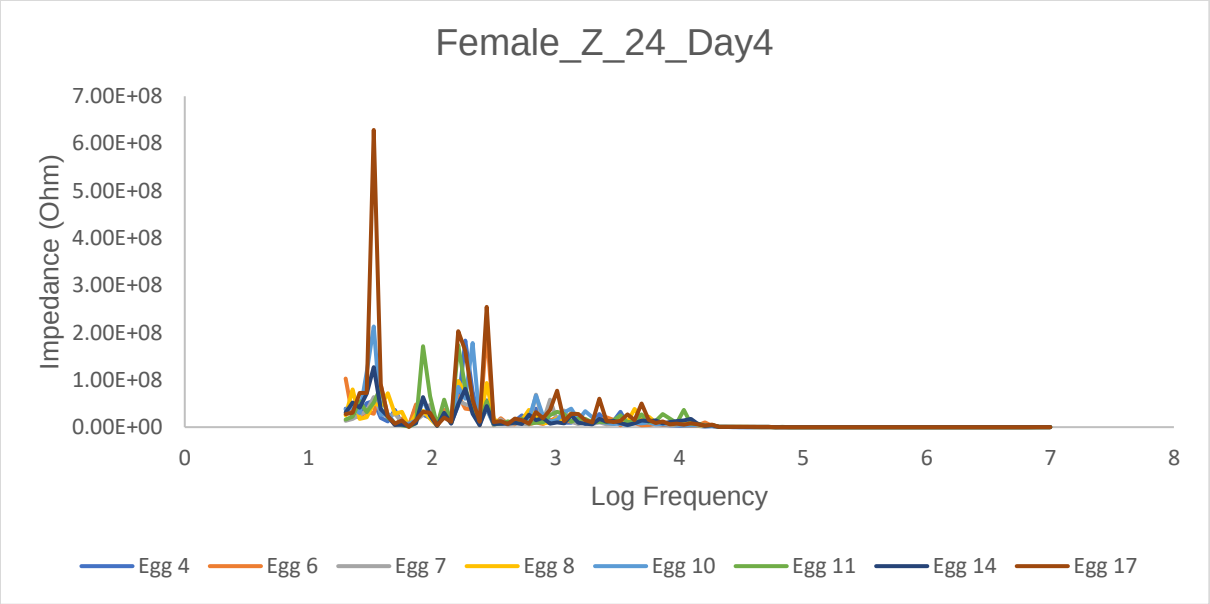
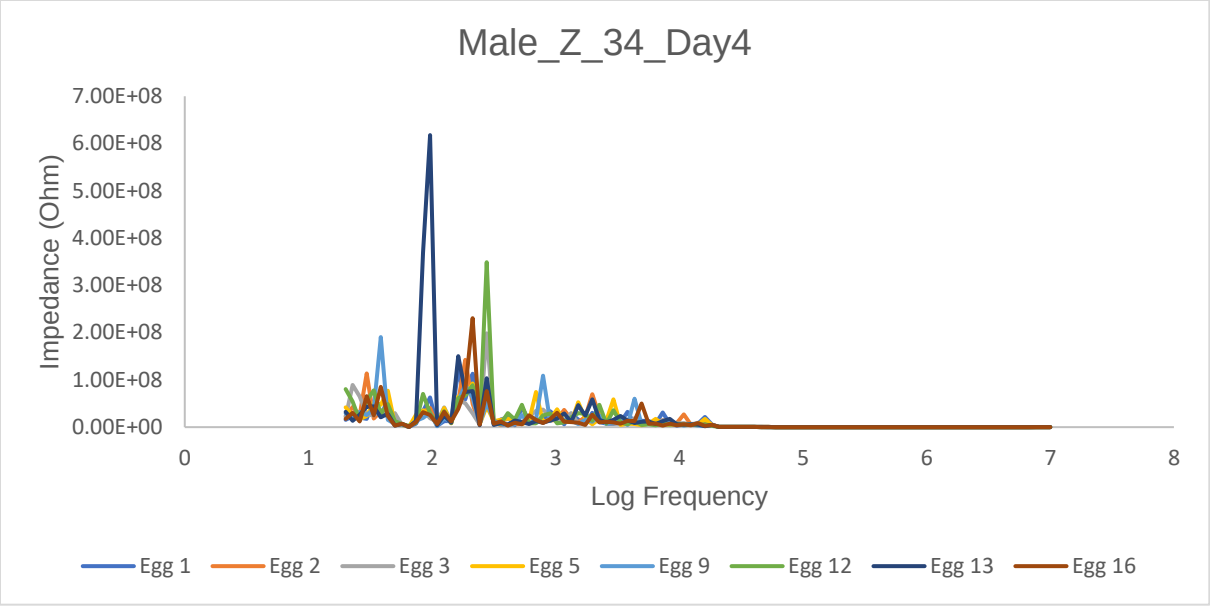


Figure A.15. Impedance measurement of electrode pair 2 and 4 of the eggs performed on day 4



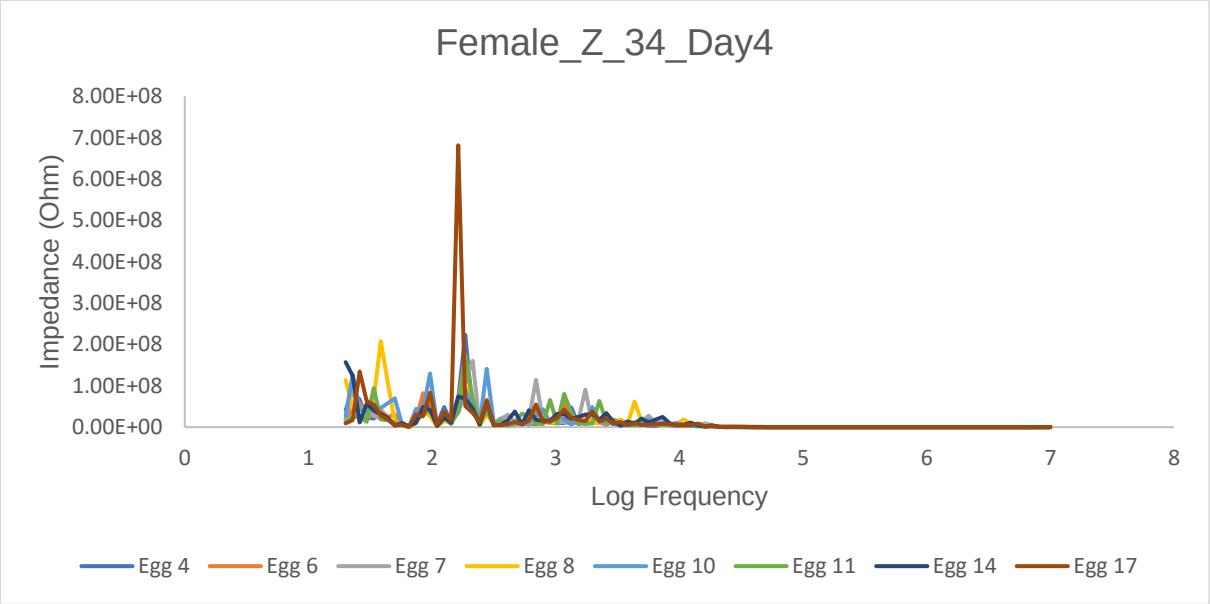
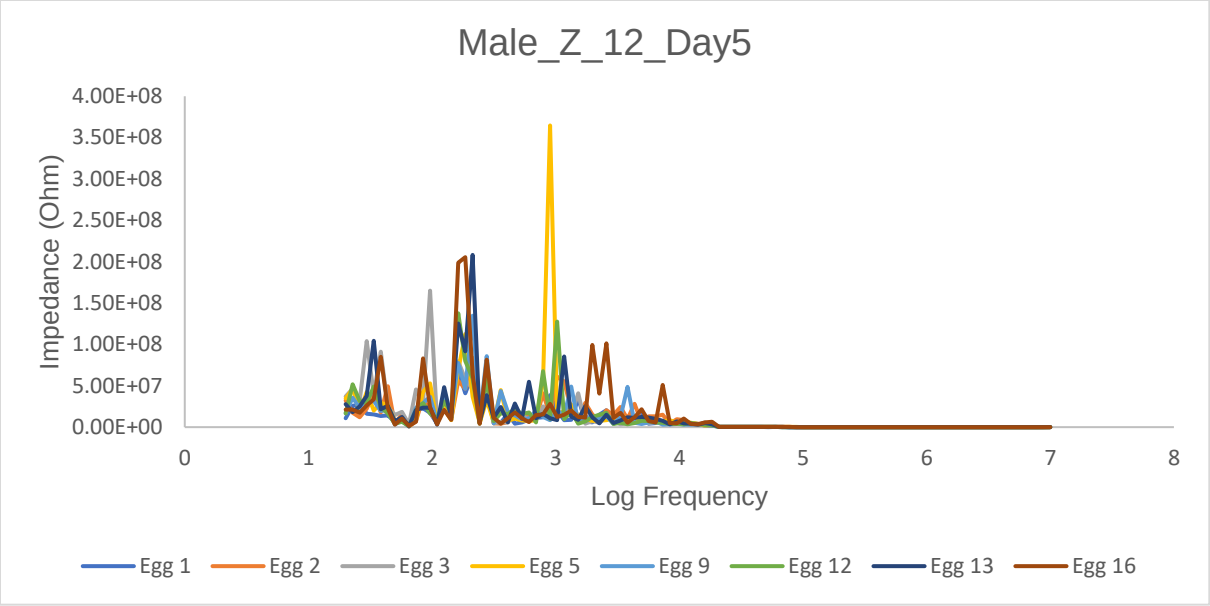


Figure A.16. Impedance measurement of electrode pair 3 and 4 of the eggs performed on day 4



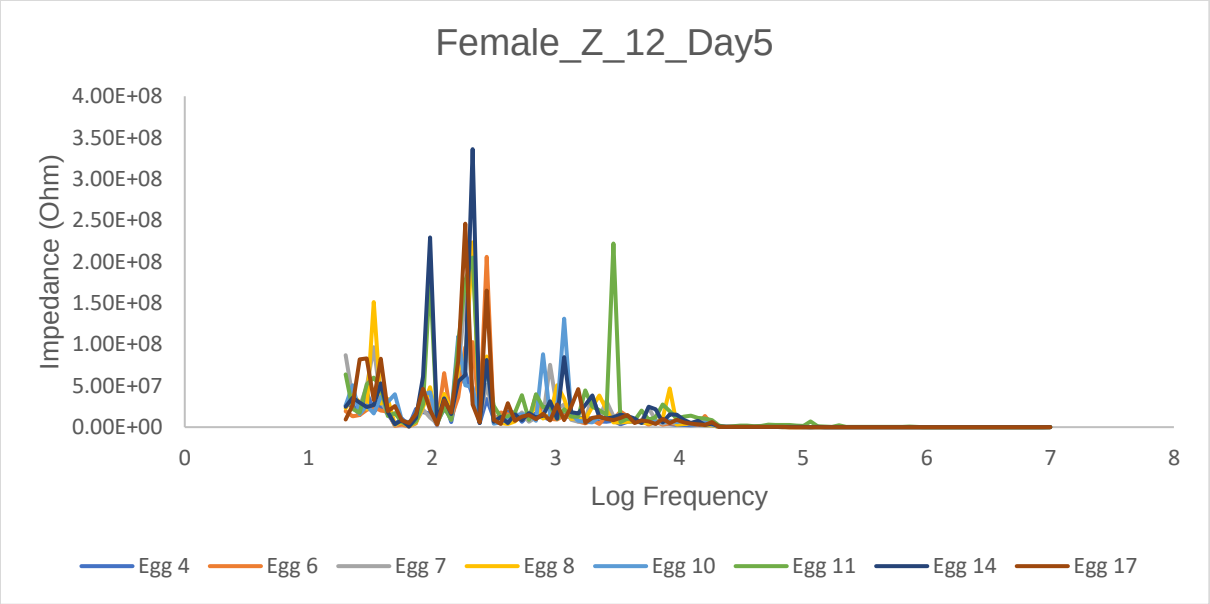


Figure A.17. Impedance measurement of electrode pair 1 and 2 of the eggs performed on day 5

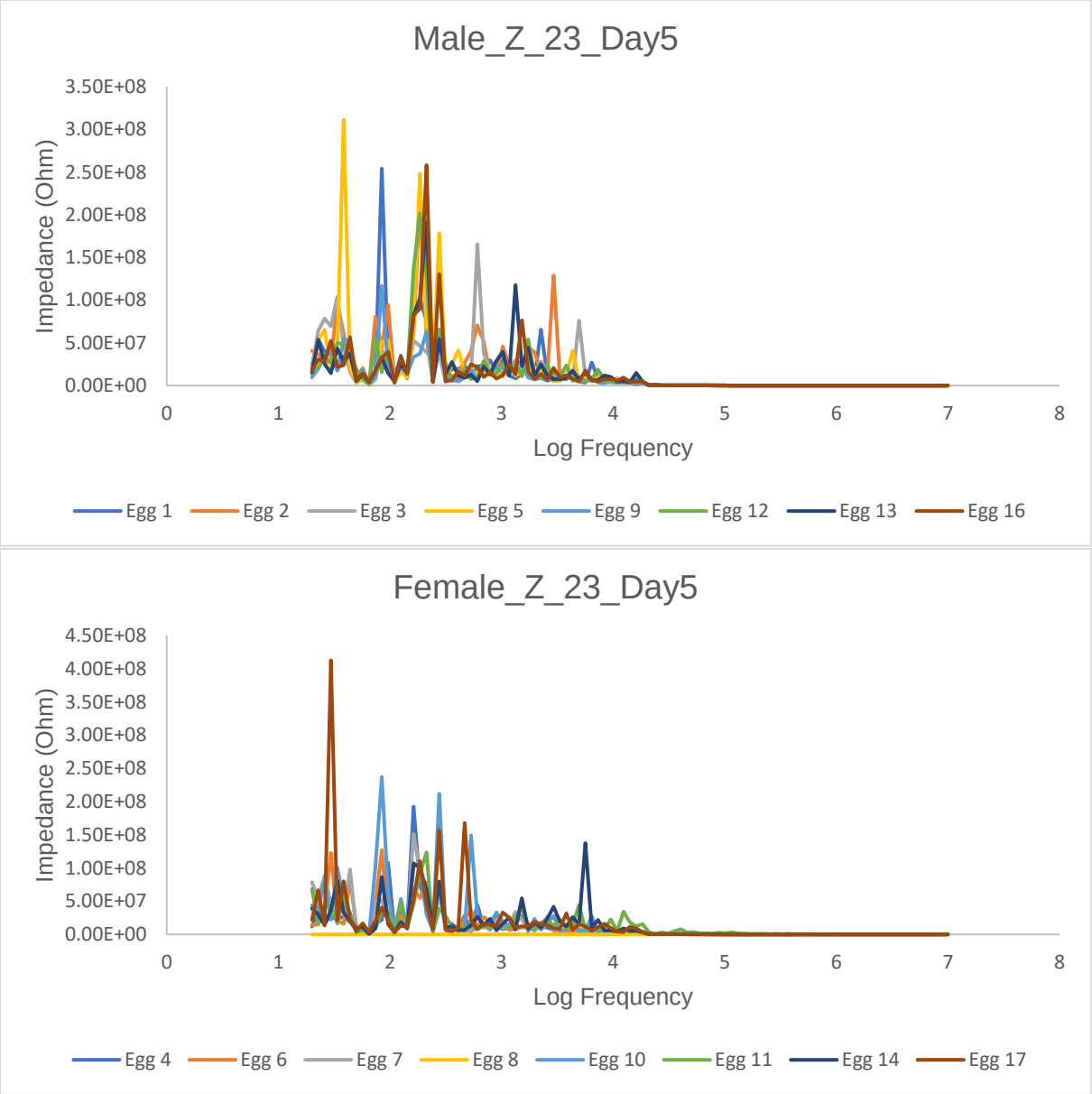
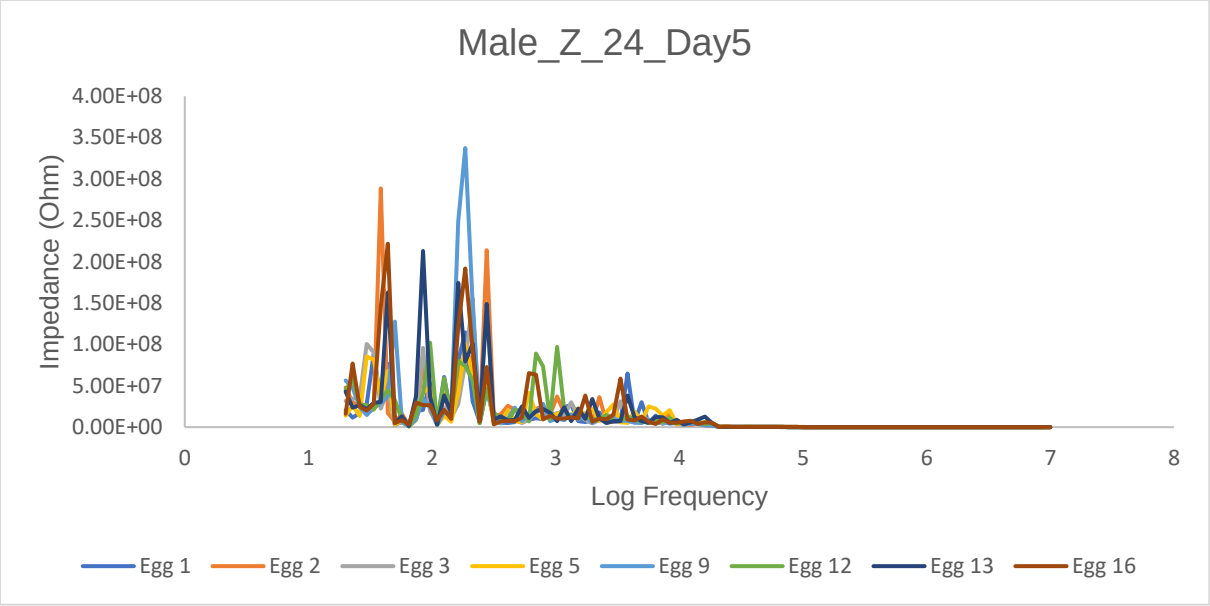


Figure A.18. Impedance measurement of electrode pair 2 and 3 of the eggs performed on day 5



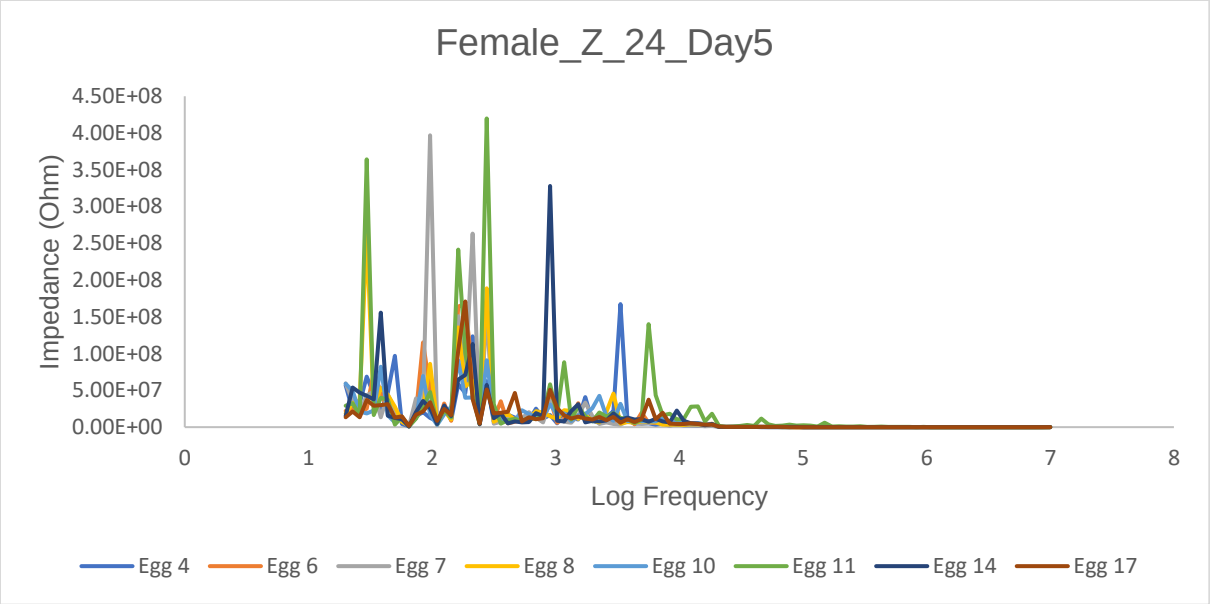


Figure A.19. Impedance measurement of electrode pair 2 and 4 of the eggs performed on day 5

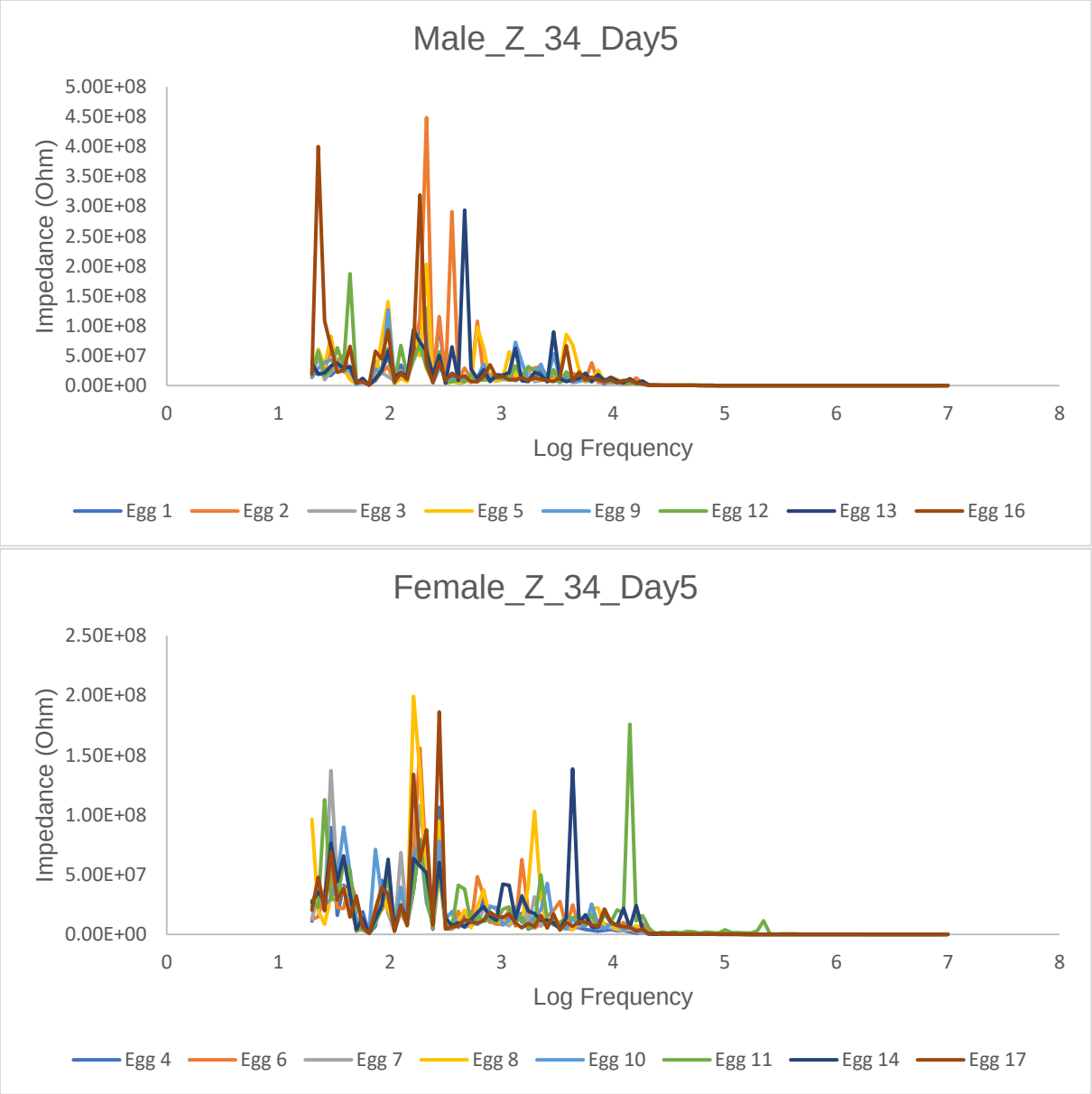


Figure A.20. Impedance measurement of electrode pair 3 and 4 of the eggs performed on day 5

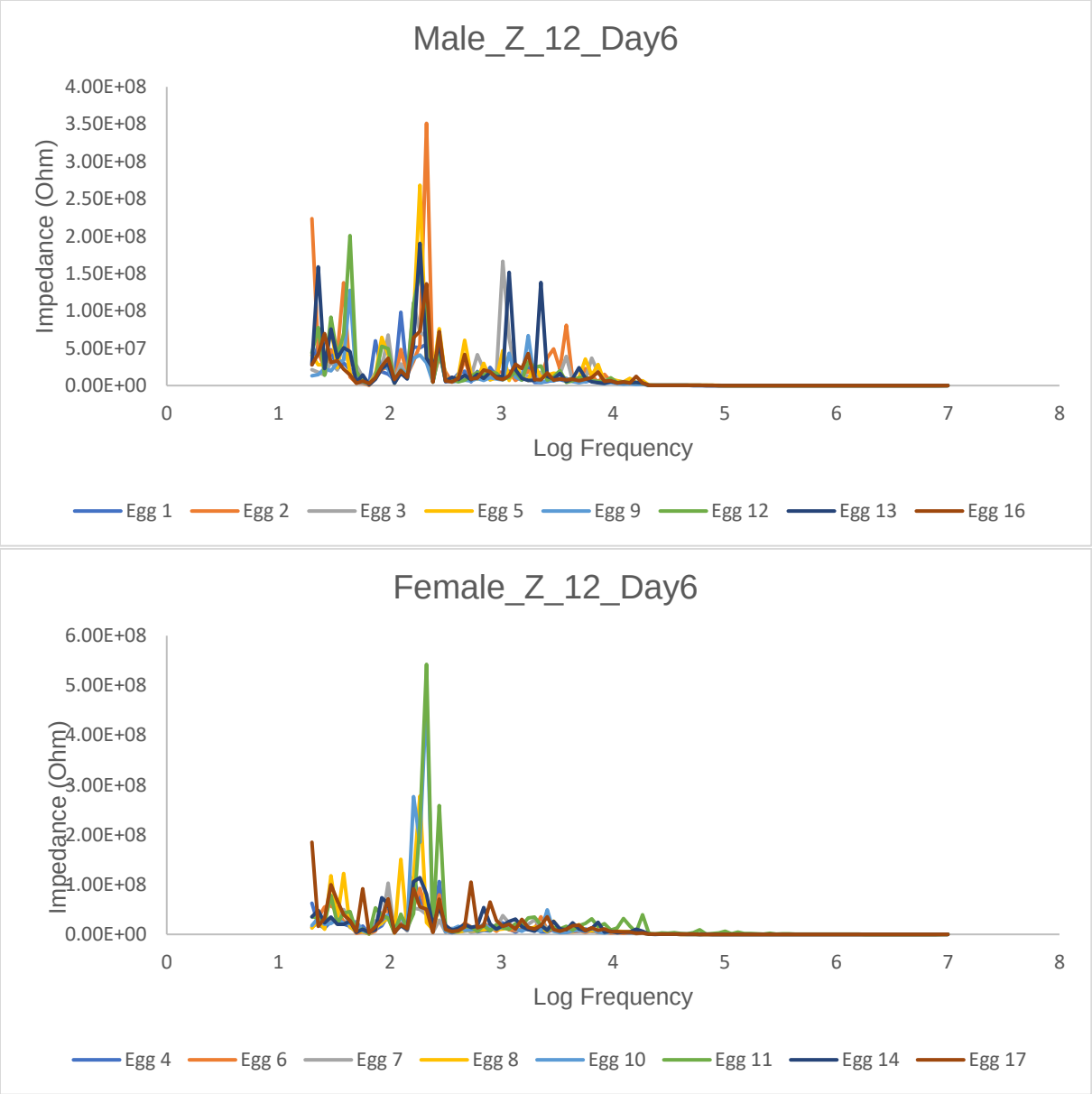


Figure A.21. Impedance measurement of electrode pair 1 and 2 of the eggs performed on day 6

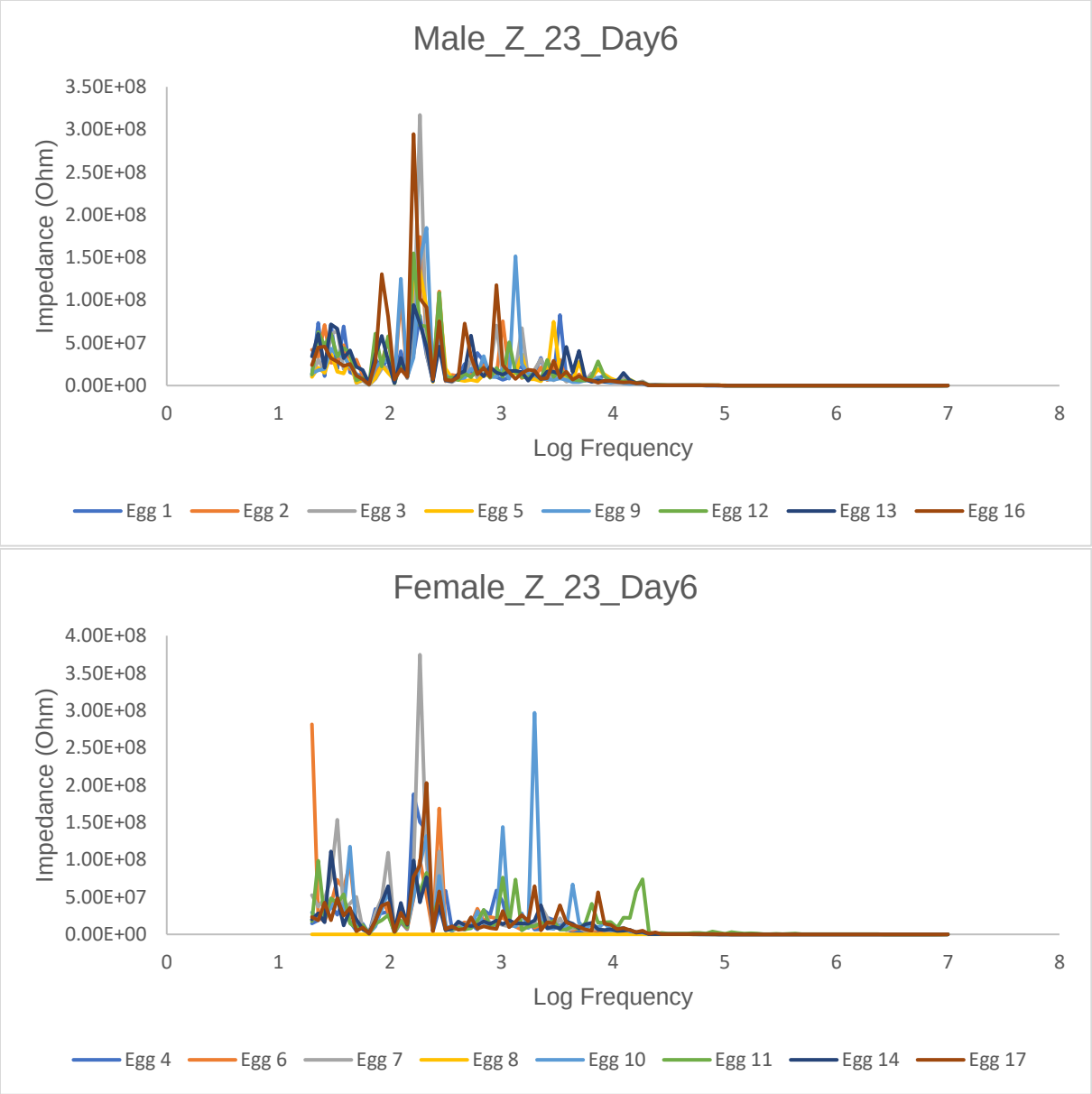
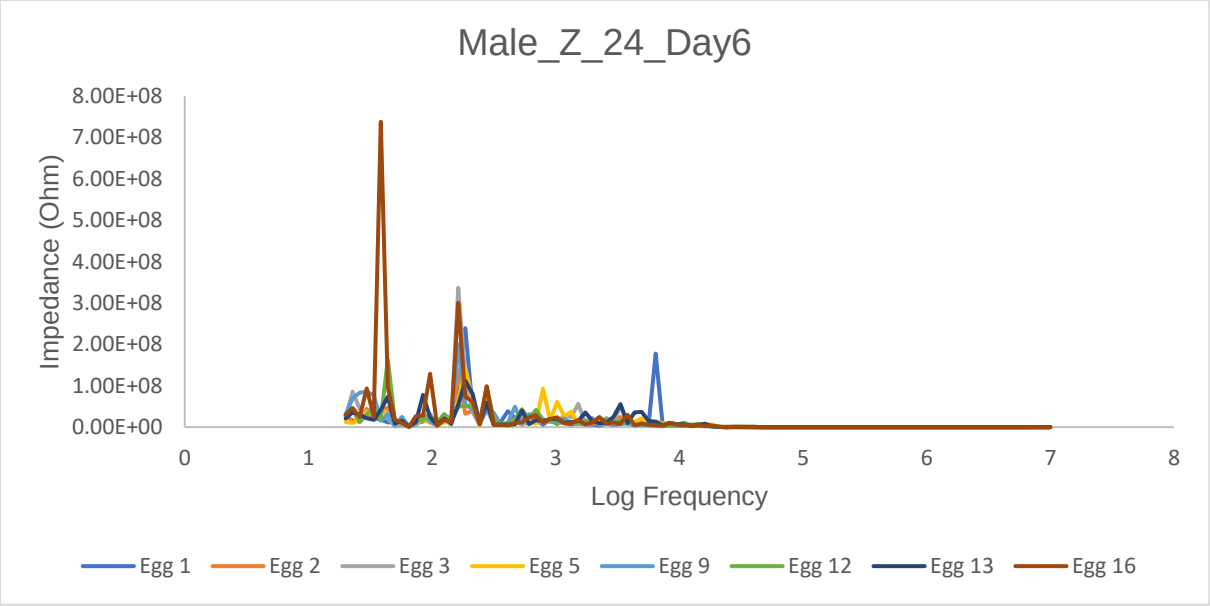


Figure A.22. Impedance measurement of electrode pair 2 and 3 of the eggs performed on day 6



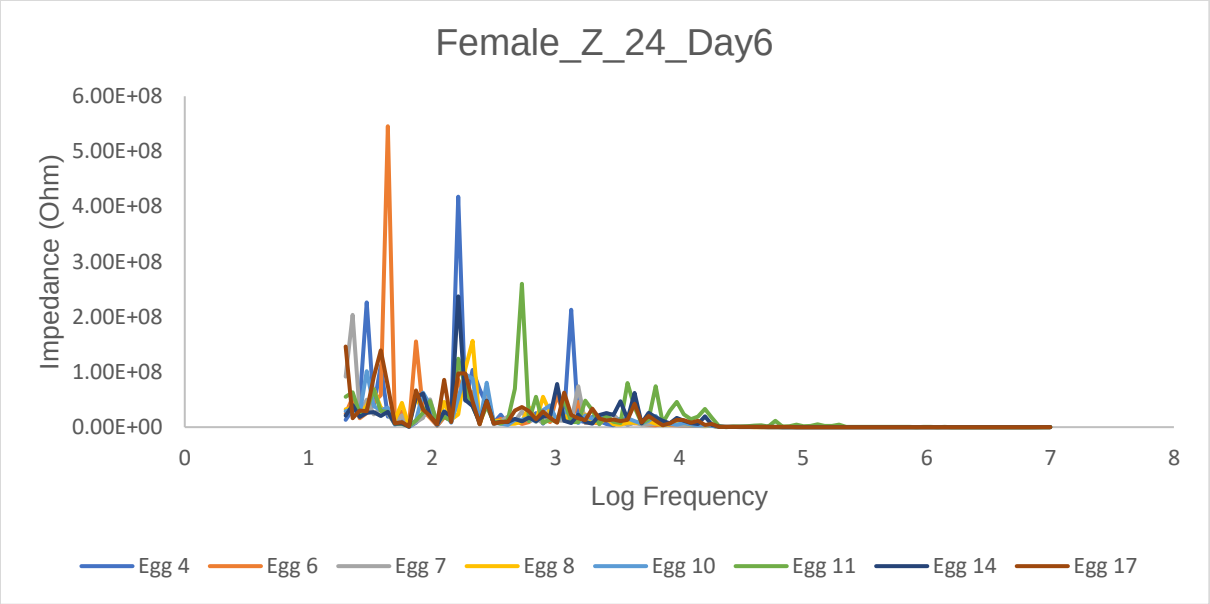
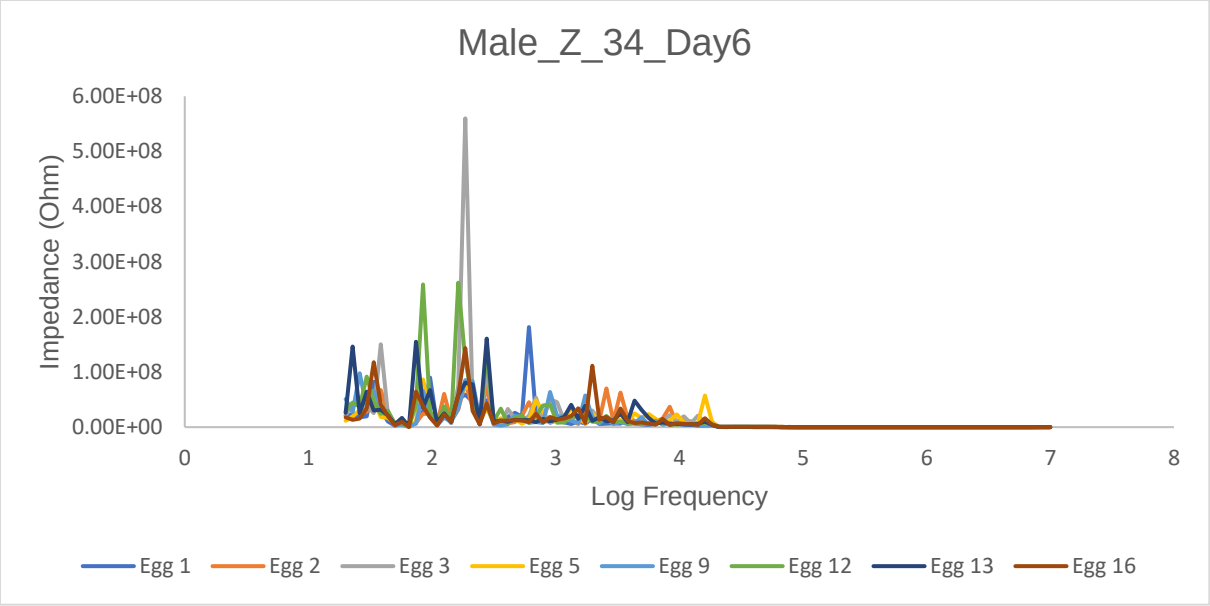


Figure A.23. Impedance measurement of electrode pair 2 and 4 of the eggs performed on day 6



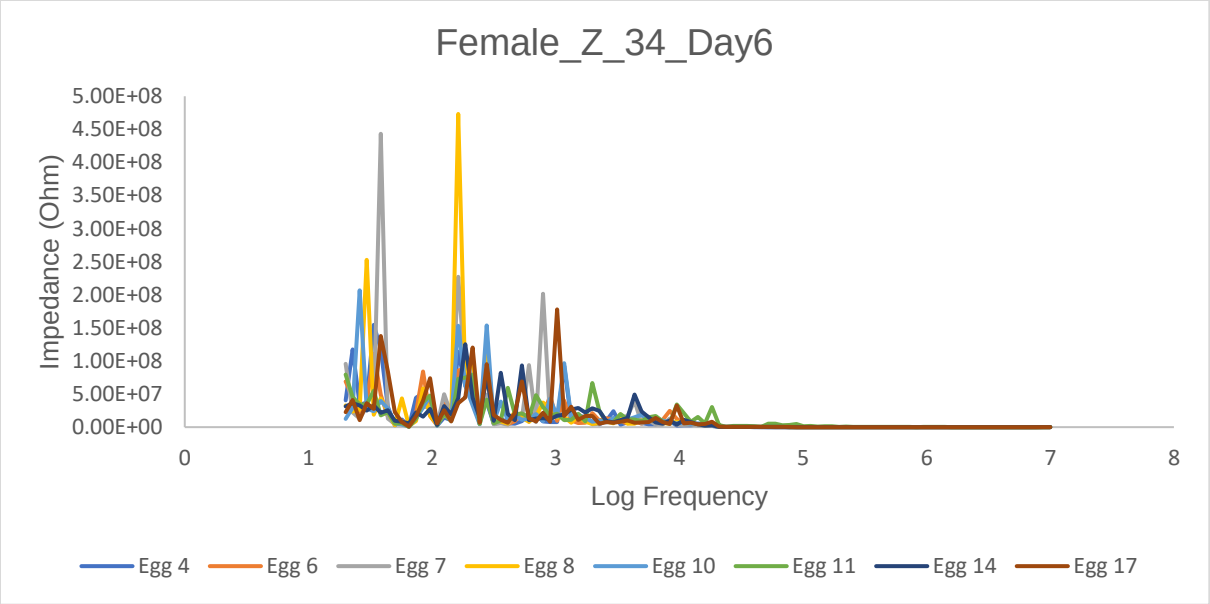
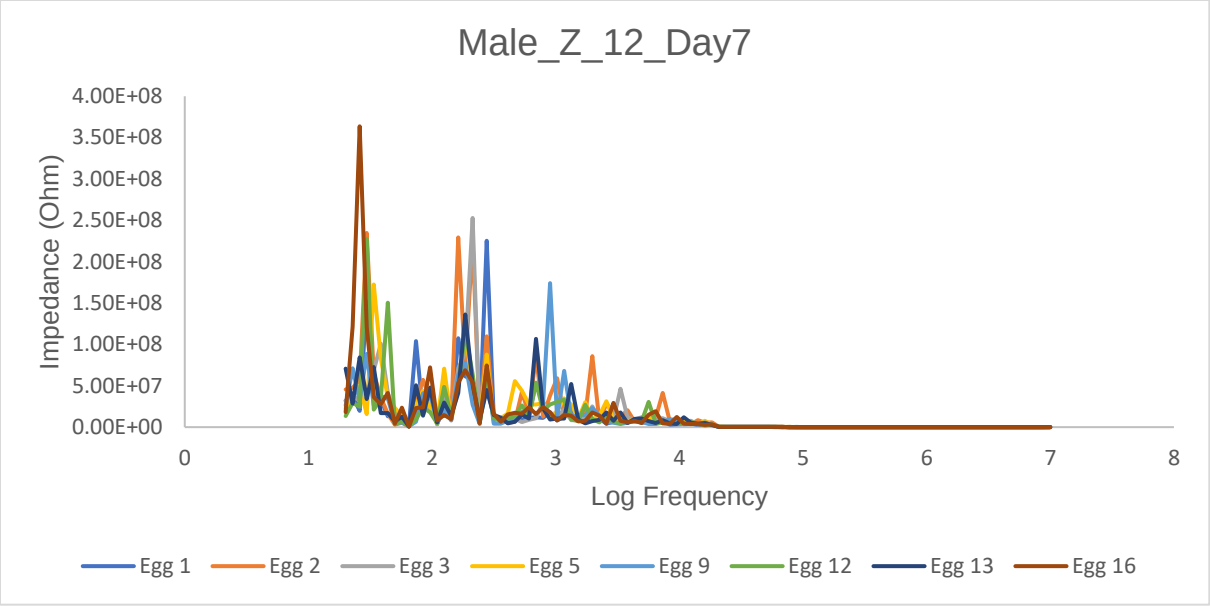


Figure A.24. Impedance measurement of electrode pair 3 and 4 of the eggs performed on day 6



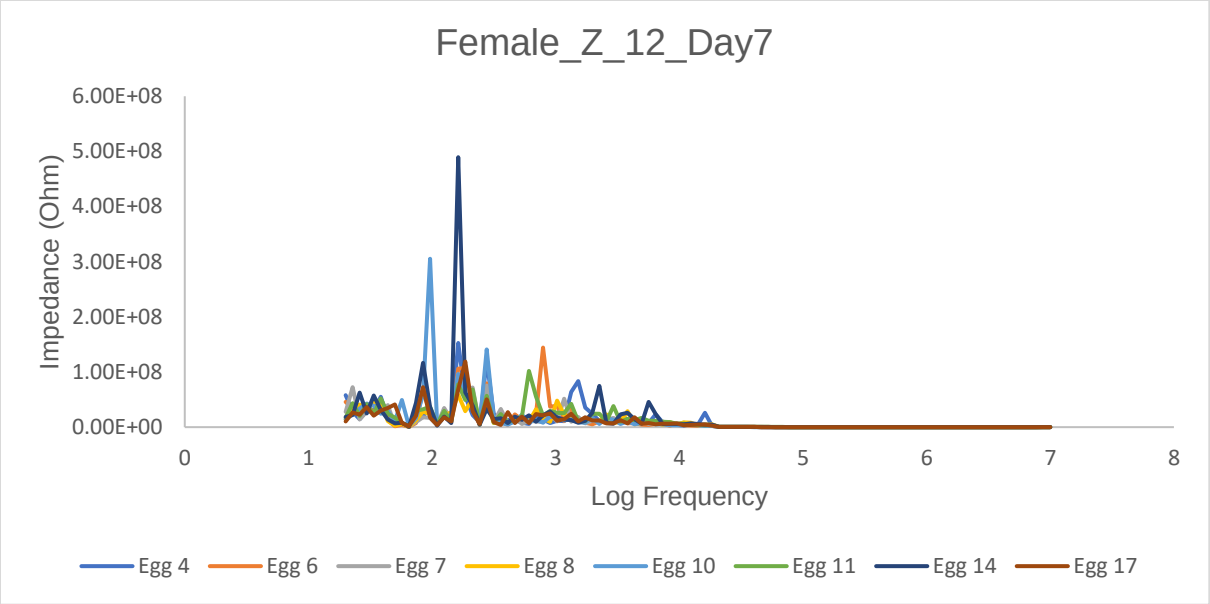


Figure A.25. Impedance measurement of electrode pair 1 and 2 of the eggs performed on day 7

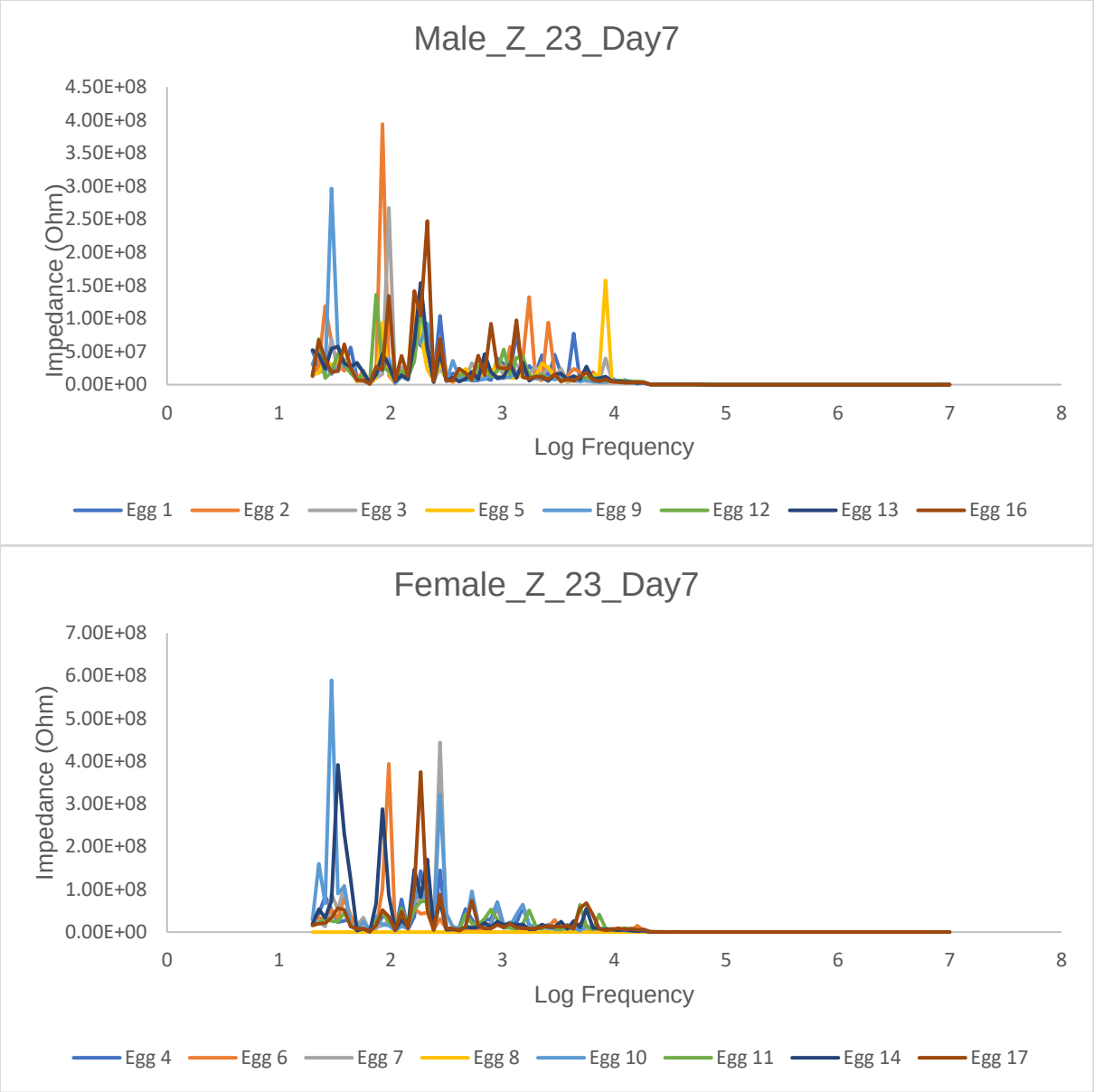
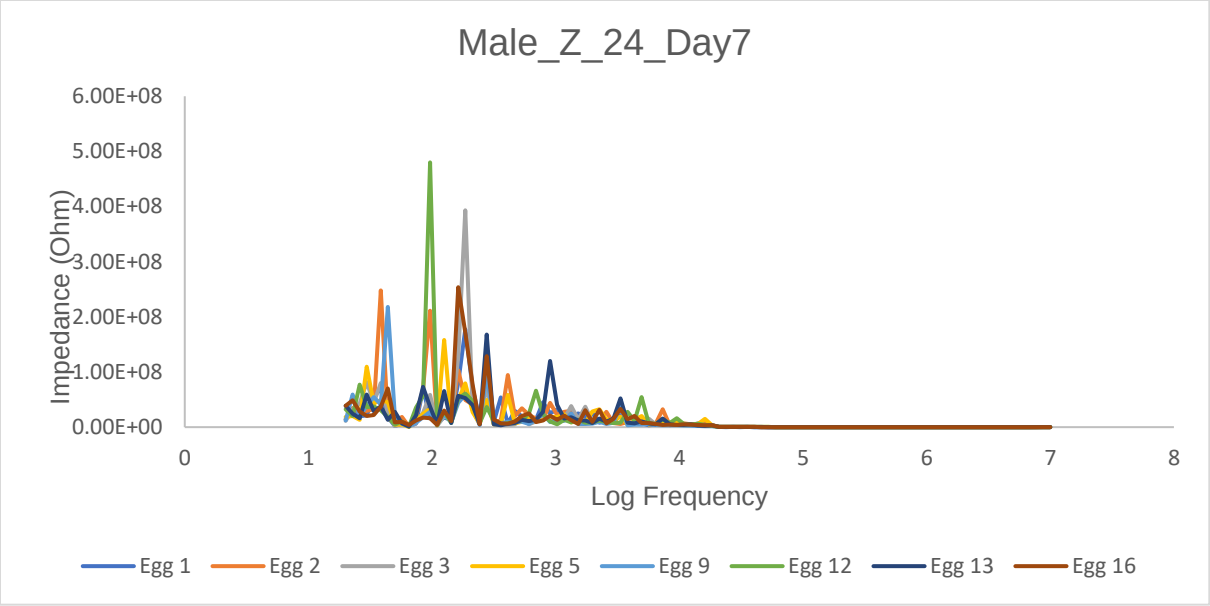


Figure A.26. Impedance measurement of electrode pair 2 and 3 of the eggs performed on day 7



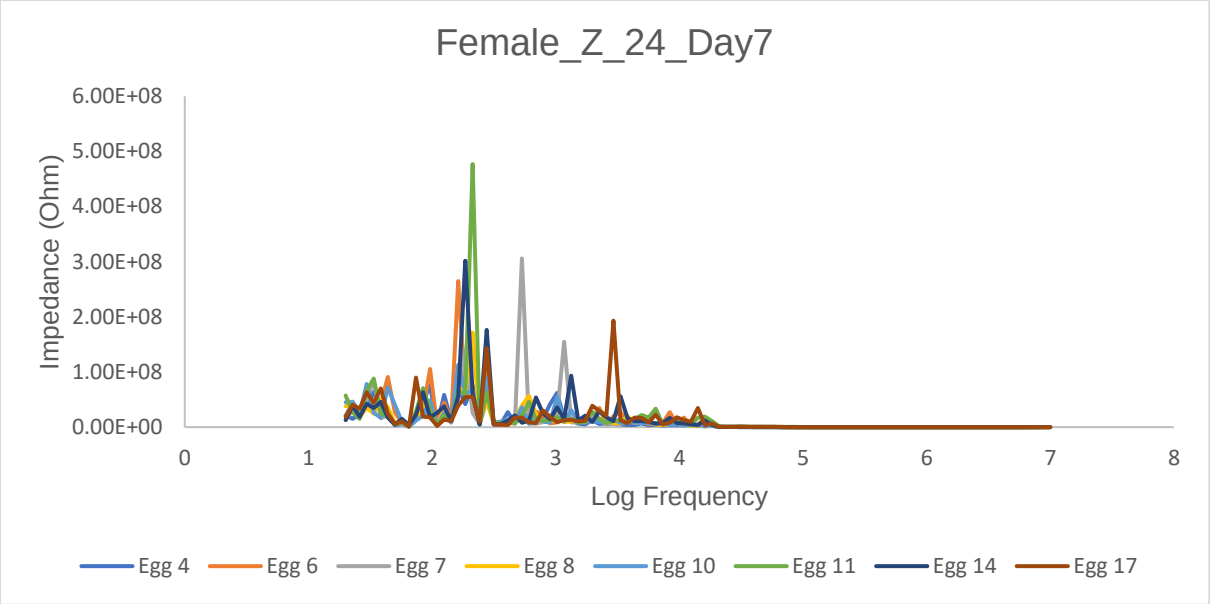


Figure A.27. Impedance measurement of electrode pair 2 and 4 of the eggs performed on day 7

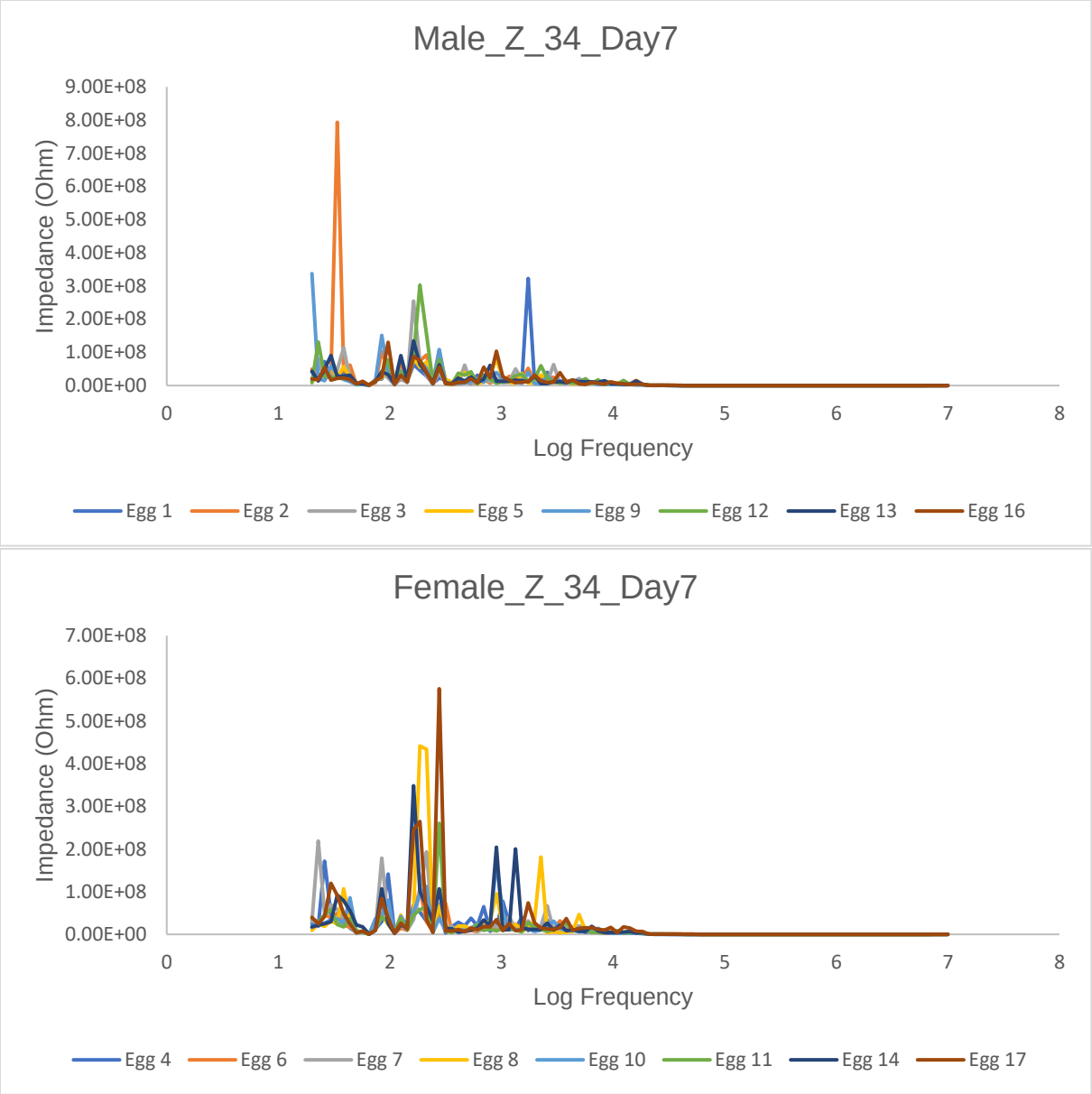
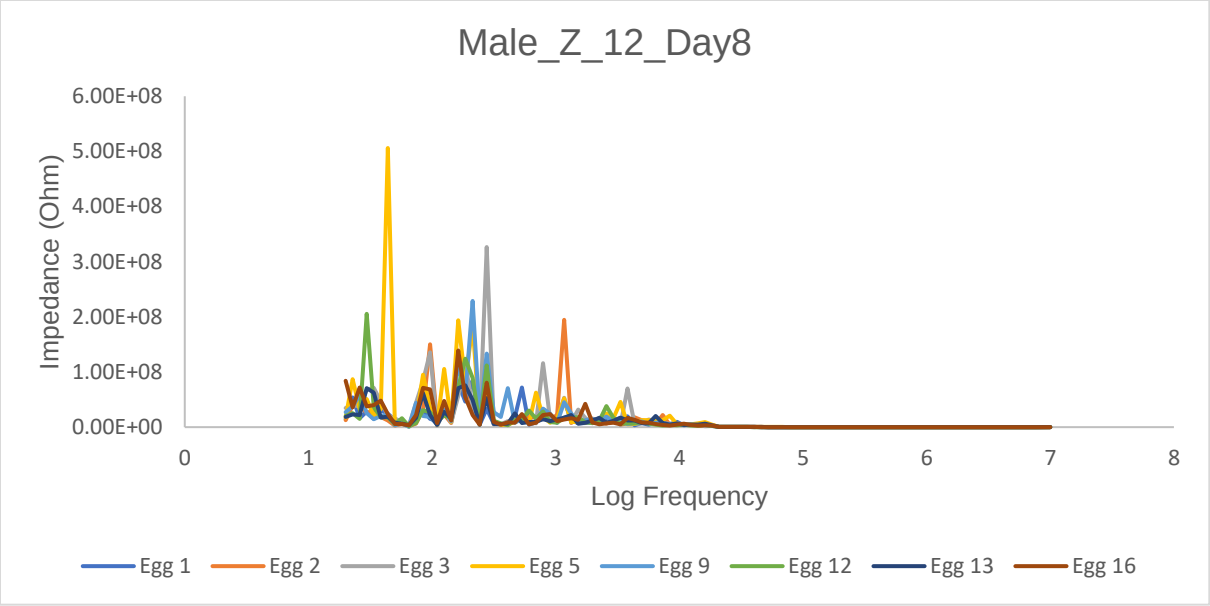


Figure A.28. Impedance measurement of electrode pair 3 and 4 of the eggs performed on day 7



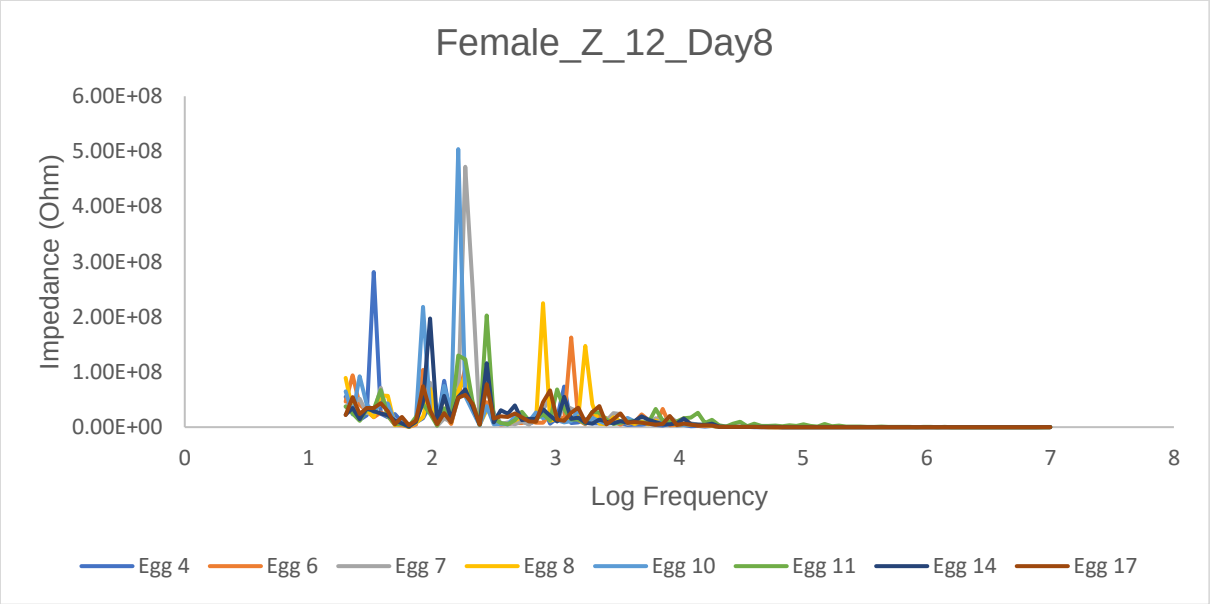
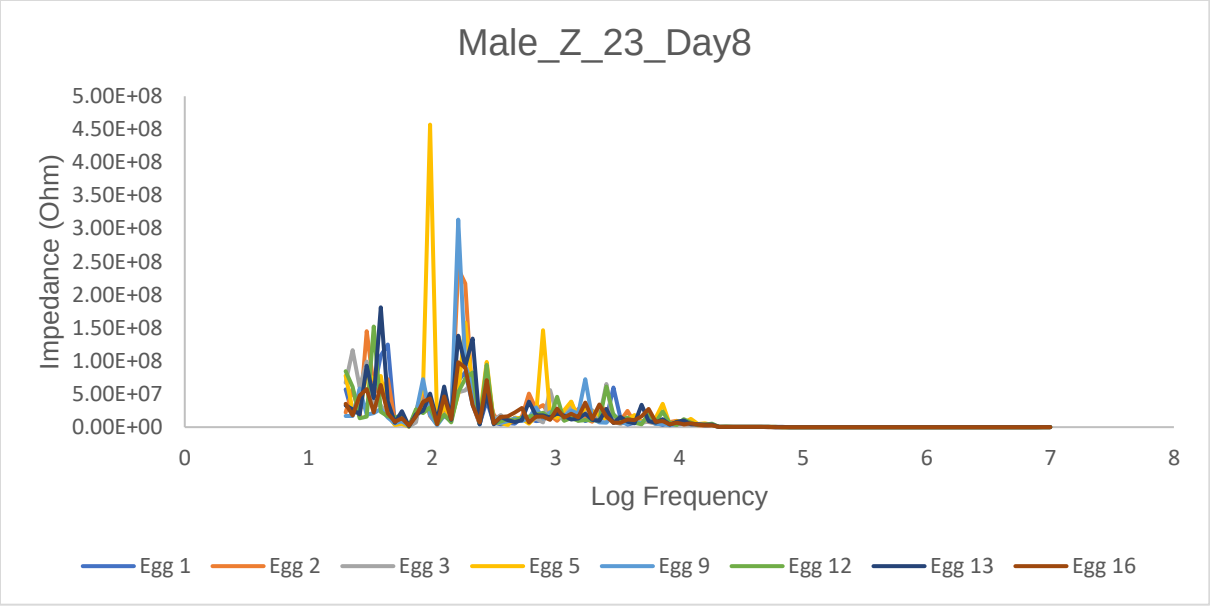


Figure A.29. Impedance measurement of electrode pair 1 and 2 of the eggs performed on day 8



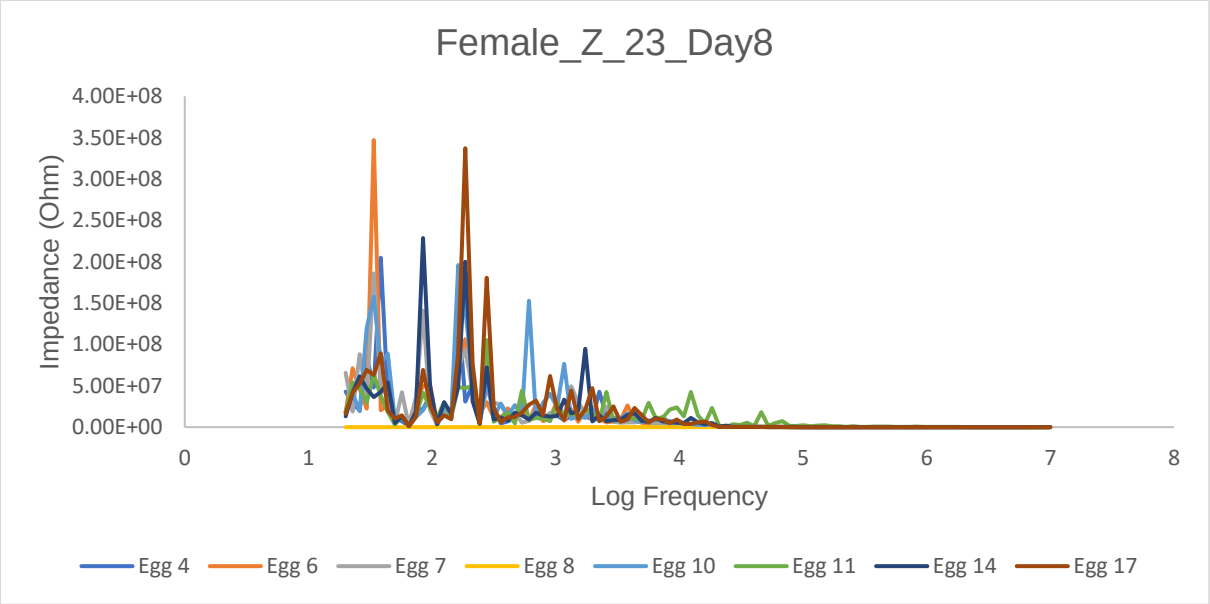
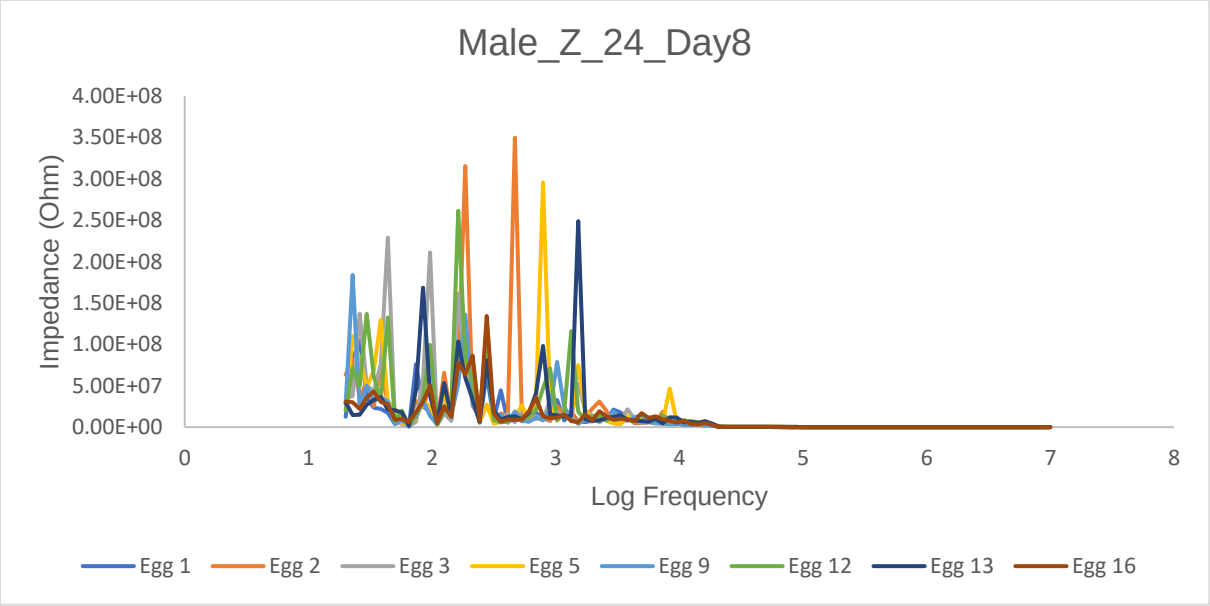


Figure A.30. Impedance measurement of electrode pair 2 and 3 of the eggs performed on day 8



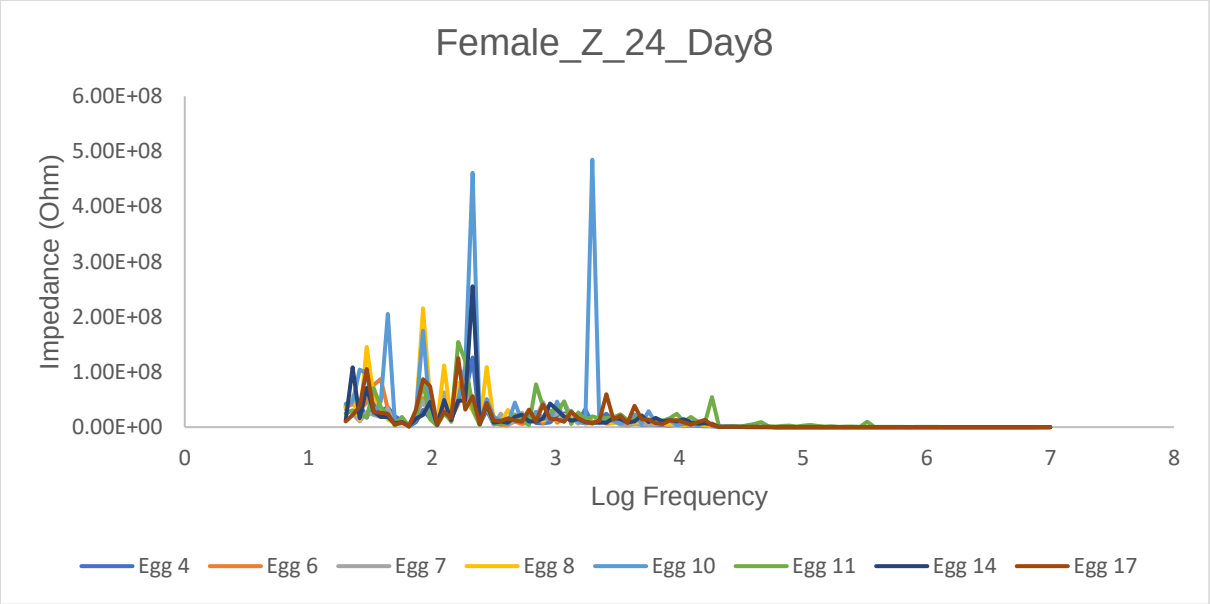
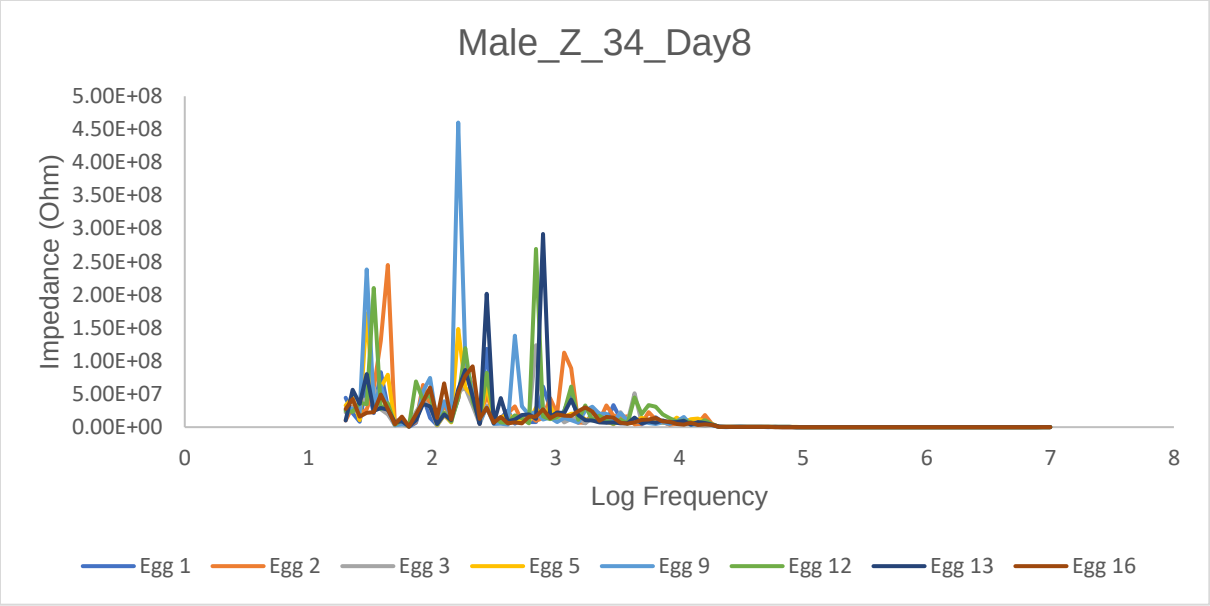


Figure A.31. Impedance measurement of electrode pair 2 and 4 of the eggs performed on day 8



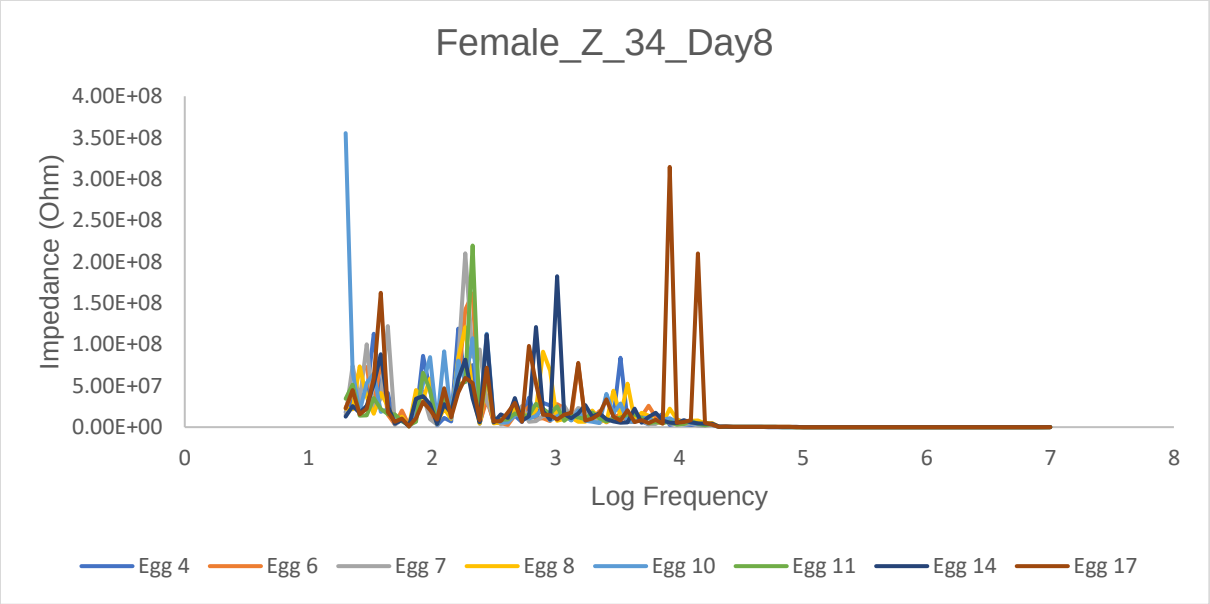
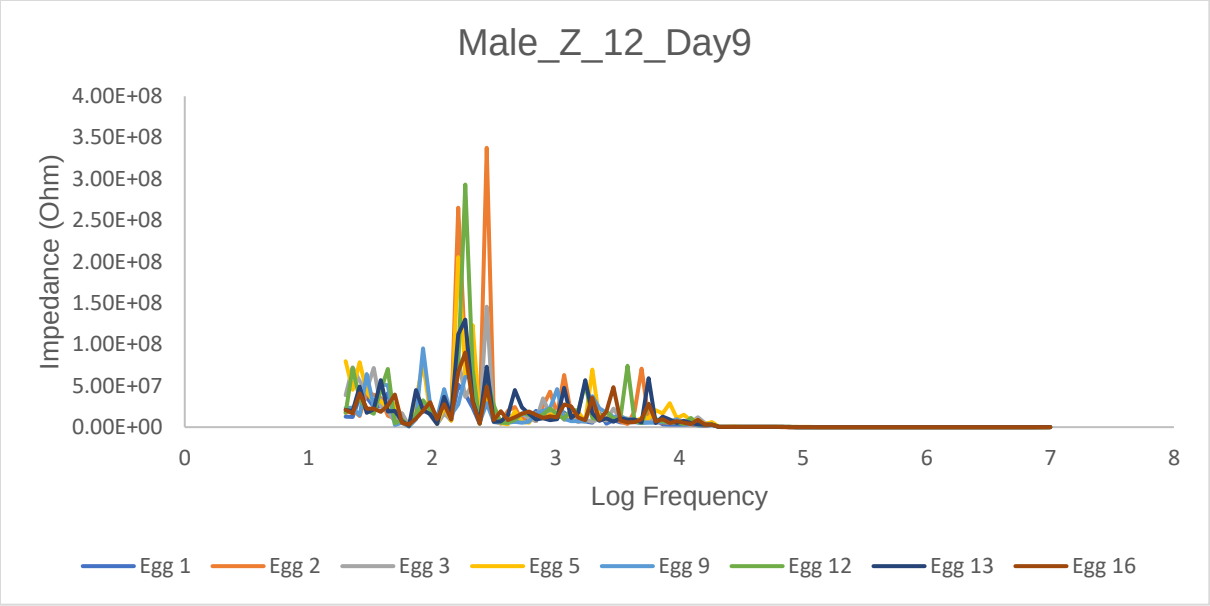


Figure A.32. Impedance measurement of electrode pair 3 and 4 of the eggs performed on day 8



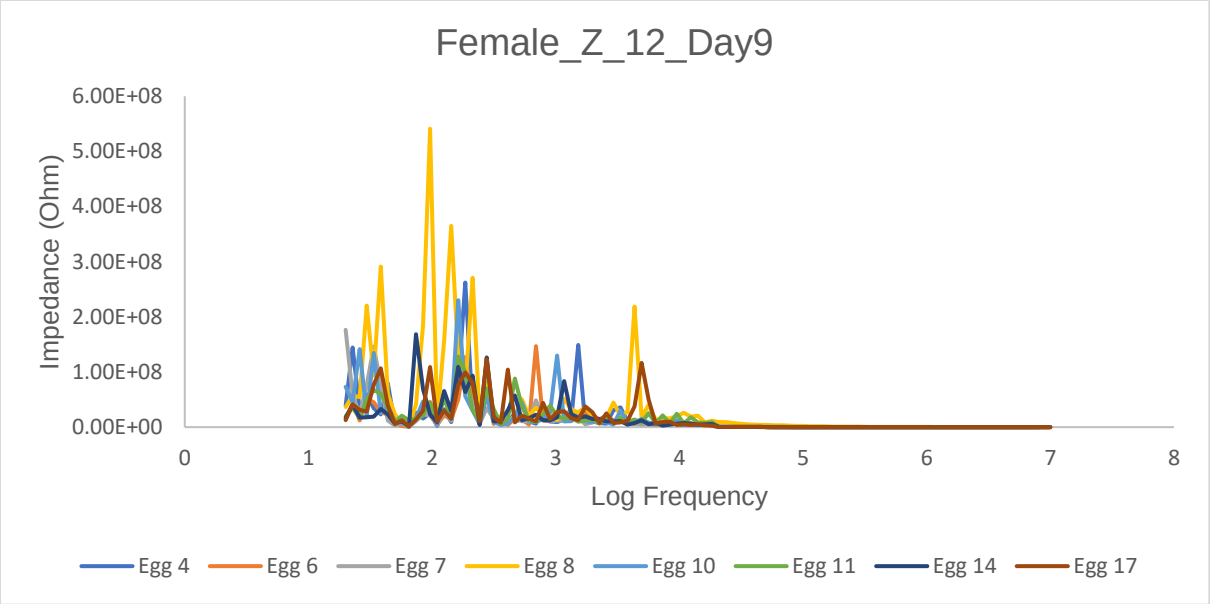
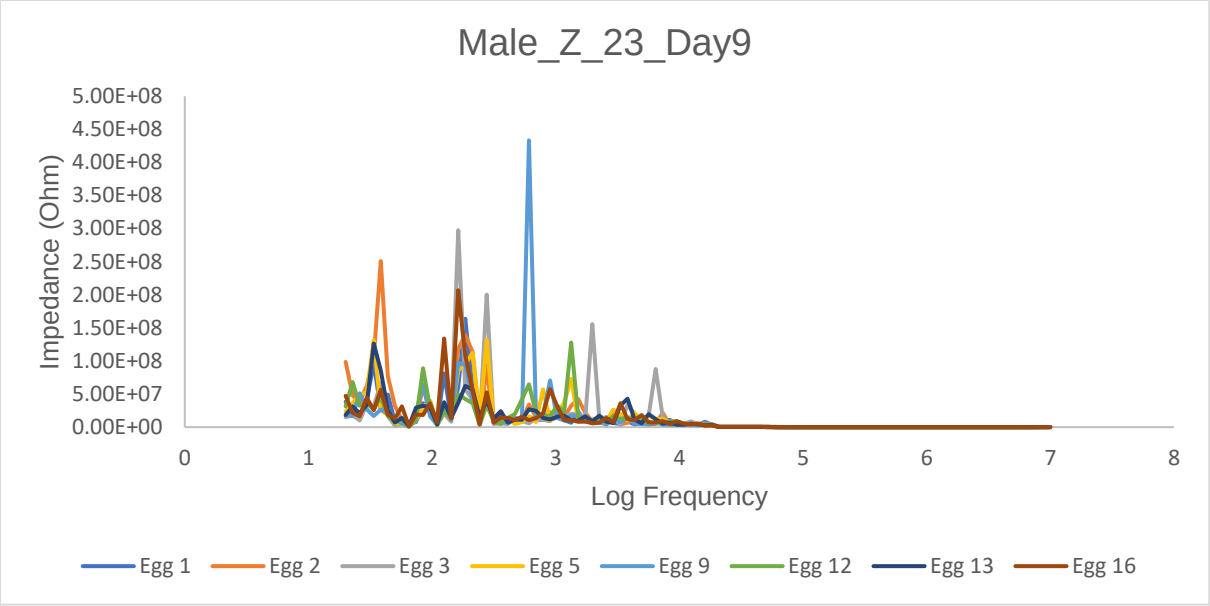


Figure A.33. Impedance measurement of electrode pair 1 and 2 of the eggs performed on day 9



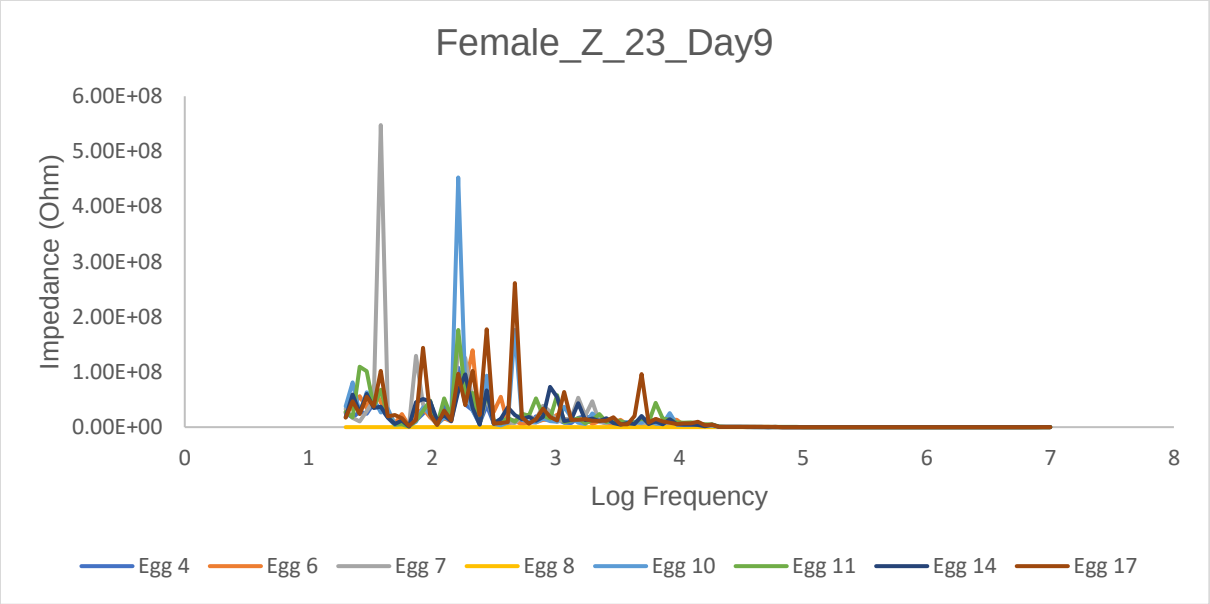
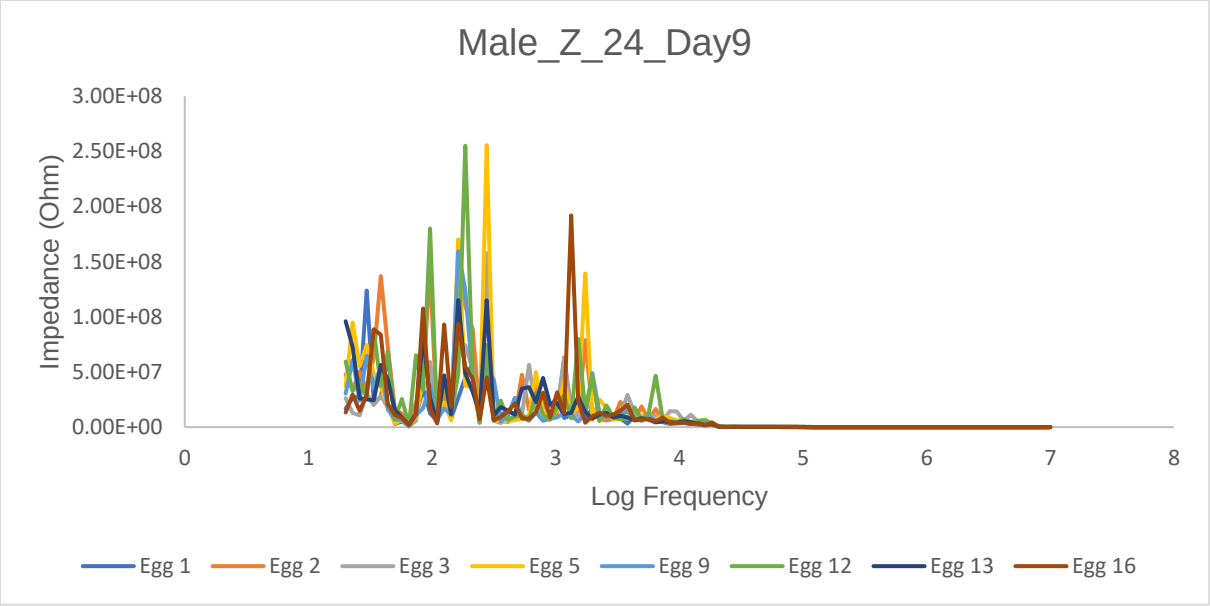


Figure A.34. Impedance measurement of electrode pair 2 and 3 of the eggs performed on day 9



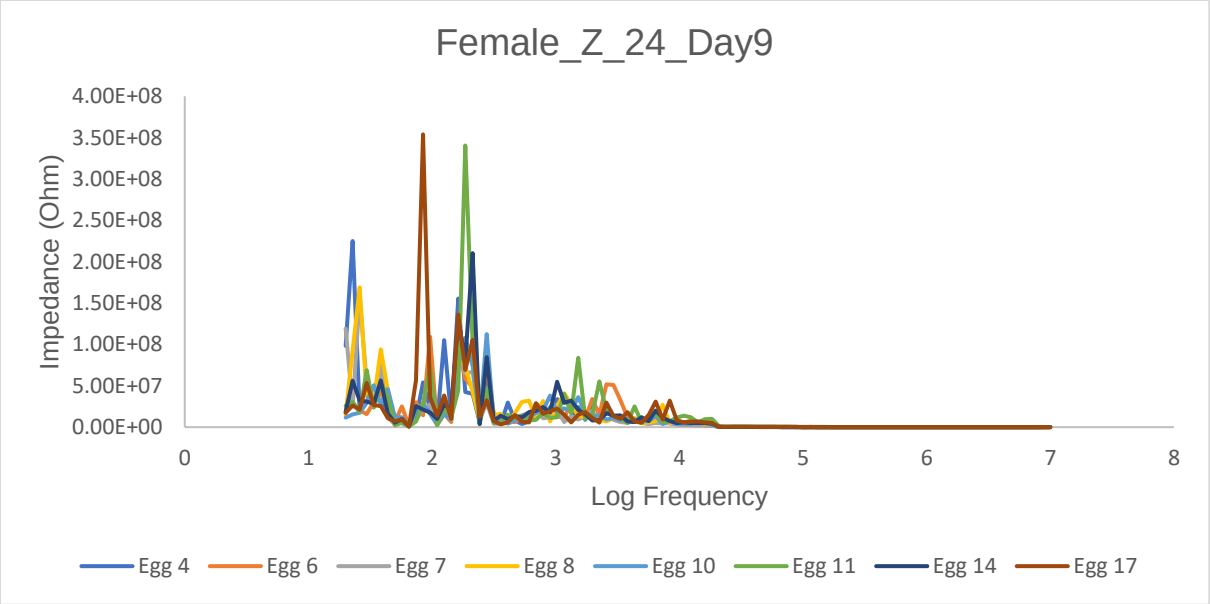


Figure A.35. Impedance measurement of electrode pair 2 and 4 of the eggs performed on day 9

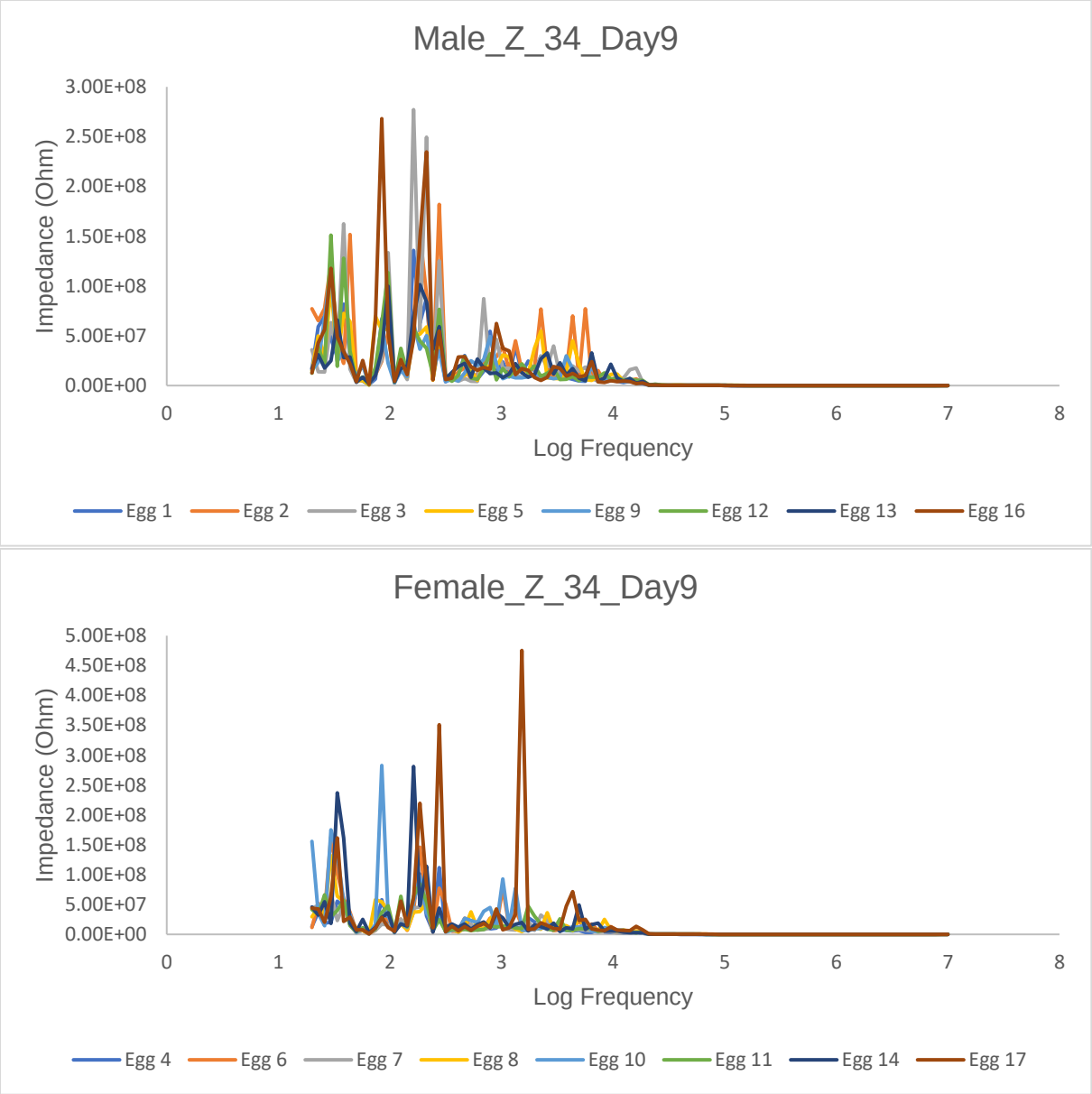
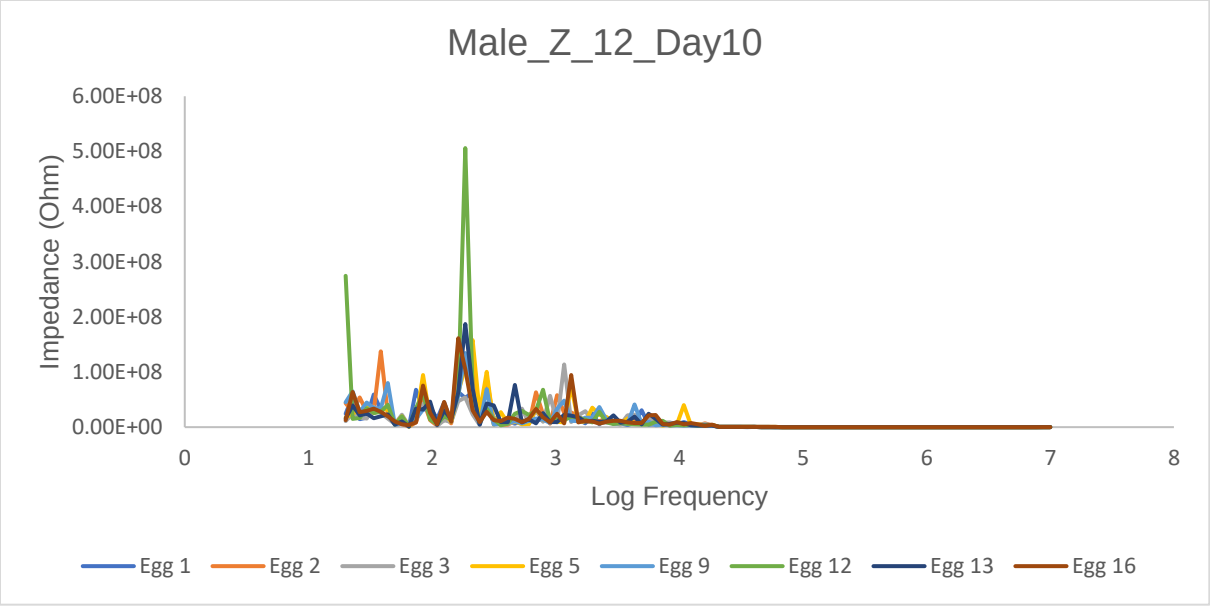


Figure A.36. Impedance measurement of electrode pair 3 and 4 of the eggs performed on day 9



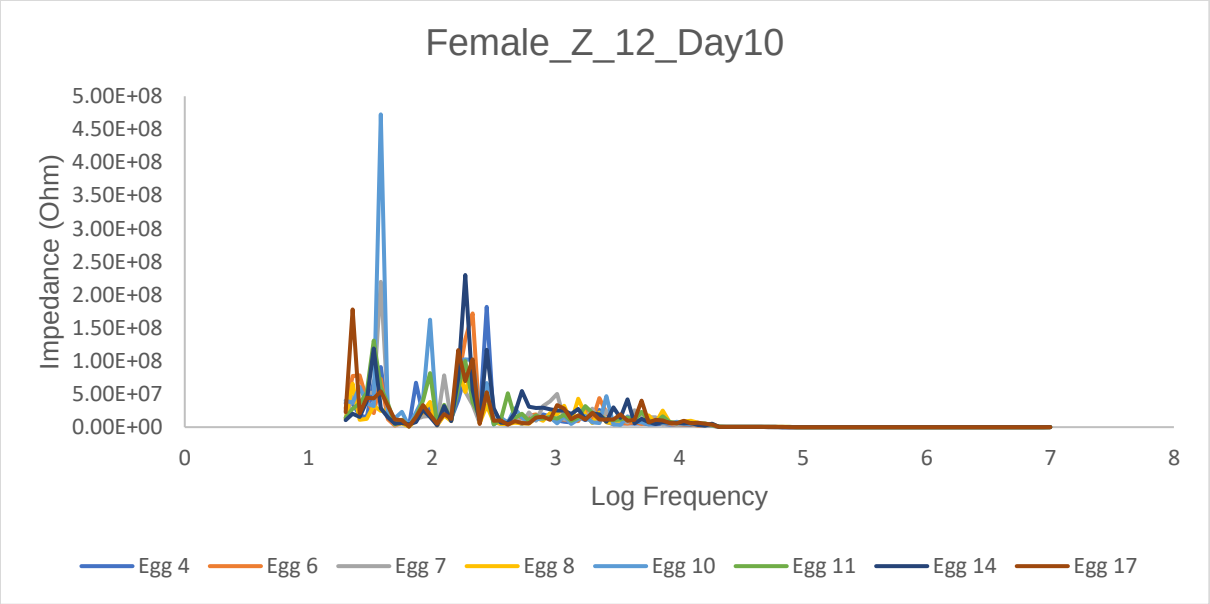
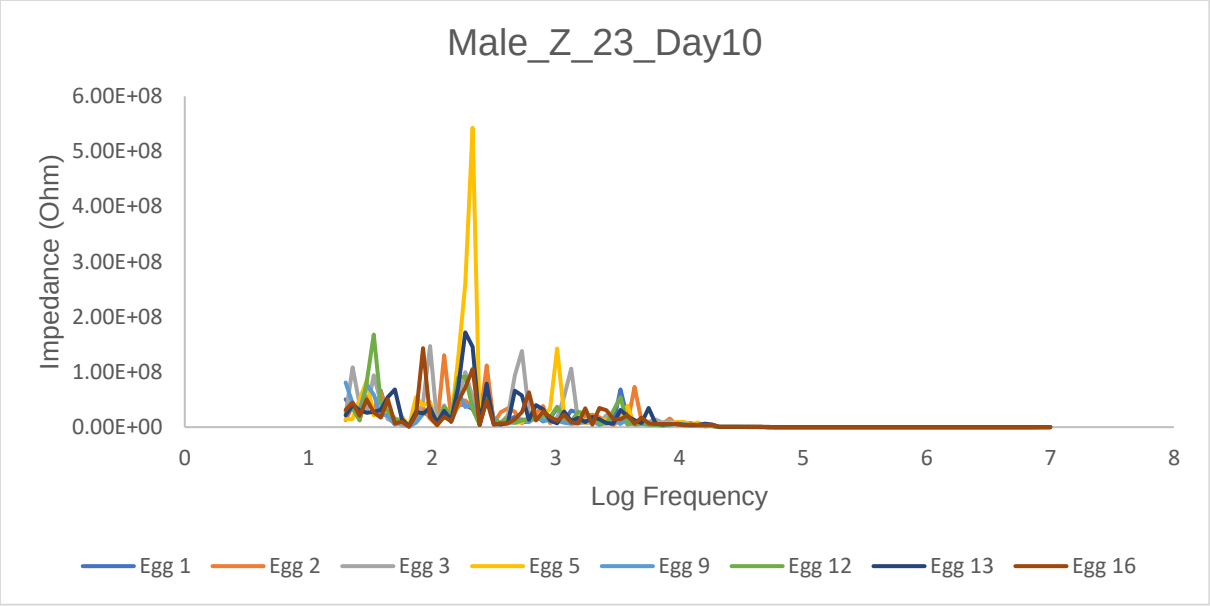


Figure A.37. Impedance measurement of electrode pair 1 and 2 of the eggs performed on day 10



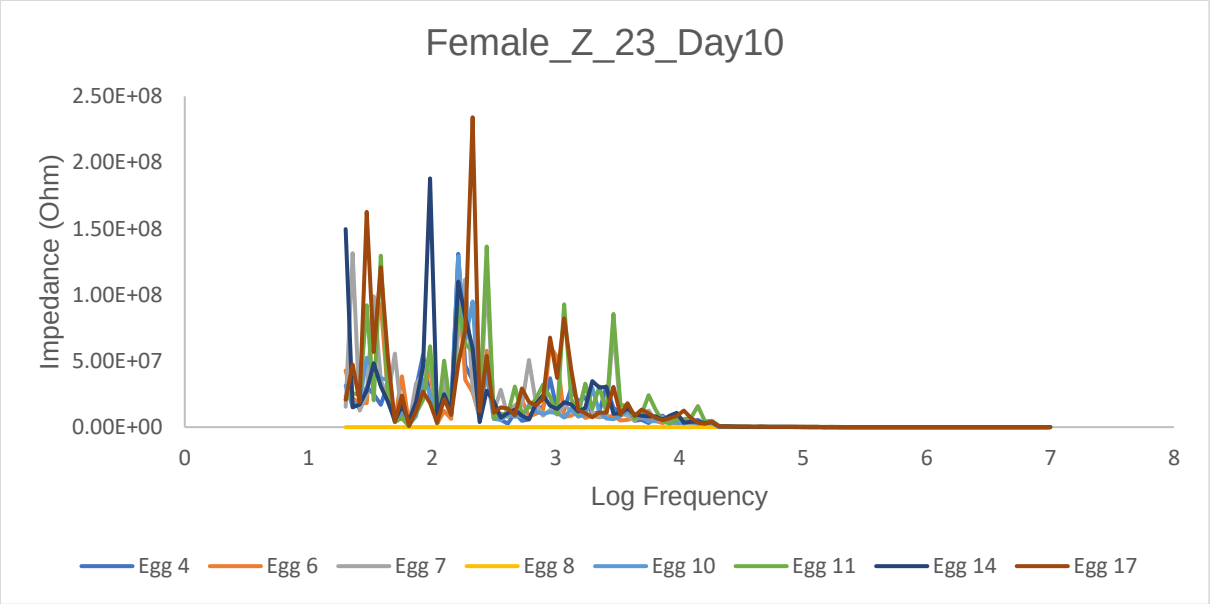
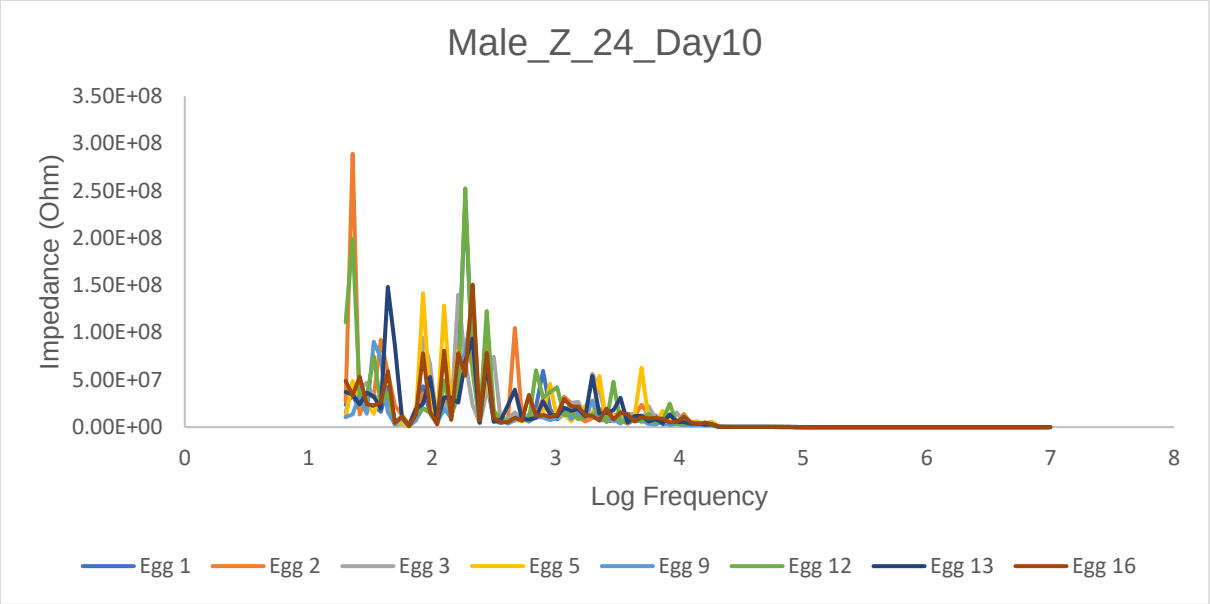


Figure A.38. Impedance measurement of electrode pair 2 and 3 of the eggs performed on day 10



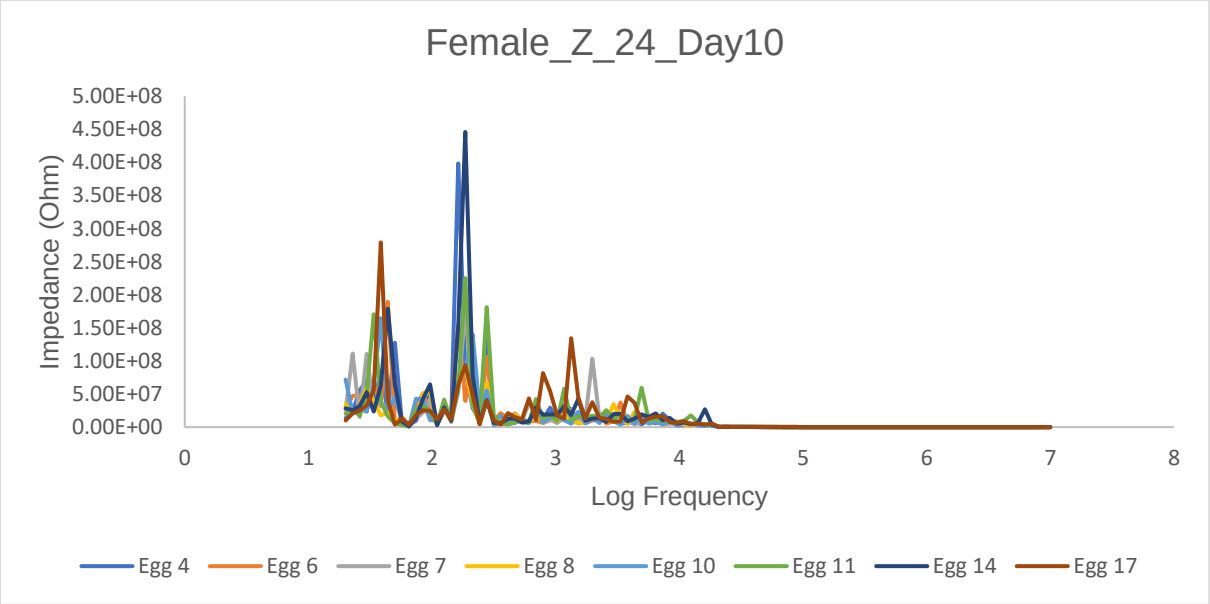
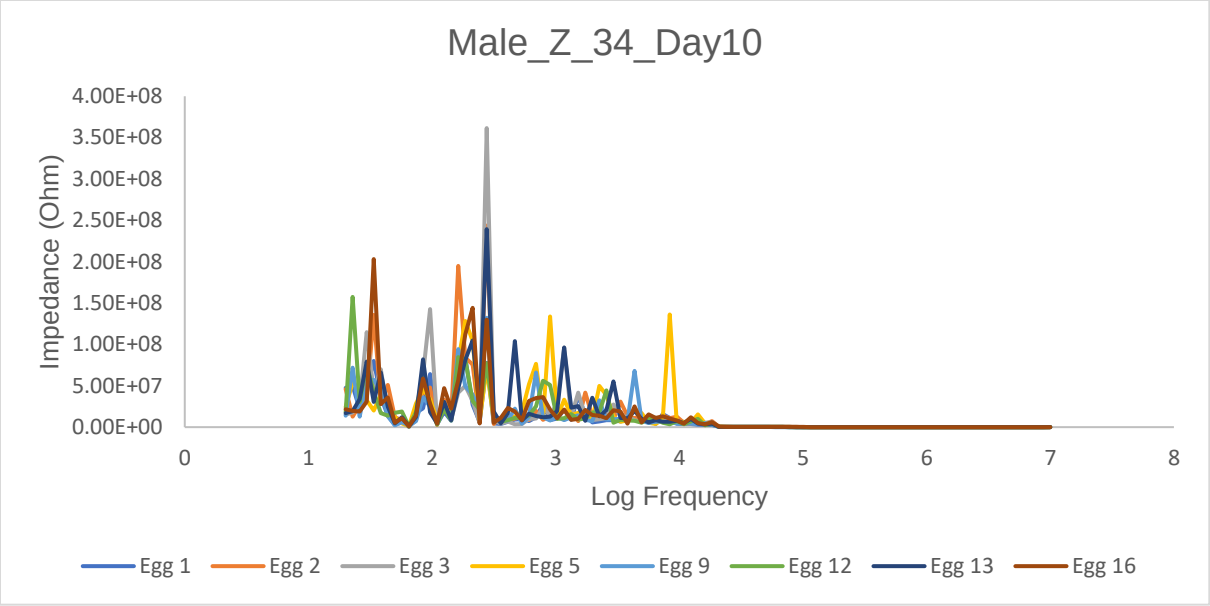


Figure A.39. Impedance measurement of electrode pair 2 and 4 of the eggs performed on day 10



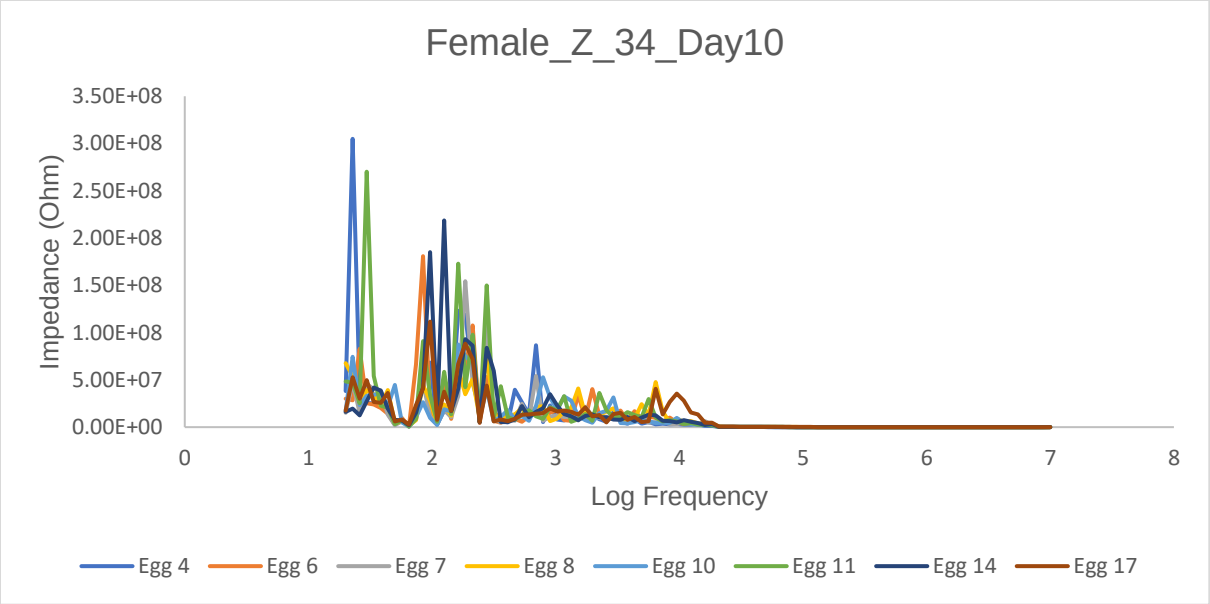
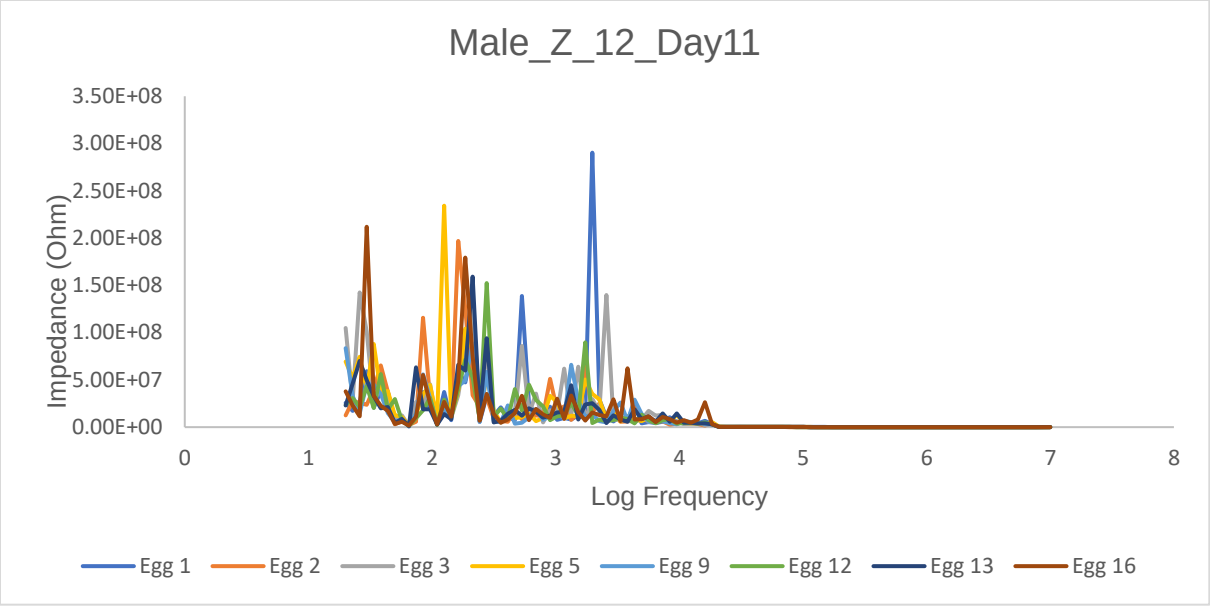


Figure A.40. Impedance measurement of electrode pair 3 and 4 of the eggs performed on day 10



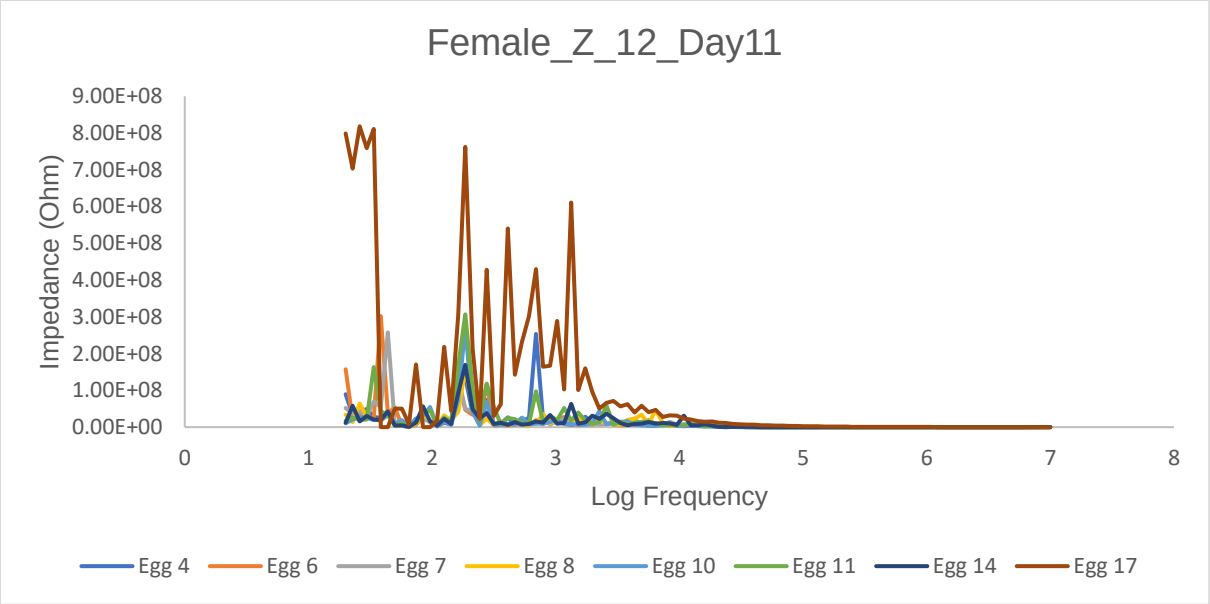
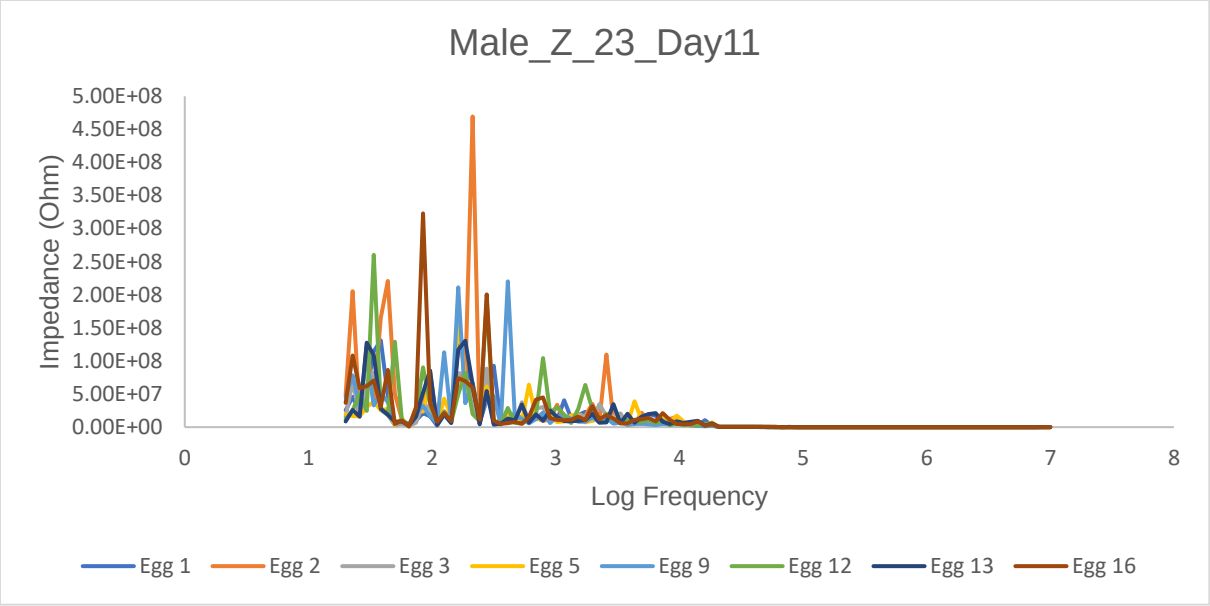


Figure A.41. Impedance measurement of electrode pair 1 and 2 of the eggs performed on day 11



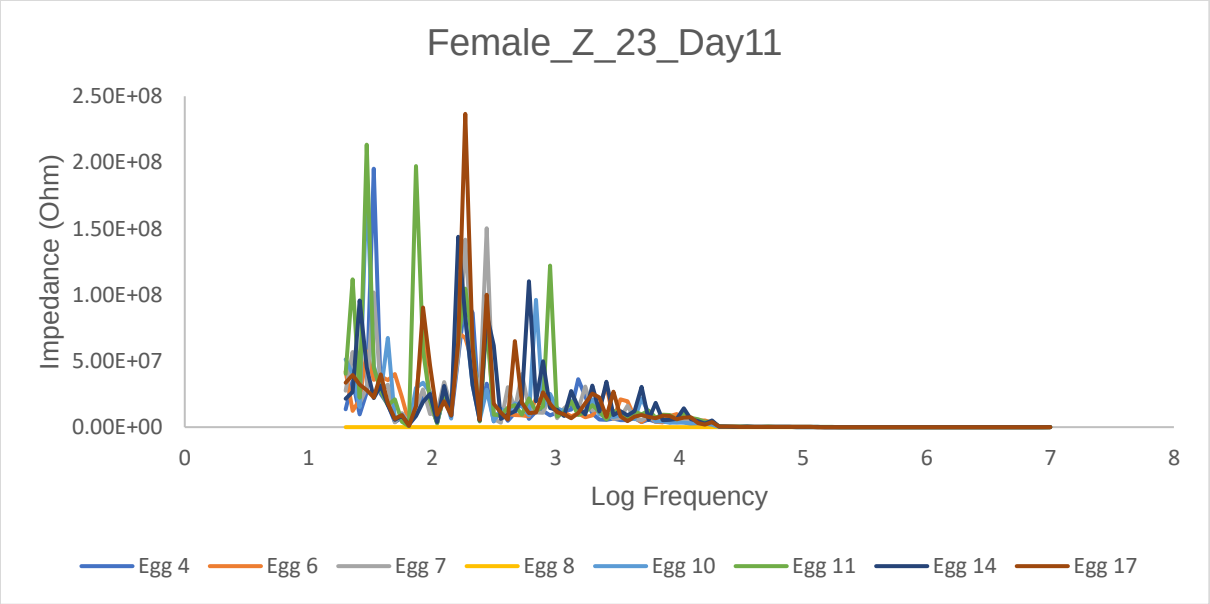
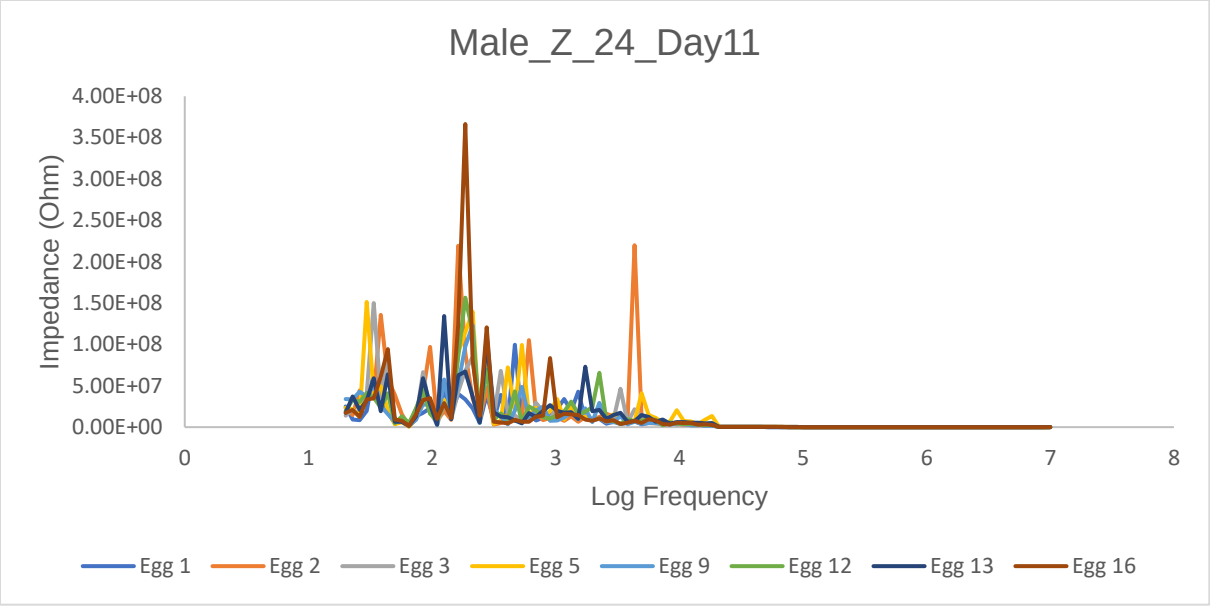


Figure A.42. Impedance measurement of electrode pair 2 and 3 of the eggs performed on day 11



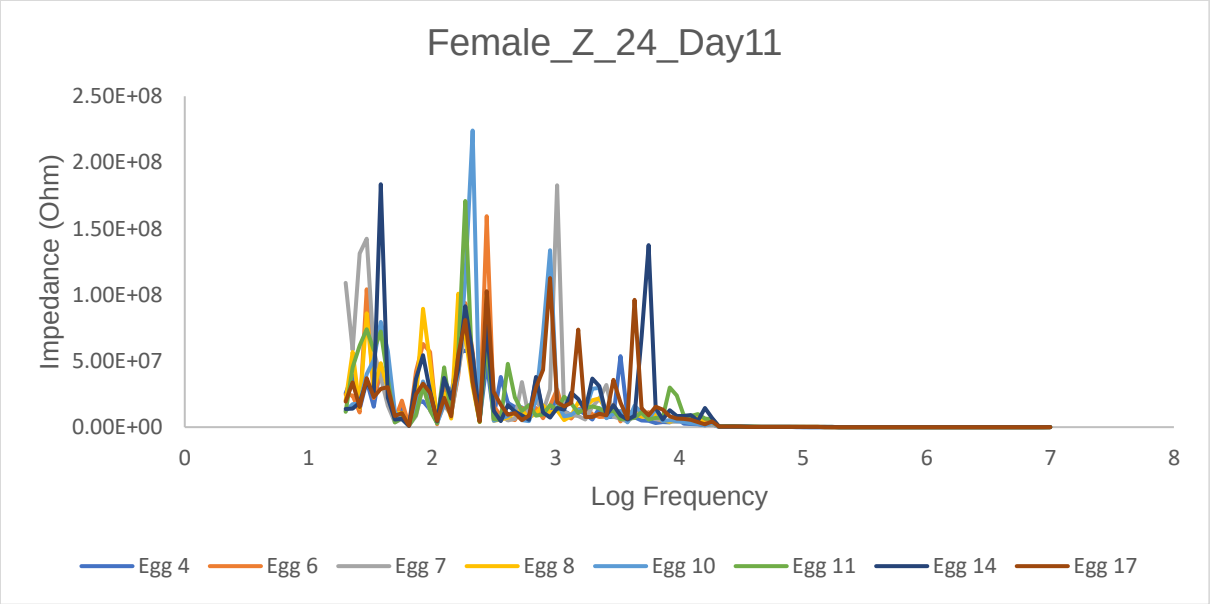
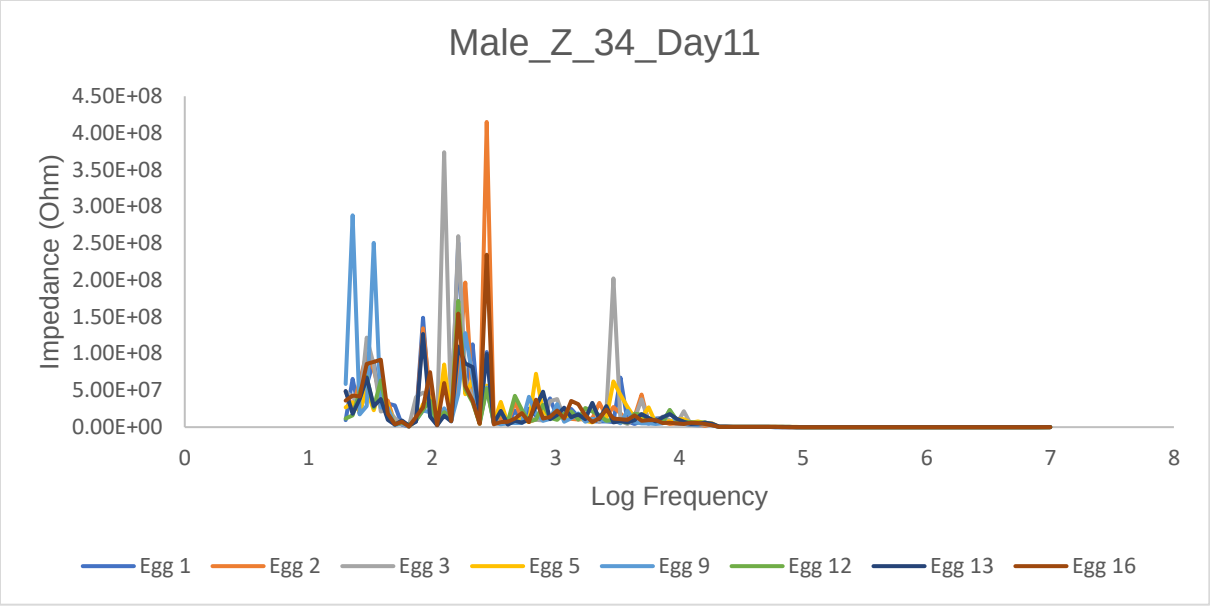


Figure A.43. Impedance measurement of electrode pair 2 and 4 of the eggs performed on day 11



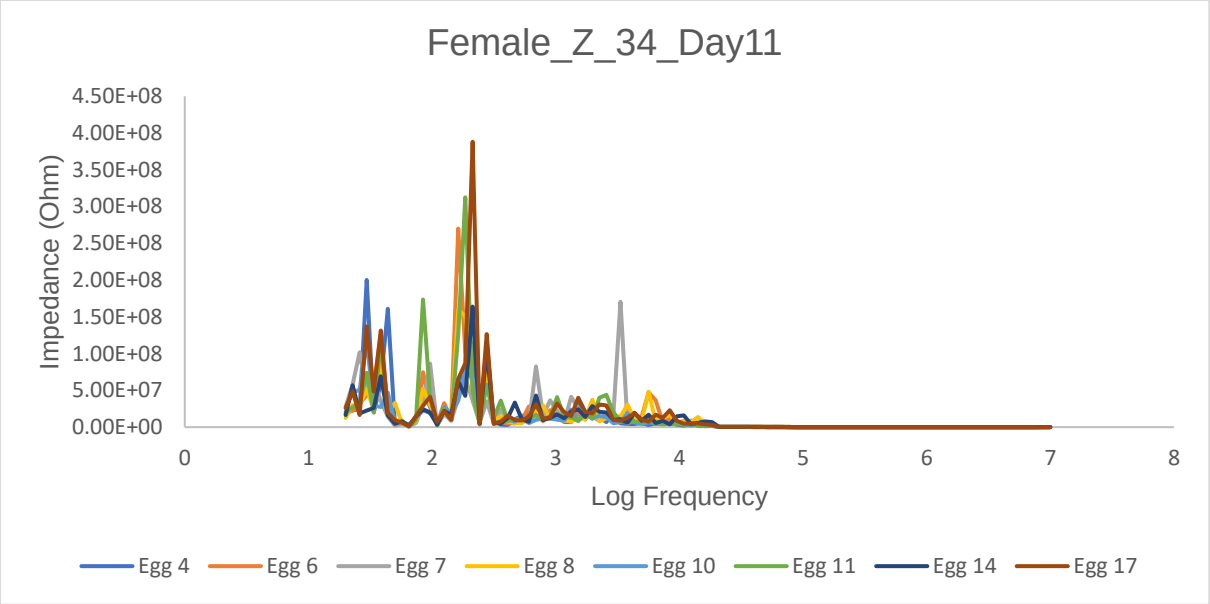


Figure A.44. Impedance measurement of electrode pair 3 and 4 of the eggs performed on day 11

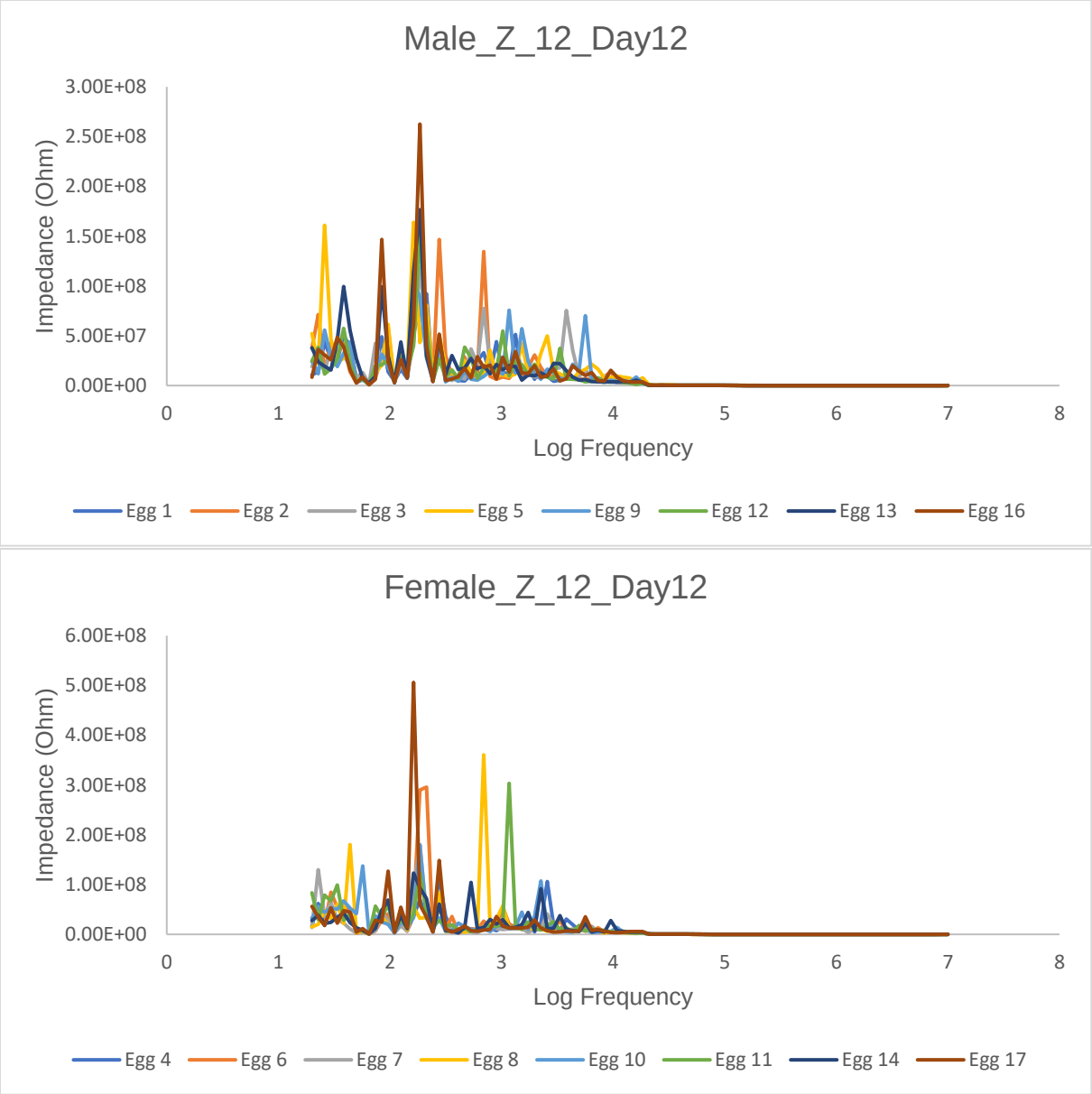


Figure A.45. Impedance measurement of electrode pair 1 and 2 of the eggs performed on day 12

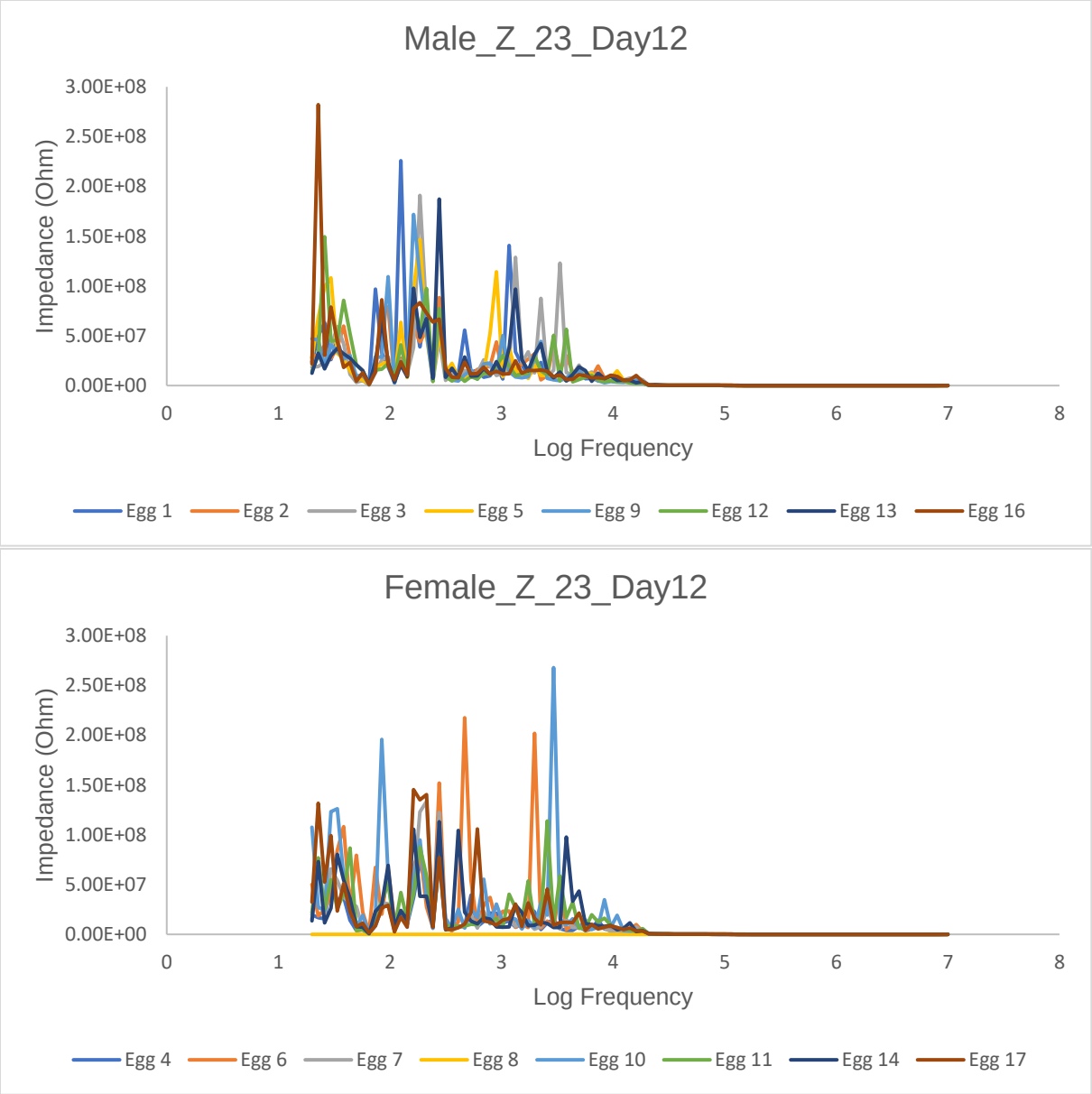


Figure A.46. Impedance measurement of electrode pair 2 and 3 of the eggs performed on day 12

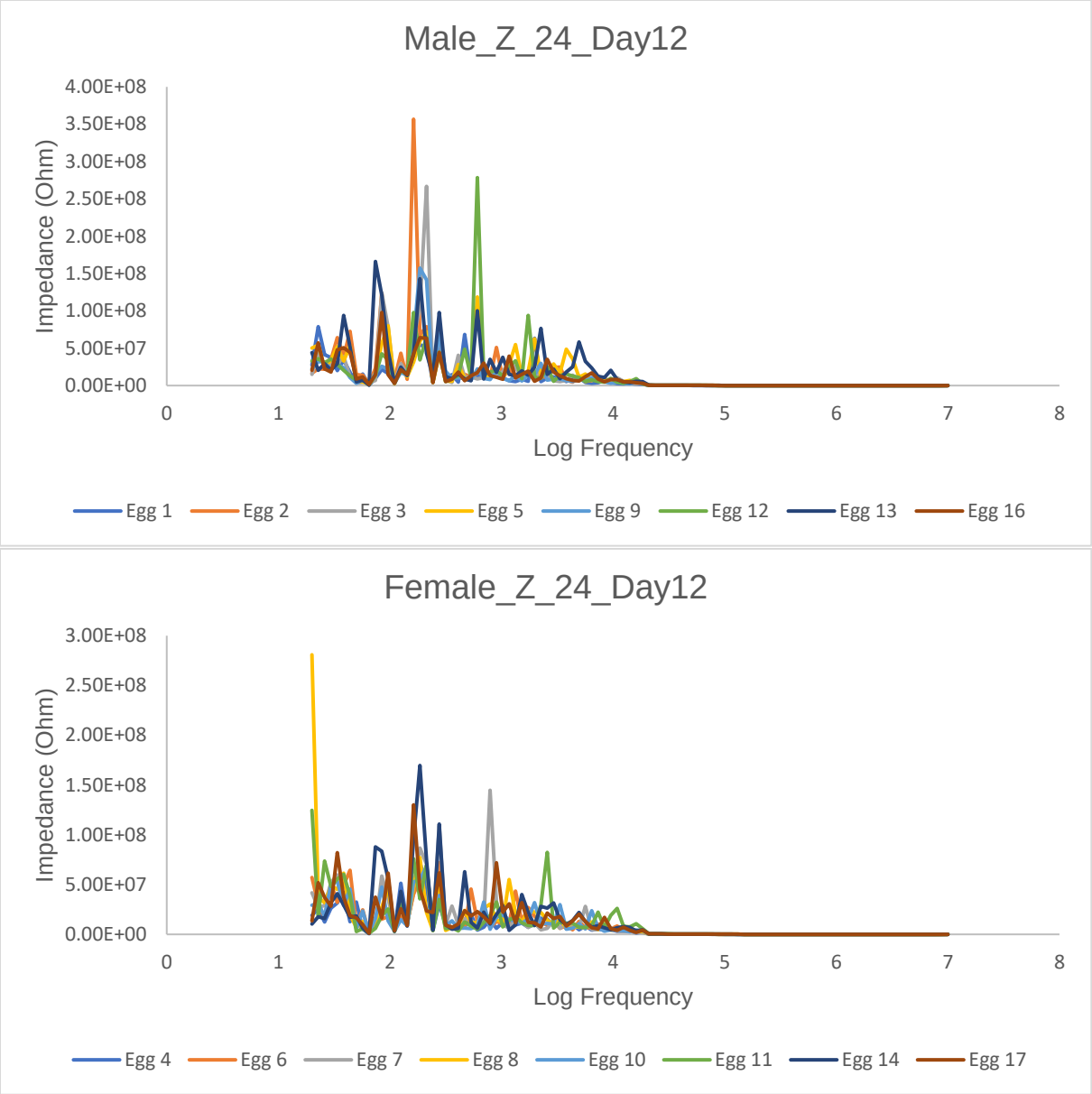


Figure A.47. Impedance measurement of electrode pair 2 and 4 of the eggs performed on day 12

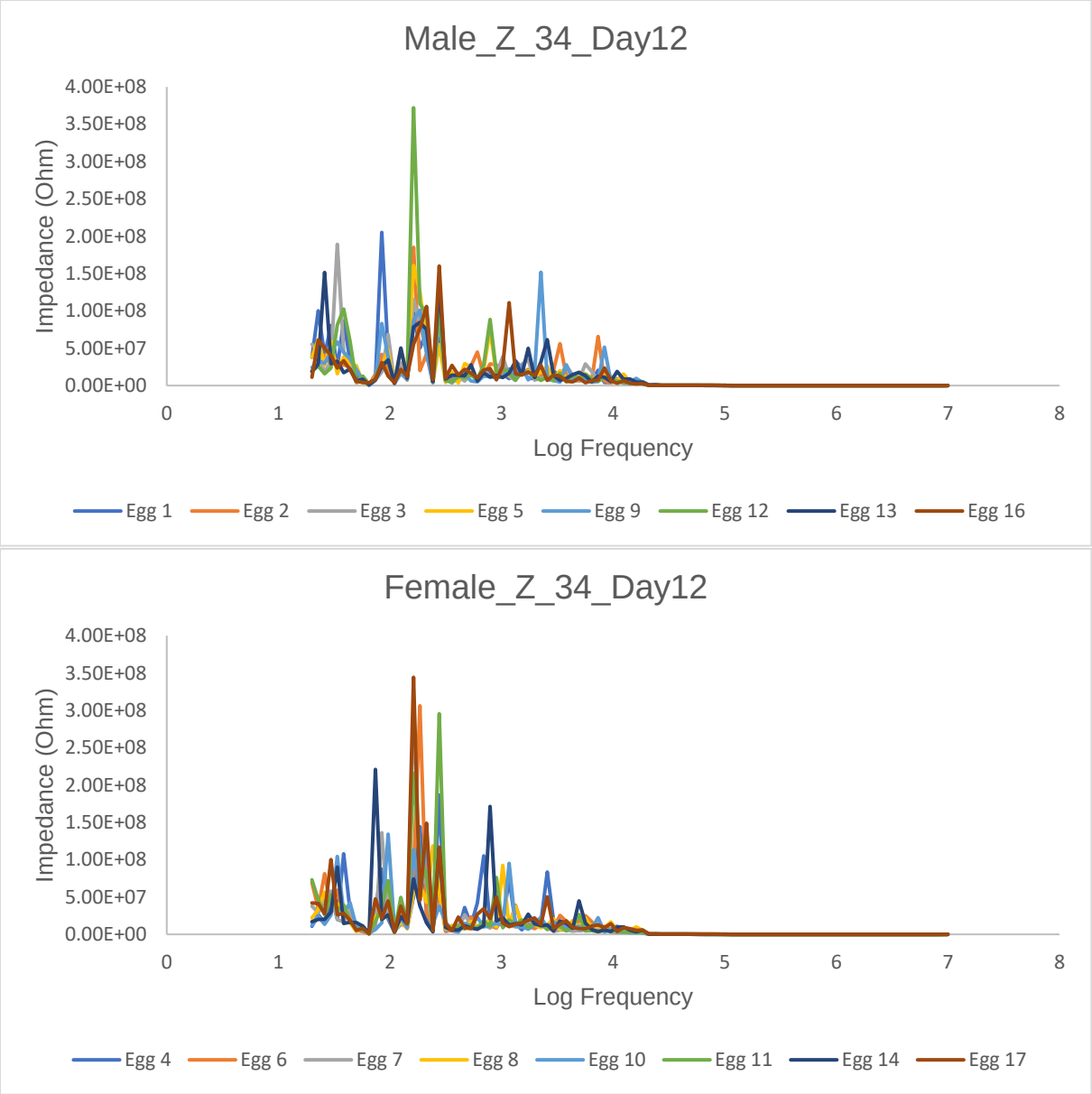


Figure A.48. Impedance measurement of electrode pair 3 and 4 of the eggs performed on day 12

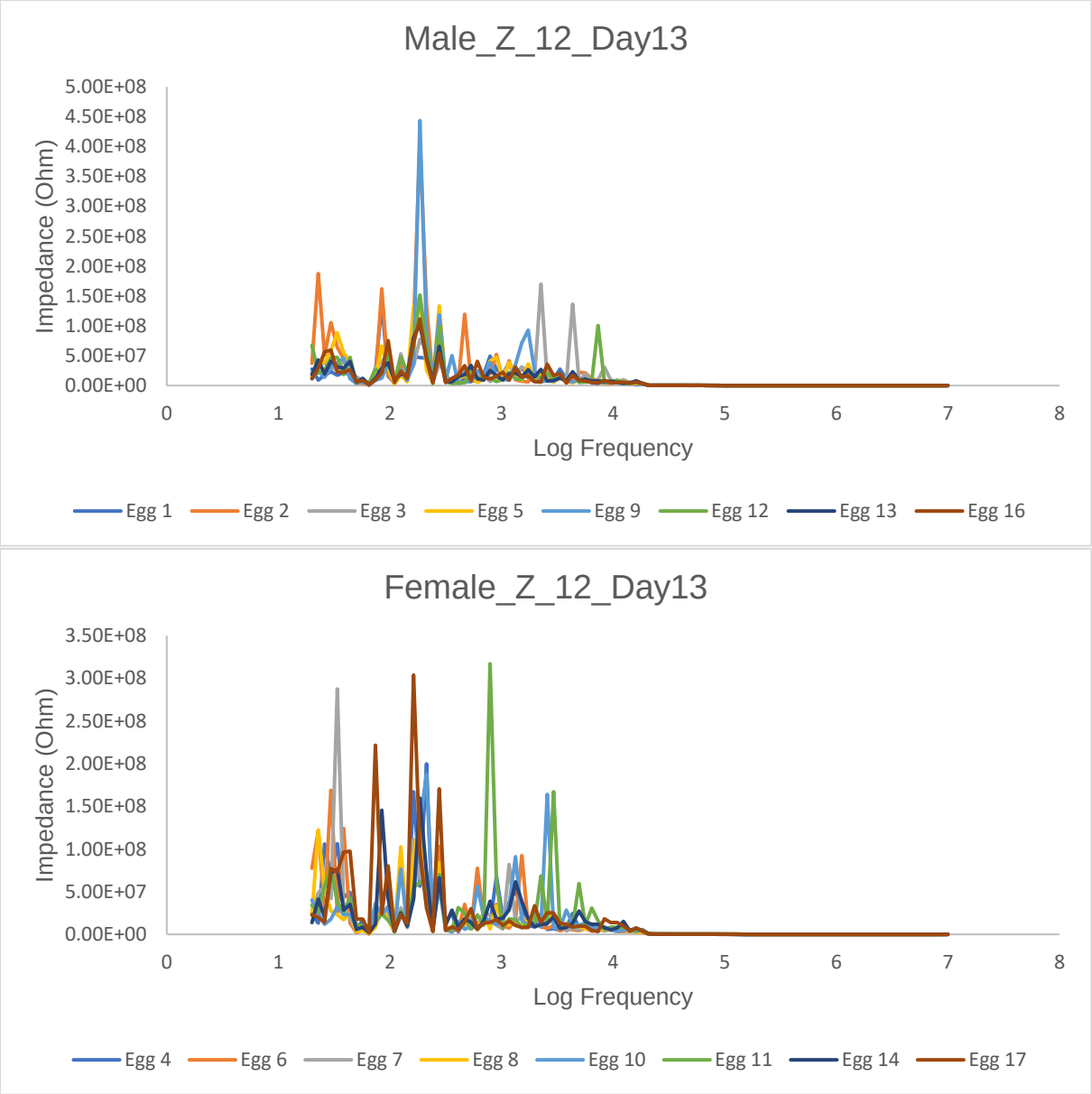


Figure A.49. Impedance measurement of electrode pair 1 and 2 of the eggs performed on day 13

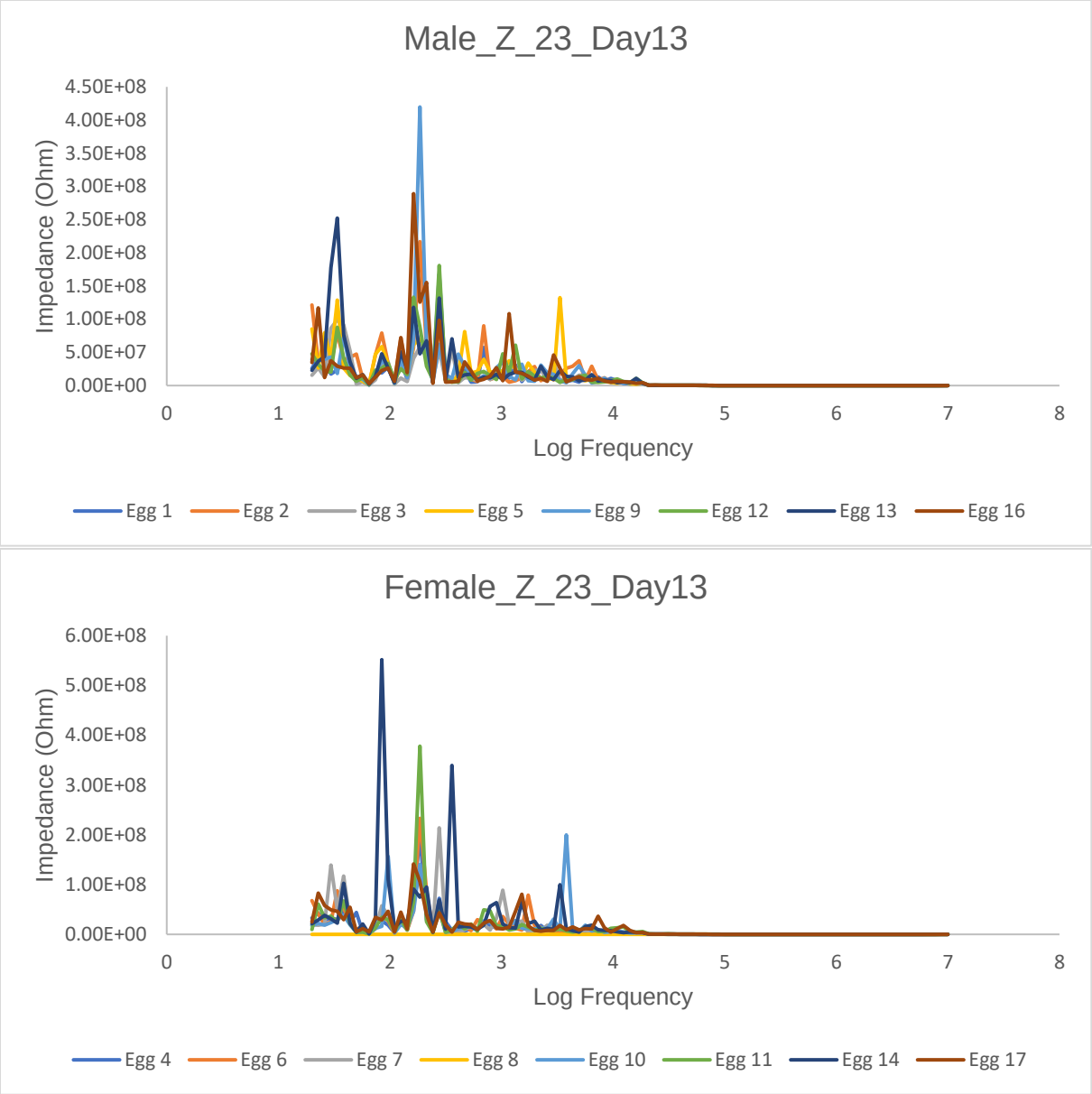
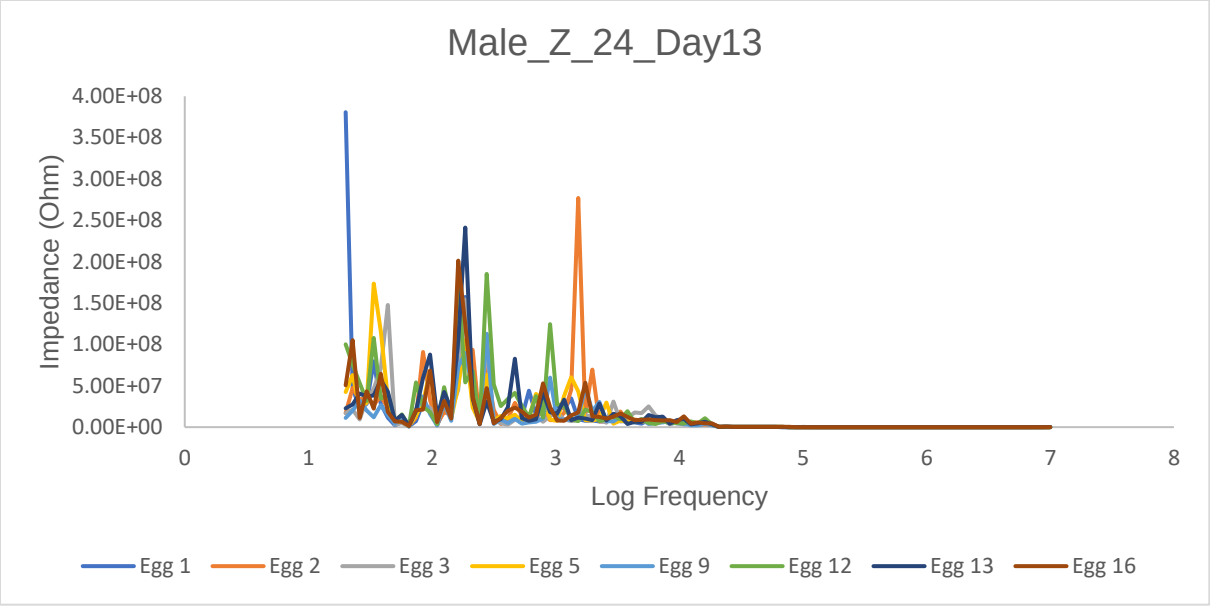


Figure A.50. Impedance measurement of electrode pair 2 and 3 of the eggs performed on day 13



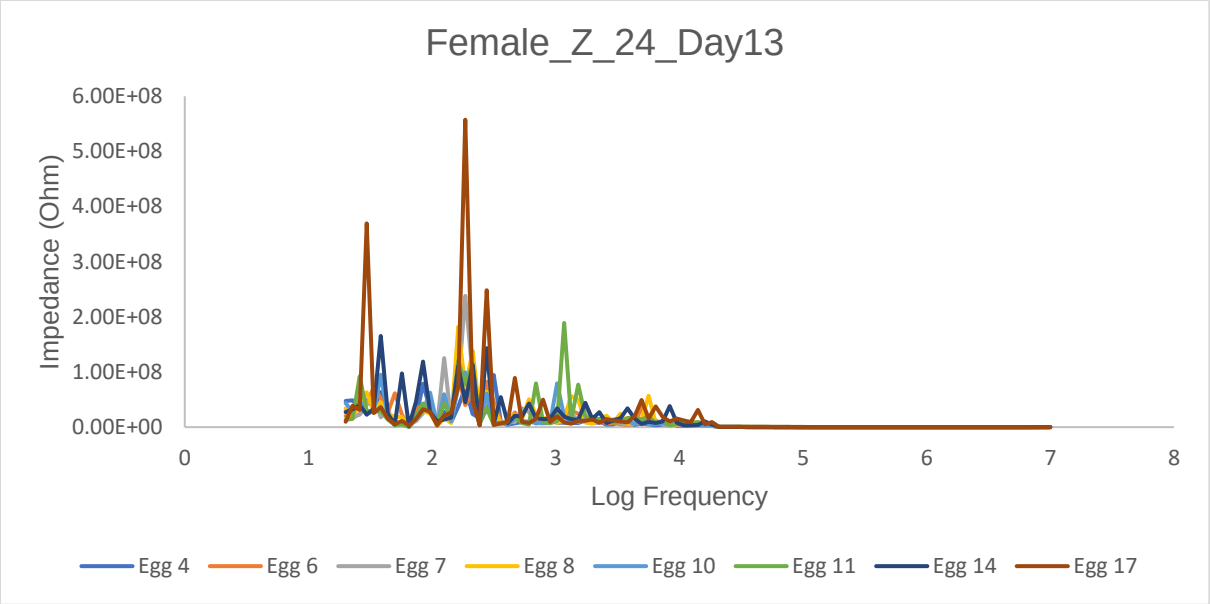
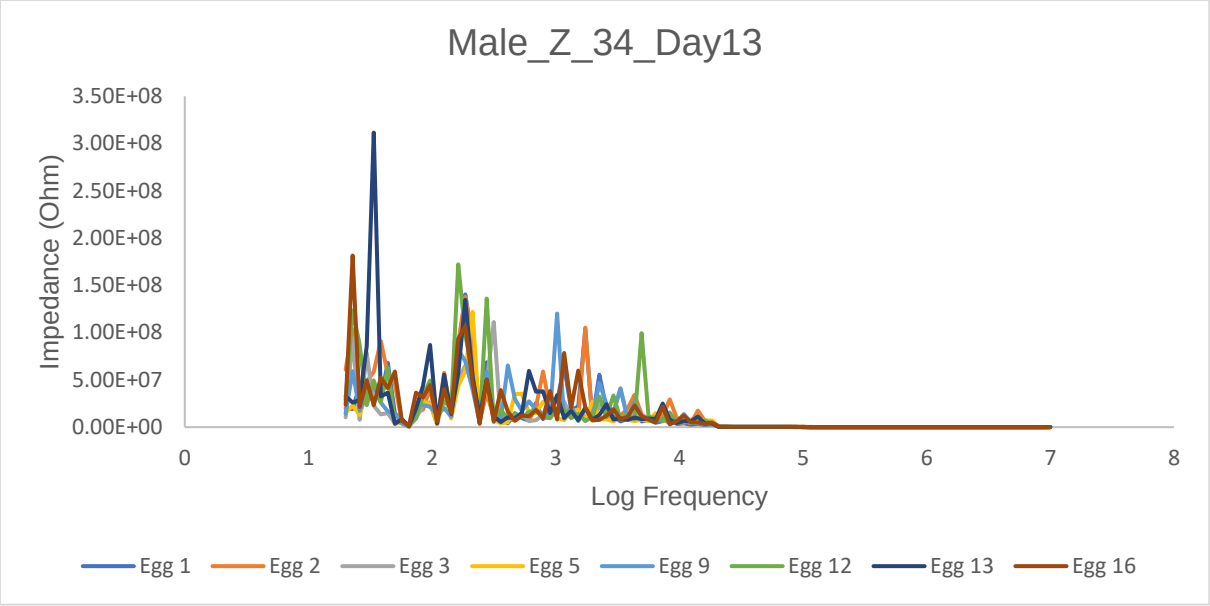


Figure A.51. Impedance measurement of electrode pair 2 and 4 of the eggs performed on day 13



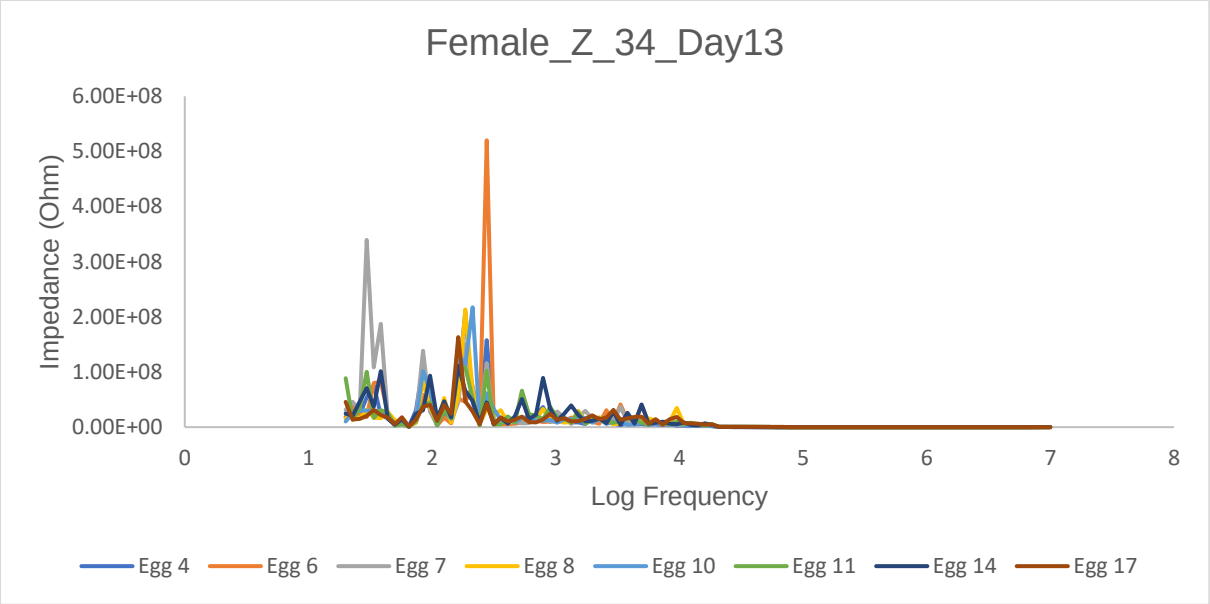


Figure A.52. Impedance measurement of electrode pair 3 and 4 of the eggs performed on day 13